

Source Diff for the Substantive Revision

Baseline commit 6b8d85e to the final revision working tree

September 2026

This change-marked artifact is a unified source diff. It covers the main manuscript, online appendix, generated table sources, and revision documents. Binary artifacts are not embedded in the patch. The canonical editable files and generated PDFs accompany this document.

```
diff --git a/paper/appendix/appendix.tex b/paper/appendix/appendix.tex
index 30bcb19..0f3c066 100644
--- a/paper/appendix/appendix.tex
+++ b/paper/appendix/appendix.tex
@@ -3,12 +3,12 @@
\usepackage{styles/working-paper}
\usepackage{styles/yax-common}
\graphicspath{{figures/}}
-\hypersetup{pdftitle={Online Appendix to AI Exposure or Occupational Composition?},pdfauthor={Lily Luo}}
+\hypersetup{pdftitle={Online Appendix to Occupational AI Exposure and Young-Worker Employment},pdfauthor={Lily Luo}}
\begin{document}
\begin{center}
{\Large\bfseries Online Appendix to\par}
\vspace{0.6em}
- {\Large ‘‘AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment’’\par}
+ {\Large ‘‘Occupational AI Exposure and Young-Worker Employment: Support and Comparisons in the CPS’’\par}
\vspace{0.8em}
Lily Luo\[\[0.25em]
September 2026
@@ -17,16 +17,15 @@
\tableofcontents
\clearpage
\appendix
-\input{appendix/sections/appendix_A_measurement.tex}
-\input{appendix/sections/appendix_B_mapping.tex}
-\input{appendix/sections/appendix_C_support.tex}
-\input{appendix/sections/appendix_D_inference.tex}
-\input{appendix/sections/appendix_E_stock.tex}
-\input{appendix/sections/appendix_F_dynamics.tex}
-\input{appendix/sections/appendix_G_flows.tex}
-\input{appendix/sections/appendix_H_reallocation.tex}
-\input{appendix/sections/appendix_I_FG_AE.tex}
-\input{appendix/sections/appendix_J_LOCO.tex}
-\input{appendix/sections/appendix_K_power.tex}
-\input{appendix/sections/appendix_L_reproducibility.tex}
+\input{appendix/sections/r3_A_data.tex}
+\input{appendix/sections/r3_B_mapping.tex}
+\input{appendix/sections/r3_C_occupational_support.tex}
+\input{appendix/sections/r3_D_characteristics.tex}
+\input{appendix/sections/r3_E_inference.tex}
+\input{appendix/sections/r3_F_dynamics.tex}
+\input{appendix/sections/r3_G_flows.tex}
+\input{appendix/sections/r3_H_bcc_architecture.tex}
+\input{appendix/sections/r3_I_reproducibility.tex}
+\bibliographystyle{styles/michailat-paper}
+\bibliography{bibliography/references}
\end{document}
diff --git a/paper/appendix/sections/appendix_A_measurement.tex b/paper/appendix/sections/appendix_A_measurement.tex
index 9d569c0..59f76f5 100644
--- a/paper/appendix/sections/appendix_A_measurement.tex
+++ b/paper/appendix/sections/appendix_A_measurement.tex
@@ -70,7 +70,7 @@
For the task family, the raw primitives satisfy  $\mathbb{E}[\alpha] = D\$,  $\mathbb{E}[\beta] = D+S/2\$, and  $\mathbb{E}[\gamma] = D+\$. On strict common support, the maximum absolute gap between raw beta and  $(\alpha + \gamma)/2\$ is  $7.0 \times 10^{-13}\$. Mapping and aggregation preserve the identity. Separate employment-weighted standardization changes the relative location and scale and creates a maximum numerical midpoint gap of 0.5112; it does not create a third primitive.

- The external-architecture admission rule was written before fitting external outcomes. Webb AI and the reversed OECD capability gap meet the declared construct, public-source, non-title mapping, full-component, distinct-cut, and 80-percent coverage criteria. Webb software remains a computerization control; its source correlation with Webb AI is 0.7021. Frey--Osborne remains an automation-risk comparator, not an AI architecture.
+ The external-architecture admission rule was written before fitting external outcomes. Webb AI and the reversed OECD capability gap meet the declared construct, public-source, non-title mapping, full-component, distinct-cut, and 80-percent coverage criteria. Webb software remains a computerization control; its source values are occupation-level patent--task overlap percentages on a 0--100 scale, while every fitted Webb interaction uses a preperiod employment-weighted standardized score. Its source correlation with Webb AI is 0.7021. Frey--Osborne remains an automation-risk comparator, not an AI architecture.

\subsection{Dependence and candidate census}

diff --git a/paper/appendix/sections/appendix_K_power.tex b/paper/appendix/sections/appendix_K_power.tex
index c898049..e4db1c9 100644
--- a/paper/appendix/sections/appendix_K_power.tex
+++ b/paper/appendix/sections/appendix_K_power.tex
@@ -1,6 +1,6 @@
\section{Prospective precision, realized resolution, and failed diagnostics}\label{app:power}

- The prospective primary SE was .01217 and the realized occupation-cluster SE is .04441, a ratio of 3.649. For beta minus alpha, .01167 versus .03759 gives a ratio of approximately 3.22 on the updated paired fit. The frozen paired design had 80-percent power to detect coefficient differences of approximately 3.27 percentage points under its planning DGP. No verified published estimate matched the YAX age groups, stock estimand, Q5--Q1 contrast, and scale, so binding equivalence inference was retired rather than assigned an arbitrary smallest effect of interest.
+ The prospective primary SE was .01217 and the realized occupation-cluster SE is .04441, a ratio of 3.649. For beta minus alpha, .01167 versus .03759 gives a ratio of approximately 3.22 on the updated paired fit. The frozen paired design had 80-percent power to detect coefficient differences of approximately 3.27 percentage points under its planning DGP. No verified published estimate matched the paper's age groups, stock estimand, Q5--Q1 contrast, and scale, so binding equivalence inference was retired rather than assigned an arbitrary smallest effect of interest.

At realized precision, the primary normal-theory MDEs(80) is .1244 log points. Architecture-pair MDEs range from .0609 to .1689. Those values belong beside the architecture comparisons because they explain why point-estimate variation and statistical distinguishability are different claims.

diff --git a/paper/appendix/sections/r3_A_data.tex b/paper/appendix/sections/r3_A_data.tex
new file mode 100644
index 0000000..426dc65
--- /dev/null
+++ b/paper/appendix/sections/r3_A_data.tex
@@ -0,0 +1,53 @@
+ \section{Data Reconstruction and Treatment Contract}\label{app:data}
+
+ \subsection{Source files and sample flow}
+
+ The stock analysis uses two restricted IPUMS CPS extracts. The wide file supplies January 2017--July 2026 records; a second file supplies the five March Basic Monthly samples that replace incorrectly selected March ASEC samples in 2017--2021. The replication manifest records extract identifiers, creation metadata where available, byte sizes, and SHA-256 hashes without redistributing microdata. The March replacement occurs before eligibility filters or occupation routing. The superseded rows contribute no positive-final-weight analysis records; the replacement adds 252,862 eligible records.
+
+ The wide file is IPUMS extract 9: its request was submitted on August 25, 2026, and the completed-file receipt was recorded four minutes later. Separate hashes authenticate its submitted specification, data, DDI, and basic codebook. The March repair is extract 11; its actual request, returned API response, data, DDI, and basic-codebook hashes are verified, but its surviving API response contains no submission or completion timestamp, so none is inferred from filesystem metadata. Flow models additionally use extract 10 for \texttt{LNKFWMWT}; it was recorded complete when checked on August 31, 2026, but the check time is not called an API completion time. Exact timestamps and hashes appear in the input manifest and provenance receipt. Restricted files, account metadata, download links, and direct identifiers are not redistributed.$$$$$ 
```

+

`\begin{table}[\hbtbp]`
`\centering`
`\caption{Calendar and cell reconstruction audit}\label{tab:app_calendar}`
`\begin{tabular}{lrr}`
`\toprule`
`Object & Count & Audit rule \\`
`\midrule`
`Expected calendar months & 115 & January 2017--July 2026 \\`
`Observed months & 114 & October 2025 not collected \\`
`Static months & 113 & December 2022 transition excluded \\`
`Primary occupations & 468 & named fixed support \\`
`Complete occupation--month grid & 52,884 &  $468 \times 113$  \\`
`Positive-total fitted cells & 51,891 & includes one-sided zeros \\`
`Both-age-zero cells & 993 & no likelihood term \\`
`One-sided zero cells retained & 12,965 & finite boundary contributions \\`
`\bottomrule`
`\end{tabular}`
`\tabnote{October 2025 is absent, not coded zero or interpolated. December 2022 is observed but excluded from the static pre/post fit.}`
`\end{table}`

+

Eligibility requires age 18--65, employment, a valid occupation, and positive `\texttt{WTFINL}`. The analysis scans 9,843,021 source rows, retains 5,322,047 eligible source records, and creates 6,188,956 route-expanded descendants. The last count is fractional before 2020 and is not a count of distinct respondents. The principal age groups are formed after routing. Of the complete grid, 13,883 young cells and 1,068 older cells have zero respondent-equivalent records. These sparse-cell facts motivate the finite-sample audit; they do not justify selecting cells on realized young counts.

+

`\subsection{Reconstructed preperiod treatment}`

+

The substantive treatment is rebuilt using only January 2017--November 2022 stock. On the fixed 468-occupation support, the employment-weighted beta mean is 0.37520 and standard deviation is 0.21041. The tie-preserving cut values are 0.153846, 0.324615, 0.456522, and 0.537037. Webb software exposure is standardized over the same 71-month construction window. Table~\ref{tab:app_treatment} compares the rebuilt contract with the historical production contract.

+

`\begin{table}[\hbtbp]`
`\centering`
`\caption{Historical and reconstructed treatment contracts}\label{tab:app_treatment}`
`\begin{tabular}{lrrr}`
`\toprule`
`Contract & Construction window & Post stock used & Q5--Q1 coefficient \\`
`\midrule`
`Historical production & full static panel & Yes &  $-0.134554$  \\`
`Reconstructed & Jan. 2017--Nov. 2022 & No &  $-0.132109$  \\`
`\bottomrule`
`\end{tabular}`
`\tabnote{Both coefficients use the corrected 113-month outcome calendar and identical occupation support. Nine occupations change quintile. The reconstructed-minus-historical paired difference is 0.002445 with 95-percent interval  $[-0.001269, 0.006158]$  .}`
`\end{table}`

+

The reconstructed contract contains 129, 99, 82, 67, and 91 occupations in Q1--Q5. Corresponding preperiod stock shares are 20.06, 19.96, 20.16, 20.31, and 19.51 percent. The complete membership file identifies all nine occupations whose historical and rebuilt labels differ. All revised substantive models use the reconstructed labels unless explicitly described as a historical audit.

+

`\subsection{Survey-production dates}`

+

Official BLS documentation establishes population-control discontinuities in January 2025 and January 2026. BLS did not revise the preceding official months onto the new bases and does not publish counterfactual age-by-occupation weights. October 2025 was not collected; November collection began late, used reduced overlap and a modified weighting procedure, and had a lower response rate. IPUMS later processed the reissued January 2026 file and corrections to longitudinal identifiers. The stock and flow manifests preserve their respective extract dates and hashes. Endpoint and exclusion checks diagnose reliance on these dates, but no aggregate adjustment factor is imposed on detailed cells.

```
diff --git a/paper/appendix/sections/r3_B_mapping.tex b/paper/appendix/sections/r3_B_mapping.tex
new file mode 100644
index 00000000..2bdf34e
--- /dev/null
+++ b/paper/appendix/sections/r3_B_mapping.tex
@@ -0,0 +1,106 @@
\section{Occupational Mapping and Analysis Support}\label{app:mapping}
+
\subsection{Bridge construction}
+
The early panel reports Census-2010 occupations and the later panel Census-2018 occupations. Exposure inputs additionally span SOC vintages. The versioned bridge first routes 503 early source codes into 568 target codes. Fifty-six sources have more than one target, with maximum multiplicity seven. Official shares partition each source record's final weight; separate young and older routing would require validation data that the public adjacent-vintage files do not provide. The routed early and current stocks reconcile to their sources with relative error below  $2 \times 10^{-16}$ .
```

+

In the frozen mapping-only lookup, the equal-administrative AIOE variant has full-route coverage for 480 of the 503 pre-2020 source codes. This 480-code mapping count is not the regression sample: intersecting the rebuilt beta and Webb inputs with positive young preperiod stock yields the 468-occupation primary analysis support below.

+

The primary joint beta/Webb universe contains 468 occupations and 87.19 percent of stock in the audited candidate universe. Table~\ref{tab:app_exclusions} gives mutually exclusive exclusion counts. A machine-readable file lists each included and excluded occupation, its preperiod stocks, scores, and reason. This file, rather than a hard-coded count in the paper, defines support.

+

`\begin{table}[\hbtbp]`
`\centering`
`\caption{Primary-support exclusions}\label{tab:app_exclusions}`
`\begin{tabular}{lr}`
`\toprule`
`Reason & Occupations \\`
`\midrule`
`Missing Rule-A beta only & 41 \\`
`Missing Webb software only & 22 \\`
`Missing both exposure inputs & 8 \\`
`Nonpositive young preperiod stock & 2 \\`
`\midrule`
`Total excluded & 73 \\`
`Primary retained & 468 \\`
`\bottomrule`
`\end{tabular}`
`\end{table}`

+

`\subsection{Separating score correction from population expansion}`

+

A literal exact-code merge between SOC-2010 AIOE and post-2018 employment data leaves only 3.33 percent of computer-and-mathematical employment. The version-aware route reaches 97.7 percent. The four-row continuous comparison separates the two operations that an aggregate "crosswalk effect" would conflate:

+

The source workbook standardizes AIOE without employment weights across 774 source occupations: its mean is approximately zero and its sample standard deviation is approximately one. A raw AIOE point is therefore one unit of that source index, not a probability or percentage. As a common descriptive increment, the source-workbook interquartile difference is 1.879 raw units. The historical four-row audit below uses a different, explicitly fixed unit: one full-window employment-stock-weighted standard deviation of repaired equal-administrative AIOE on the expanded 495-occupation AIOE-plus-Webb support. The retained artifact does not store the numerical weighted standard deviation, so I do not manufacture a raw-point conversion.

+

`\begin{table}[\hbtbp]`
`\centering`
`\caption{AIOE taxonomy repair and support expansion}\label{tab:app_crosswalk}`
`\begin{tabular}{llrr}`
`\toprule`
`Score & Support & Occupations & Coefficient per fixed SD \\`
`\midrule`
`Literal original & original & 410 &  $-0.01885$  \\`
`Repaired & same original & 410 &  $-0.01920$  \\`
`Repaired & expanded & 495 &  $-0.03156$  \\`
`Repaired & expanded, excluding SOC15 & --- &  $-0.02940$  \\`
`\bottomrule`
`\end{tabular}`

+\\tabnote{The first row is an invalid vintage merge retained as an implementation audit, not a defensible specification. Row 1 to 2 isolates score repair on fixed support; row 2 to 3 changes the population; the last row tests whether readmitted computer-and-mathematical occupations alone account for that expansion movement. Every coefficient uses the fixed weighted-SD unit defined in the text; none is a percentage or a coefficient per raw AIOE point.}

+\\end{table}

+

+The invalid merge is documented as a failure encountered in this project. I do not claim that a published paper made the same error without direct verification of its code and occupational universe.

+

+\\subsection{Split-route and age-allocation sensitivity}

+

+Within the primary support, split sources contribute 14.93 percent of young and 17.75 percent of older early stock. Ninety-nine target occupations can receive a split route. A one-to-one-target restriction and an aggregation to stable Census-2010 categories both retain negative historical-contract estimates, but each changes the target population. They are mapping sensitivities, not repairs of the primary bridge.

+

+Using the same route proportions by age is transparent but unvalidated. Accounting ranges that allocate every ambiguous source in all feasible ways place the 2017--2019 young-to-older stock ratio between 0.1031 and 0.1480 in Q1 and between 0.0790 and 0.1175 in Q5. These are stock-accounting bounds, not bounds on the nonlinear regression coefficient. An earlier historical-contract tilt involved 6.88 percent of early stock and is retained only as an implementation checkpoint; the current-contract sensitivity follows below. No adjacent coding vintage is treated as a same-person validation sample.

+

+\\subsection{Current-contract route-allocation sensitivity}

+

+The current 113-month contract repeats the exercise on the 468-occupation support. Thirty-one split source occupations, accounting for 14.86 percent of supported early-period routed stock, receive jointly feasible young-versus-older high/low allocation-odds tilts $\$K\\in\\{0.25, 0.5, 2/3, 1, 1.5, 2, 4\\}$. Each grid point preserves source--age--month mass and official structural route zeros. Fixed-label rows hold the no-tilt exposure groups constant; rebuilt-treatment rows recompute them after allocation. Table-\\ref{tab:app_mapping_tilts} keeps those objects separate.

+

+\\input{tables/r3_appendix_mapping_tilts.tex}

+

+All pooled paired occupation-clustered intervals relative to $\$K=1$ contain zero under both label rules. For rebuilt treatment in the family-month model, however, the coefficient moves from $\$-0.0217$ at $\$K=1$ to $\$0.0439$ at $\$K=0.25$; the paired movement is $\$0.0656$ with interval $\$[0.0031, 0.1280]$. Five of the six rebuilt-treatment family-month comparisons exclude zero, even though rebuilding changes at most two occupations at a grid point. Thus the pooled pattern is stable over this declared grid while the family-conditioned estimand is sensitive to route allocation and resulting membership. Null intervals are nondetections, not equivalence.

+

+A separate 25-point grid allows young and older allocation odds to vary jointly while maintaining the same mass constraints. The pooled coefficient ranges from $\$-0.1599391$ to $\$-0.0702179$. No outcome-directed optimization followed inspection of the grid, but its extrema remain an explored feasible-grid envelope rather than sharp or global bounds.

+

+Finally, excluding every current occupation that can receive an inbound one-to-many route leaves 369 occupations. On this clean-route support, pooled coefficients are $\$-0.1642$ ($\$[-0.2665, -0.0618]$) with fixed labels and $\$-0.14688$ ($\$[-0.2415, -0.0521]$) with rebuilt labels; family-month coefficients are $\$-0.0871$ ($\$[-0.2325, -0.0584]$) and $\$-0.0524$ ($\$[-0.1897, 0.0848]$). On a separate stable-Census-2010 support of 408 occupations, the estimate is $\$-0.1522$ ($\$[-0.2438, -0.0606]$). On the full 468-occupation support restricted to the 77 months from 2020 onward, the current-label estimates are $\$-0.1181$ ($\$[-0.1941, -0.0420]$) pooled and $\$-0.03048$ ($\$[-0.1716, 0.1108]$) with family-by-month effects. These three exercises change the sample or calendar as well as the labeling rule and are not a single nested robustness ladder; none validates one coding vintage against another.

+

+The inherited optional symmetric coding-error sensitivity is not executed: no authenticated dual-coded CPS sample or validated external misclassification matrix is available, immediate occupation reversals mix true mobility with reporting and coding changes, and the route-allocation exercises above are not relabelled as a coding-error correction.

diff --git a/paper/appendix/sections/r3_C_occupational_support.tex b/paper/appendix/sections/r3_C_occupational_support.tex

new file mode 100644

index 0000000..95f26d8

--- /dev/null

+++ b/paper/appendix/sections/r3_C_occupational_support.tex

@@ -0,0 +1,126 @@

+\\section{Broad-Family Support, Profiles, and Influence}\\label{app:families}

+

+\\subsection{Family-by-quintile support}

+

+Table-\\ref{tab:app_family_support} reports, for every SOC two-digit family, the number of retained occupations, preperiod stock share, exposure range, quintiles present, direct-tail overlap, and nuisance-adjusted Q5 information share. Only arts/media (SOC27), health practitioners (SOC29), health support (SOC31), and sales (SOC41) span Q1 and Q5 under the reconstructed comparison. Other families connect the Q5--Q1 regression coefficient through intermediate quintiles and the imposed common Q2--Q5 slopes.

+

+\\input{tables/r3_appendix_family_support.tex}

+

+Table-\\ref{tab:app_family_information} makes the loss and concentration of identifying information explicit. Family-by-month conditioning retains 29.7 percent of the pooled Q5 target's nuisance-adjusted information, while the direct-tail comparison has only 6.2 effective occupations and its five largest occupation contributions supply 77.8 percent of information. Its normal-theory $MDE_{\{80\}}$ is 0.4576 log point. The continuous within-family companion is more diffuse and has $MDE_{\{80\}}=0.0313$ per one within-family standard deviation, but it imposes a common slope across families.

+

+\\input{tables/r3_appendix_family_information_precision.tex}

+

+The direct-tail population retains only occupations in the four spanning families that belong to Q1 or Q5. It contains 29 occupations and 5.03 percent of primary preperiod stock. In the SOC2-by-calendar-month model its coefficient is 0.1494, with interval $\$[-0.1694, 0.4682]$. This benchmark is valuable because it exposes direct overlap; it is not a preferred estimate for the original population, and the interval includes large effects of either sign.

+

+Table-\\ref{tab:app_direct_tail_occupations} names every occupation supplying that comparison. The imbalance is economically visible: for example, the health-support Q5 endpoint consists only of medical transcriptionists, while its Q1 endpoint includes nursing assistants and home health aides. Naming the membership clarifies what the direct-tail estimate compares; it does not make the sparse endpoints representative of their broad families.

+

+\\input{tables/r3_appendix_direct_tail_occupations.tex}

+

+\\subsection{Supported edges and broader beta support}

+

+The fully interacted within-family model yields 89 supported pairwise quintile-edge functionals. All 89 estimates and their standard errors are retained in the machine-readable results. They are distinct family-specific comparisons rather than repeated national Q5--Q1 estimates. Their joint covariance has rank 50, so an 89-degree-of-freedom Wald test is invalid and is blocked in the results rather than replaced by a selected list of pointwise $\$p$ -values. The associated 50-dimensional family-specific coefficient system is reported separately with rank-aware inference.

+

+Table-\\ref{tab:app_pairwise_aggregates} gives a transparent summary of the supported edges. Each row averages only family-specific contrasts for which both endpoint quintiles are observed, using fixed preperiod endpoint-stock weights. Because the eligible family set changes across rows, these aggregates do not impose a common national quintile profile and cannot be read

```

+as ten estimates of one population parameter.
+
+\input{tables/r3_appendix_pairwise_aggregates.tex}
+
+Table-\ref{tab:app_broader_support} asks whether sparse tails arise solely from
+requiring Webb software exposure. Relaxing that requirement expands the
+population from 468 to 490 occupations, but the added occupations represent
+only 1.88 percent of broader-support preperiod stock. Direct tails expand from
+29 to 62 occupations and from 5.03 to 12.61 percent of stock because SOC43
+joins the four original tail-spanning families. The overlap therefore improves
+at the margin but remains confined to five of 23 families. Coefficients change
+little in this descriptive population expansion; this is not a bound and does
+not make the two populations equivalent.
+
+\input{tables/r3_appendix_broader_support.tex}
+
+\subsection{Full exposure profile}
+
+The pooled Q2--Q5 coefficients are jointly different from zero
+($\chi^2_{2,4}=11.1383$, asymptotic $p=.0251$, multiplier $p=.0212$), but pairwise
+movements do not establish a monotone dose response. After SOC2-by-calendar-
+month conditioning, the four coefficients are jointly indistinguishable from
+zero ($\chi^2_{2,4}=2.0708$, asymptotic $p=.7227$, multiplier $p=.7422$). The
+four-coefficient paired movement likewise does not reject
+($\chi^2_{2,4}=5.4169$, asymptotic $p=.2471$, multiplier $p=.2368$), despite the
+single Q5 movement's pointwise interval excluding zero. Table-\ref{tab:app_profile}
+reports pointwise and simultaneous intervals and the common-draw movement for
+every group.
+
+\input{tables/r3_appendix_profile.tex}
+
+The continuous companion demeans beta within family using preperiod employment weights and imposes a common linear slope. With SOC2-by-calendar-month conditioning, the rebuilt coefficient is $-0.0025$
+$ per 0.1084 raw beta units. Its interval is $[-0.0240, 0.0191]$ in that standardized within-family unit. The unconditioned continuous coefficient is essentially zero as well. This is a
different functional form from the tail contrast.
+
+\subsection{Exact numerator, denominator, and family accounting}
+
+Table-\ref{tab:app_stock_accounting} separates the model-free aggregate
+Q5--Q1 stock-ratio movement into its young numerator and older denominator.
+The older tail difference reinforces the negative young movement but is
+smaller in absolute magnitude. A second exact accounting separates changes
+in within-family young-to-older ratios from changes in older family weights.
+Neither accounting is forced to equal the nonlinear regression target, and no
+component is interpreted as a causal percentage explained.
+
+\input{tables/r3_appendix_stock_accounting.tex}
+
+An outcome-free design audit explains why this accounting is not presented as
+two saturated single-age regressions. On the 52,884 single-age
+occupation--month rows, unrestricted occupation-by-month indicators span the
+entire row space and absorb all five requested exposure-by-post columns. An
+additive occupation-plus-calendar-month companion retains rank five for those
+columns, but no outcome-bearing companion coefficients were estimated. The
+table therefore reports exact stock identities under the paper's normalization,
+not a regression decomposition of the nonlinear grouped-binomial target.
+
+\subsection{Leave-one-family and occupation influence}
+
+Leave-one-family-out estimates are computed from the same family-conditioning program and reported with the omitted family's stock and target-information contribution. These rows answer whether a
broad comparison is concentrated; they are not used to choose which families belong in the main estimate. Table-\ref{tab:app_lofo} displays the eight largest absolute movements. Omitting
office/administrative support or food preparation moves the coefficient in opposite directions by about 0.04 log point, and every paired deletion interval includes zero.
+
+\input{tables/r3_appendix_lofo.tex}
+
+Figure-\ref{fig:influential_between_family_paths} displays the economically
+relevant endpoints behind those two largest leave-family-out movements: Q5
+office/administrative-support stocks and Q1 food-preparation stocks. These
+families do not themselves span both tails, which is precisely why their paths
+inform the broad between-family comparison rather than the sparse direct-tail
+benchmark. The selection is outcome-informed and the figure is diagnostic;
+it neither identifies a pandemic mechanism nor supplies a deletion rule.
+
+\begin{figure}[htbp]
+\centering
+\includegraphics[width=0.96\textwidth]{r3_figure4_influential_family_paths.pdf}
+\caption{Employment paths for the two largest leave-family-out movers}\label{fig:influential_between_family_paths}
+\tabnote{The panels show Q1 food-preparation and serving occupations (SOC35) and Q5 office and administrative-support occupations (SOC43), the family-tail endpoints relevant to the broad Q5--Q1
comparison. Young and older CPS-weighted employment stocks are indexed to their own 2019 means; October 2025 remains missing. The families were selected after outcomes were observed because
they generate the two largest absolute leave-family-out coefficient movements, so these paths are influence diagnostics rather than confirmatory evidence or a causal pandemic decomposition.
The red dashed line marks January 2023.}
\end{figure}
+
+Figure-\ref{fig:family_paths} shows the four direct-tail families, selected by descending nuisance-adjusted Q5 information rather than coefficient sign. Within each exposure tail, the young and older
employment stocks are shown separately and indexed to their own 2019 means. The figure therefore exposes which age series drives a tail's relative path without treating either age group as
untreated.
+
+\begin{figure}[htbp]
+\centering
+\includegraphics[width=0.96\textwidth]{r3_figure3_family_paths.pdf}
+\caption{Young and older employment stocks in direct-tail families}\label{fig:family_paths}
+\tabnote{Families are selected by direct-tail, SOC2-by-calendar-month Q5 nuisance-adjusted information, with ties broken by SOC2 code; selection does not use coefficient sign. Gray and blue
distinguish Q1 and Q5; solid and dashed series distinguish ages 22--25 and 26--65. Each series equals 100 at its own 2019 mean. The symmetric-log vertical scale is linear from zero to 10 and
logarithmic above 10 so that sparse-cell spikes do not obscure the other series; exact zero-stock months remain visible. The red dashed line marks January 2023. These are descriptive weighted
-stock paths without design-based bands.}
\end{figure}
+
+Occupation leave-one-out calculations are interpreted in the same way. Fast-food and counter workers have high signed influence because low-exposure service employment recovered differently after the
pandemic. Other occupations push in the opposite direction. Historical-contract diagnostics show that deleting the five largest absolute prior influences moves the coefficient toward zero,
whereas deleting ten or twenty can make it more negative. Symmetric influence trimming stays near the pooled estimate. Because every such deletion is outcome adaptive, none replaces the broad
-family regression.
+
+Economically defined service exclusions are cleaner diagnostics but still change the population. Removing Q1 food-service occupations attenuates the historical-contract absolute coefficient by less
than one tenth; removing the broader SOC35/37/39 set of food, building-and-grounds, and personal-care service occupations does not attenuate it. Protective-service occupations (SOC33) are not
part of that exclusion. These results reject a one-occupation explanation, not hypotheses about reopening, immigration, wages, or labor supply that the CPS design does not separately identify.
diff --git a/paper/appendix/sections/r3_D_characteristics.tex b/paper/appendix/sections/r3_D_characteristics.tex
new file mode 100644
index 0000000..516a906
--- /dev/null
+++ b/paper/appendix/sections/r3_D_characteristics.tex
@@ -0,0 +1,141 @@
+\section{Occupational Characteristics and Heterogeneity}\label{app:characteristics}
+
+\subsection{Inputs, timing, and common-support comparisons}
+
+The characteristic exercise uses one variable at a time before considering a parsimonious joint model. Every row first re-estimates the unaugmented exposure model on exactly the support available for
that characteristic. It then adds the characteristic interacted with young status and the post indicator, reuses the same occupation multipliers, and forms the augmented-minus-baseline
difference within each draw. Sample loss and coefficient movement are therefore distinct objects. The corresponding paired minimum detectable effect is the two-sided, five-percent normal-
theory quantity $(1.96+0.84)SE(\Delta)$; it describes resolution and is never used as a second rejection rule.

```

+The leading computer-use variable is O*NET item 4.A.3.b.1, "Interacting With Computers," from release 24.3 (May 2020). Remotability is the Dingel--Neiman occupation classification. Wage and observed education composition are calculated from the CPS preperiod, while externally coded education requirements are kept conceptually separate. Routine and manual indices use preperiod O*NET inputs. All scales, source files, checksums, weighted correlations, and support counts are stored in the characteristic result directory.

+The primary matched comparison uses all 455 occupations with computer-use ratings and crosses computer conditioning with family-month paths. The four Q5-by-post coefficients are \$-0.09575\$ without either adjustment, \$-0.19660\$ with computer use only, \$0.03539\$ with family paths only, and \$-0.04665\$ with both. Relative to the pooled baseline, the corresponding paired movements are \$-0.10086\$ (\$[-0.17723,-0.02448]\$), \$0.13114\$ (\$[0.03517,0.22710]\$), and \$0.04910\$ (\$[-0.06418,0.16237]\$). The combined-minus-computer movement is \$0.14995\$ (\$[0.06039,0.23951]\$), while combined-minus-family is \$-0.08204\$ (\$[-0.15461,-0.00947]\$) under occupation clustering. These distinct conditional projections are not additive components.

+The computer-use-by-post coefficient itself is \$0.04602\$ per weighted standard deviation in the pooled computer-only model, with occupation interval \$[0.01171,0.08032]\$, and \$0.04868\$ (\$[0.01053,0.08684]\$) in the combined family-month model. The weighted standard deviation is 0.97290 O*NET points, and the beta--computer correlation is 0.7970. Adding computer use reduces Q5 information retention from 69.1 to 29.0 percent in the pooled model and from 42.5 to 30.0 percent in the family-month model. Thus coefficient movement and inflation occur in a thin residual comparison; the computer coefficient does not make computer use exogenous or purify the exposure coefficient as AI.

+I also evaluated formal omitted-confounder sensitivity rather than inserting the grouped-binomial pseudo-\$R^2\$ into a linear formula. The coefficient--stability calculations of `\citester{2019selection}`, the partial-\$R^2\$ framework of `\citetcinelli{2020sensitivity}`, and the endogenous-control selection model of `\citetdiegert{2022assessing}` are derived for linear regression targets with specific strength restrictions. The present target is one component of a vector-valued grouped-binomial fixed-effect projection with generated exposure groups; no verified mapping supplies those scalar inputs. A linear-probability companion would answer a different estimand. I therefore report no arbitrary breakdown value. This applicability ruling is not evidence that unobserved confounding is weak.

+The separate common-characteristic block uses 347 occupations and holds its support and primary-universe assignments fixed. Relative to its \$-0.12372\$ baseline, computer use moves the coefficient by \$-0.12475\$ (\$[-0.20934,-0.04016]\$). The paired movements for remotability, wages, education requirements, routine intensity, and manual/physical content each include zero. Adding all six gives \$-0.23331\$ and a paired movement of \$-0.10958\$ (\$[-0.22069,0.00152]\$). Family paths alone give \$-0.00348\$; adding all six characteristics to that model gives \$-0.10506\$, with a paired movement of \$-0.10158\$ (\$[-0.20057,-0.00258]\$). The common-support block covers 76.78 percent of rebuilt treatment-construction stock and likewise does not identify a purified AI contrast.

+`\subsection{Expanded support-specific and cumulative audit}`

+The separate characteristic module in `Tables-\ref{tab:appendix_characteristic_estimates}` and `\ref{tab:appendix_characteristic_information}` reports the support-specific comparisons and the uncontaminated portion of the cumulative sequence. Its one-at-a-time rows use each characteristic's maximal finite support, whereas its corrected joint rows use the literal 347-occupation common support. An audit found that an earlier version of this module had retained historical full-static-panel exposure groups and a 341-occupation historical intersection. The reported module now authenticates and uses the canonical corrected-preperiod membership and Webb normalization throughout; its 468-occupation baseline exactly reproduces \$-0.1321\$. The support-specific computer comparison retains 455 occupations and holds those rebuilt primary-universe groups fixed, while the focused exercise requires both computer-use and remotability data and rebuilds groups on its 408-occupation support. Their coefficients therefore differ by design, not by transcription. Rows at and after the superseded unbounded shortfall control are removed rather than mixed with the corrected shortfall population. Every addition and this corrective rerun are labeled post-outcome exploratory.

+`\input{tables/r3_appendix_characteristic_estimates.tex}`

+`\input{tables/r3_appendix_characteristic_information.tex}`

+`\subsection{Pandemic shortfall}`

+The corrected exercise uses ages 22--65 throughout and admits only one-to-one occupation mappings with sufficient 2017--2019 histories. It retains 363 occupations, rather than the central 468. Linear trend predictions for total employment are bounded below at zero; predicted young employment shares are bounded to $[0,1]$. This is consequential: before bounding there are 174 negative total predictions and 1,258 out-of-range share predictions. The characteristics are the average bounded trend-prediction-minus-realized gaps from January 2020 through November 2022; December 2022 remains the transition month.

+On this narrower common support, the pooled and family-month baseline coefficients are \$-0.16417\$ and \$-0.08708\$. Total-stock conditioning moves them by 0.00040 and \$-0.00050\$; young-relative conditioning moves them by 0.00114 and 0.00613. `Table-\ref{tab:appendix_shortfall}` reports linked-household sampling sensitivities that regenerate the shortfall inside each draw. All four conditioning-movement intervals include zero. Jointly regenerating exposure labels changes an average of 1.99 occupations per draw and leaves the young-relative family-month movement at 0.00613 with sampling-sensitivity standard error 0.00695.

+All 399 linked-household draws finish with zero failed joint refits. The largest binding bootstrap-of-bootstrap endpoint Monte Carlo standard error is +0.00979 log point, below the declared 0.01 threshold, so the predeclared expansion to 799 draws is not triggered.

+The controls are generated from related CPS outcomes and may share sampling error and regression-to-the-mean with the dependent stocks. A deterministic household split gives a similar average but heterogeneous directions; it does not eliminate PSU-level common error. The bounded functional form and changed population also remain consequential. The exercise therefore does not establish that pandemic dynamics are absent, and the linked-household intervals are not complete CPS design-based inference.

+`\input{tables/r3_appendix_shortfall.tex}`

+`\subsection{Industry, education, and exact age}`

+The industry analysis is rebuilt at the occupation-by-broad-industry-by-age-by-month level rather than assigning each occupation a dominant industry. The risk set covers all 468 occupations and 3,127 preperiod-connected occupation--industry cells. Relative to the stratified baseline coefficient of \$-0.1376\$, adding 13 broad-industry-specific young-by-post shifts gives \$-0.0987\$. The paired movement is \$0.0389\$, with interval $[0.0082,0.0697]$ and paired $\mathrm{MDE}_{(80)}=0.0444$. The conditioned coefficient remains below zero. These are statements about the microcell objective, not firm-level adoption.

+Education-specific models are estimated on a common occupational support. The BA-or-higher estimate is \$-0.1380\$ and the non-BA estimate is \$-0.1226\$; their paired difference is \$-0.0154\$, with interval $[-0.1557,0.1250]$ and paired $\mathrm{MDE}_{(80)}=0.2014$. The comparison is too imprecise to distinguish moderate education differences. Exact-age estimates for ages 22, 23, 24, and 25 are reported as one jointly estimated exploratory collection with simultaneous intervals. No single age is selected after inspection.

+`\input{tables/r3_appendix_heterogeneity.tex}`

+`\subsection{Older comparison bands, age composition, and enrollment}`

+`Table-\ref{tab:app_cohort_enrollment}` varies the older comparison group while holding the current exposure labels and occupational support fixed. The pooled coefficient remains negative across all five bands, but its magnitude is not constant. Against the 26--65 benchmark, paired movements for the 26--30, 31--40, and 41--50 comparisons have occupation-multiplier intervals $[-0.0412,0.0644]$, $[-0.0610,0.0056]$, and $[-0.0413,0.0271]$. The 51--65 movement is 0.0350 (\$[0.0036,0.0663]\$). Under family-month conditioning, every coefficient interval and all four corresponding paired movement intervals include zero when shocks are assigned at the occupation level. Under 22-family shock assignment the ages 22--25 versus 26--30 paired movement instead has interval $[-0.1712,-0.0041]$; the two procedures answer different stochastic questions and are not combined, and neither is evidence of a causal age effect.

+Fixing the older group's exact-age composition at its preperiod shares gives a
+pooled coefficient of $\$-0.131\$$ ($\$[-0.2202,-0.0425]\$$) and a family-month
+coefficient of $\$-0.0218\$$ ($\$[-0.1597,0.1161]\$$). Relative to observed age
+composition, the paired movements are 0.0008 ($\$[-0.0011,0.0027]\$$) and
+ $\$-0.0001\$$ ($\$[-0.0026,0.0023]\$$).
+
+Enrollment restrictions change the numerator rather than assigning exposure to
+nonemployed people. With all ages 26--65 in the denominator, restricting ages
+22--25 to nonenrolled workers changes the pooled coefficient from $\$-0.1321\$$ to
+ $\$-0.1239\$$; the paired movement is 0.0082 ($\$[-0.0125,0.0289]\$$). On the common
+22--64 enrollment-question universe, restricting both age groups to nonenrolled
+workers changes the pooled coefficient from $\$-0.1407\$$ to $\$-0.1354\$$, a paired
+movement of 0.0053 ($\$[-0.0144,0.0250]\$$). The analogous family-month movements
+are $\$-0.0068\$$ ($\$[-0.0393,0.0257]\$$) and $\$-0.0059\$$
+($\$[-0.0377,0.0260]\$$). Enrollment may itself respond to labor-market conditions,
+so these comparisons describe composition rather than a causal correction.
+Older workers throughout are comparison groups, not untreated controls.
+
+\input{tables/r3_appendix_cohort_enrollment.tex}
+
+\subsection{Young-worker composition and national cohort supply}
+
+Table-\ref{tab:app_young_composition} keeps employment-conditioned composition
+separate from national cohort supply. The Q1--Q5 columns summarize employed
+ages 22--25 in occupations on the fixed 468-occupation support. The national
+profile uses every positive-weight CPS record ages 22--25 and assigns no
+occupation or exposure quintile. Across Q1--Q5, the pre-to-post enrolled-share
+changes are $\$-0.10\$$, $\$-0.74\$$, $\$-2.00\$$, $\$-1.46\$$, and $\$-1.80\$$ percentage points;
+BA-or-higher shares change by 1.79, 1.71, 2.69, 1.48, and 3.81 points. No
+employed-quintile mean-age change exceeds 0.017 year in absolute value, and the
+largest exact-age-share change is 1.62 points. Nationally, enrollment changes
+from 0.2336 to 0.2211, BA-or-higher from 0.2941 to 0.3121, employment from
+0.7110 to 0.7247, and mean age by $\$-0.0085\$$ year; average monthly weighted
+population rises from 16.94 to 17.60 million. These are descriptive
+composition and selection patterns. Current enrollment may respond to
+labor-market conditions, and the profiles do not identify a cohort correction
+or assign exposure to nonworkers.
+
+\input{tables/r3_appendix_young_composition.tex}
diff --git a/paper/appendix/sections/r3_E_inference.tex b/paper/appendix/sections/r3_E_inference.tex
new file mode 100644
index 0000000..c96af84
--- /dev/null
+++ b/paper/appendix/sections/r3_E_inference.tex
@@ -0,0 +1,67 @@
+\section{Inference and Finite-Sample Audit}\label{app:inference}
+
+\subsection{Stochastic targets}
+
+The principal wild-score interval treats occupation as the shock cluster and conditions on the constructed stocks, exposure labels, occupational bridge, and retained support. It accommodates
arbitrary serial dependence within an occupation but not unrestricted dependence across occupations. A 22-family Webb-weight exercise instead allows common shocks within broad SOC families. A
separate household/sample-unit multiplier rebuilds cell stocks from microdata, preserves all observed records of the sampled unit across months, and assigns the same multiplier to every
fractional descendant of a routed source record. That exercise describes repeated-sample sensitivity conditional on released CPS final weights; the public Basic Monthly extracts do not supply
the design variables needed to call it complete CPS design-based inference.
+
+These variances are not mechanically added. Occupation-shock and household-resampling procedures can contain overlapping components, so summing them without a derived orthogonal decomposition could
double count uncertainty. They are reported as separate sensitivity targets.
+
+\subsection{Score and bootstrap implementation}
+
+The conditional likelihood retains one-sided zero cells and drops only occupation--month cells with zero total stock. Scores use the fitted grouped-binomial objective, and nuisance-adjusted target
scores use the Schur-complement projection implied by the fitted information matrix. The same 9,999 Rademacher occupation multipliers are reused for paired comparisons. Each draw first forms
the two refitted or score-updated targets and then their difference; intervals for movement therefore retain covariance. The finite-cluster correction, random seed, multiplier matrix hash,
convergence checks, and score-conservation checks are written to the execution receipts.
+
+For the reconstructed pooled comparison, the analytic standard error is 0.04517 and the 9,999-draw 95-percent interval is $\$[-0.2206,-0.0437]\$$. After SOC2-by-calendar-month conditioning, the
coefficient is $\$-0.0217\$$, with standard error 0.07132 and interval $\$[-0.1607,0.1173]\$$. The paired movement is 0.11043, with standard error 0.05189 and interval $\$[0.0107,0.2102]\$$. The
corresponding two-sided normal-theory paired $\mathbf{MDE}_{\{80\}}$ is 0.14539. The detected movement remains smaller than this planning quantity; that is not a contradiction because an MDE is a
power summary, not an acceptance threshold for an observed draw.
+
+Table-\ref{tab:appendix_precision_summary} consolidates the principal precision diagnostics. It states the scale and whether the quantity is a coefficient or a paired difference; the same numerical
formula is not reinterpreted as an equivalence margin.
+
+\input{tables/r3_appendix_precision_summary.tex}
+
+\subsection{Numerical existence and separation}
+
+All eleven declared grouped-binomial specifications pass the treatment-rank,
recession-direction, binding trust-path versus independent zero-start
Newton/IRLS agreement, fitted-mean, normalized-gradient, raw and diagonally
scaled fitted-Hessian, and two-sided target-profile checks at the unchanged A1
thresholds. The pooled, family-post, family-month, dynamic, and
quintile-by-month seasonality specifications require no boundary-row profiling.
+Each post-2020 specification profiles one boundary nuisance group spanning 77
+positive-total data rows, and each occupation-by-month seasonality specification
+profiles 110 boundary nuisance groups spanning 770 rows; these rows are
+profiled on the extended-likelihood face rather than deleted. In every model
+the recession audit finds no direction capable of moving a reported target or
+treatment coefficient, and the focal target is finite on the resolved
+extended-likelihood face. The retained L-BFGS-B path is diagnostic rather than
+a passing solver: it fails the standardized-score threshold in all eleven fits
+but reveals no better independently stationary contradictory target.
+
+\subsection{Broad-family and household sensitivity}
+
+The few-family exercise uses 22 broad-family weights and reports baseline, conditioned, and paired targets from common draws. Its role is sensitivity to cross-occupation dependence, not a declaration
that SOC2 is the uniquely correct cluster. The household/sample-unit exercise performs 399 positive-weight full refits. With labels fixed, its basic interval is $\$[-0.1865,-0.0794]\$$ for the
pooled coefficient, $\$[-0.1212,0.0587]\$$ for the family-conditioned coefficient, and $\$[0.0362,0.1753]\$$ for their movement. Regenerating preperiod exposure labels within the same draws gives $\$$
 $\$[-0.1864,-0.0762]\$$, $\$[-0.1236,0.0665]\$$, and $\$[0.0355,0.1771]\$$. A median of two occupations changes quintile per regenerated-label draw. Webb restandardization by itself is an affine
transformation absorbed by the specified fixed effects, so this difference is a quintile-membership sensitivity rather than a separate normalization channel.
+
+All 399 household refits succeed. The authoritative stopping calculation
+puts the largest bootstrap-of-bootstrap interval-endpoint Monte Carlo standard
+error at 0.00820 log point, below the declared 0.01 threshold, so the
+predeclared expansion to 799 draws is not triggered.
+
+The source does not expose official Basic Monthly replicate weights or all confidential design identifiers. Household identifiers are therefore a dependence-preserving public-data approximation, not
an ASEC replication-weight procedure. Link-specific flow analyses additionally report person- and household-cluster sensitivities using their own risk sets.
+
+\subsection{Calendar-time HAC and stress validation}
+
+The time-HAC audit operates on nuisance-adjusted score sums and implements occupation clustering plus an aggregate elapsed-calendar HAC term minus within-occupation elapsed-calendar HAC terms.
Calendar lags are measured in months, so the missing October 2025 observation does not make September and November adjacent. Under rebuilt exposure labels, lag-16 standard errors are 0.04449
for the pooled target, 0.05522 for the family-conditioned target, and 0.04501 for their paired movement. None of the 15 rebuilt five-parameter covariance matrices across the three targets and
five lags is indefinite, and no projection or eigenvalue clipping is applied. Indefinite positive-lag matrices from the superseded historical-treatment/SOC2-post audit are retained only as
labeled implementation history.
+
+The completed finite-sample audit uses 11 declared designs, 799--1,599 retained outer replications per design, 9,999 common multiplier draws per joint refit, and independently computed pseudo-targets.

Absolute coefficient bias is at most 0.00102 log point in the empirical observed-effect design. Interval performance is target-specific: occupation-Rademacher coverage is 0.865 for the pooled projection, 0.921 for the family-month projection, and 0.889 for their paired movement. Family-Rademacher coverage is respectively 0.990, 0.895, and 0.992. Rademacher and Webb weights give nearly identical coverage at a fixed cluster level, so changing the multiplier law does not repair the cluster-level mismatch.

+The earlier 26.7-percent and 11.3-percent rejection figures are withdrawn rather than reinterpreted. In the clean family-variance-zero design, the family-month pseudo-target is zero to numerical precision; its five-percent rejection rates are 6.1 percent under occupation Rademacher weights, 8.9 percent under family Rademacher weights, and 5.6 percent for the cross-fit full-refit oracle. In the empirically calibrated structural-null design, removing the structural exposure coefficient still leaves a pooled pseudo-target of $\$-0.11035\$$ and a paired pseudo-target of 0.11035. Because the data-generating process retains the observed fitted family-by-month surface. The near-nominal oracle then rejects zero for the pooled projection 96.5 percent of the time. This is a restatement of the broad-family decomposition built into the calibration, not independent evidence that composition caused the empirical gap. The simulations therefore limit both the inferential and structural interpretation: none of the feasible cluster procedures is adequate for every target and significance of the pooled projection does not by itself test a structural exposure effect.

+Tables\ref{tab:appendix_simulation_core}--\ref{tab:appendix_simulation_designs_inference} make the simulation inventory and its Monte Carlo uncertainty explicit. The first compares every declared procedure on a common empirically calibrated design; the next two report estimation and interval performance for all eleven core, adverse, and factor-ablation designs while holding the occupation-Rademacher procedure fixed. This separation prevents a change in the DGP from being mistaken for a change in the cluster or multiplier rule.

+ \input{tables/r3_appendix_simulation_core.tex}

+ \input{tables/r3_appendix_simulation_designs.tex}

diff --git a/paper/appendix/sections/r3_F_dynamics.tex b/paper/appendix/sections/r3_F_dynamics.tex
new file mode 100644
index 00000000..7f72d83
--- /dev/null
+++ b/paper/appendix/sections/r3_F_dynamics.tex
@@ -0,0 +1,74 @@
+ \section{Dynamics, Endpoints, and Trend Sensitivity}\label{app:dynamics}
+ \subsection{Quarterly system}
+ The corrected dynamic model interacts every nonreference exposure quintile with calendar quarter. The omitted period is 2022Q4, which contains October and November because December 2022 is the transition month and is excluded. The resulting Q5--Q1 path has 23 preperiod and 15 postperiod coefficients. The pooled and SOC2-by-calendar-month versions are estimated on identical reconstructed support, and the archive stores the complete Q2--Q5 coefficient vector, full covariance matrix, influence matrix, and simultaneous critical values.

+ Relative to omitted 2022Q4, the joint null that all 23 Q5 preperiod coefficients equal zero is rejected in the pooled model ($\$p=.0171\$$) and the family-conditioned model ($\$p=.0053\$$). Table\ref{tab:app_pretrend_windows} also reports the reference-invariant null that the coefficients are equal to one another within each fixed window. That null rejects over the full period for the pooled model ($\$p=.0121\$$) but not for the family-conditioned model ($\$p=.0641\$$); it does not reject at five percent in 2017Q1--2019Q4, 2021Q1--2022Q3, or the full window excluding 2020Q2--2020Q4 for either structure. Thus the full pooled rejection no longer clears five percent when 2020Q2--2020Q4 are excluded, whereas the family reference-relative rejection persists even though equality within each fixed window is not rejected. In leave-one-quarter diagnostics, 2019Q2 produces the largest reduction in the pooled reference-relative Wald statistic and 2018Q4 the largest reduction in the conditioned statistic; these nonadditive deletions do not identify unique causal drivers.

+ \input{tables/r3_appendix_pretrend_windows.tex}

+ As a precision diagnostic, an OLS linear functional through all preperiod coefficients is $\$-0.0078\$$ log point per year in the pooled model and $\$0.0041\$$ after family conditioning. Its occupation-cluster standard errors are 0.0119 and 0.0292, implying normal-theory $\$ \mathit{MDE} _{f(80)} \$$ values of 0.0332 and 0.0819 log point per year. An insignificant linear slope cannot rule out economically relevant smooth drift and is not a substitute for the joint tests. Simultaneous bands exclude zero only in late pooled quarters and never in the family-conditioned path. These results do not establish which preperiod force produces the differences and do not validate parallel evolution for a causal interpretation of the January 2023 split.

+ Table\ref{tab:app_static_dynamic_reconciliation} distinguishes three targets. The postperiod functional $\$P\$$ averages the 15 post-quarter coefficients with predeclared calendar weights relative to omitted 2022Q4. Its pooled value is $\$-0.1199\$$, with interval $\$[-0.2637, 0.0239]\$$; its family-conditioned value is $\$-0.2074\$$, with interval $\$[-0.4252, 0.0103]\$$. The reference-invariant dynamic functional $\$D\$$ subtracts the corresponding weighted preperiod mean. It is $\$-0.131372\$$ pooled and $\$-0.021689\$$ after family conditioning, compared with static coefficients $\$S\$$ of $\$-0.132109\$$ and $\$-0.021675\$$. Covariance-preserving tests give $\$p=.454\$$ and $\$p=.995\$$ for $\$D-\$S\$$. Hence the static and dynamic post-minus-pre targets are numerically reconciled. Within the family model, $\$P-S=-0.1858\$$ has $\$p=.074\$$, so the design does not detect that difference at five percent. Across conditioning structures, however, the change in $\$P-\S is $\$-0.1980\$$ ($\$p=.021\$$), whereas the change in $\$D-\S is $\$-0.0008\$$ ($\$p=.693\$$). Thus the short-reference functional responds to conditioning differently from the reference-invariant comparison. The $\$D-\S result is a nondetection, not an equivalence test, and $\$D\$$ remains a linear functional rather than the nonlinear static grouped-binomial coefficient.

+ \input{tables/r3_appendix_static_dynamic_reconciliation.tex}

+ \subsection{Official HonestDiD implementation}

+ The trend-sensitivity program passes the joint event-time vector and covariance matrix to the official \texttt{HonestDiD} implementation. The environment receipt fixes HonestDiD 0.2.8 and its source commit, and records the compatible official CVXR and solver versions. The program does not feed a single static coefficient, $\$D\$$, or a standard error into an event-study procedure; its target is $\$P\$$. It reports both the complete 2017Q1--2022Q3 calibration and the prespecified consecutive 2021Q1--2022Q3 post-reopening calibration. The latter does not delete an internal pandemic quarter or compress a calendar gap. For the family-conditioned vector its $\$M=0\$$ smoothness interval excludes zero, but the interval includes zero at $\$M=0.005\$$; the pooled intervals include zero even at $\$M=0\$$. The conventional event-functional interval already includes zero in both structures. The correct positive zero-exclusion breakdown is therefore undefined: there is no positive robustness threshold to lose. Earlier installation and API failures remain in the execution log; the successful official run is not replaced with a home-built approximation.

+ \input{tables/r3_appendix_honestdid.tex}

+ \subsection{Onset, endpoint, seasonality, and survey dates}

+ Onset sensitivity reports every monthly split from November 2022 through June 2023. January 2023 is retained because December 2022 was designated as the transition month after the November 30, 2022 public release of ChatGPT; robustness across nearby dates does not identify that event as the cause. The endpoint grid includes December 2024, September and December 2025, and July 2026 wherever the calendar permits. October 2025 is never imputed. Relative to the full window, the pooled movements through December 2024, September 2025, and December 2025 are 0.0211, 0.0225, and 0.0212 log point. Their magnitudes are essentially the same even though their interval endpoints differ. The full available window remains the canonical target, while December 2024 is reported prominently as the pre-2025-production-break benchmark; the latter is not promoted after outcomes are observed because it retains a negative detected coefficient. The analogous conditioned movements remain near zero.

+ \input{tables/r3_appendix_endpoint_sensitivity.tex}

+ Exposure-group-by-month-of-year interactions and the feasible occupation-month-of-year specification both move the target little: the largest absolute movement from the corresponding full-window coefficient is $\$0.0011\$$. A coding-stable post-2020 specification gives $\$-0.1181\$$ pooled and $\$-0.0304\$$ with family-month effects; paired movements from the full-window estimates are $\$0.0140\$$ ($\$[-0.0288, 0.0569]\$$) and $\$-0.0087\$$ ($\$[-0.1053, 0.0878]\$$). These replacement fits pass the same numerical certification as the baseline. The earlier failed attempts and diagnostics remain retained in the failure registry. Excluding September and November 2025 around the collection gap does not detectably change the full-window estimate. These rows address sensitivity to calendar construction and survey production, not counterfactual reweighting for the January 2025 or January 2026 population controls.

+ \subsection{Official-weight and respondent-equivalent era comparison}

+ Table\ref{tab:appendix_population_eras} reports a post-outcome exploratory joint-model comparison of 2023--2024 with 2025--2026. The official-weight row retains the paper's employment-stock estimand

```

; the respondent-equivalent row is a deliberately different routed-count outcome used only to diagnose whether the later pattern mechanically appears only in final-weight stocks. It cannot
recover unpublished age-by-occupation counterfactual weights.
+
+ \input{tables/r3_appendix_population_eras.tex}
diff --git a/paper/appendix/sections/r3_G.flows.tex b/paper/appendix/sections/r3_G.flows.tex
new file mode 100644
index 0000000..26c74f3
--- /dev/null
+++ b/paper/appendix/sections/r3_G.flows.tex
@@ -0,0 +1,73 @@
+ \section{Linked Flows and Worker Outcomes} \label{app:flows}
+
+ \subsection{Link construction and risk sets}
+
+ \text{Person links follow official IPUMS/Census matching guidance and use the available longitudinal weight for each eligible link. Adjacent-month and twelve-month endpoints are distinct samples. A twelve-month link is not the sum of twelve observed monthly transitions and may conceal intervening changes. Table~\ref{tab:app_flow_retention} reports fixed-support retention ranges across exposure quintiles rather than collapsing different periods and risk sets into one number.}
+
+ \text{Employment exit begins with an employed person in an occupation on the analysis support. Reported occupation change requires employment at both endpoints and a valid occupation at each. Entry-destination allocation begins with a nonemployed origin, ends with observed employment, and asks whether the destination falls in Q5 rather than Q1. It therefore conditions on entry and is neither an employment-finding probability nor an employer hiring rate. Every crosswalk descendant of one source record shares its link and resampling weight.}
+
+ \input{tables/r3_appendix_flow_retention.tex}
+
+ \text{Retained and eligible-unlinked origins differ observably. Among postperiod older Q5 origins at twelve months, retained records average age 44.69 and have a 59.25-percent BA-or-higher share, versus 41.50 and 53.35 percent among eligible-unlinked records. Among postperiod young Q2 origins, the 35-or-more hours share is 60.83 percent retained versus 66.24 percent unlinked. These comparisons use origin \texttt{WTFINL}; they diagnose observable selection but do not correct selection or sign bias from unobserved outcomes.}
+
+ \subsection{Results and uncertainty}
+
+ \text{At the adjacent-month horizon, the young-relative Q5--Q1 estimate is 0.132 for employment exit, 0.003 for reported occupation change, and $-0.070$ for entry-destination allocation. Their respective 95-percent intervals are $[-0.042, 0.307]$, $[-0.118, 0.123]$, and $[-0.250, 0.110]$. At twelve months, the three estimates are 0.123, $-0.018$, and $-0.052$. Their intervals are $[-0.111, 0.357]$, $[-0.097, 0.061]$, and $[-0.251, 0.147]$. All include zero. Person- and household-cluster sensitivities, descriptive transition rates, effective risk counts, and named occupation influence appear in the machine-readable flow directory.}
+
+ \text{Usual-hours changes are estimated within linked employed persons. The adjacent estimate is $-0.119$ hours per week with interval $[-0.295, 0.057]$. Weekly earnings use the eligible outgoing-rotation sample and its earnings weight; the conditional log-earnings coefficient is $-0.0245$ with interval $[-0.1024, 0.0534]$. The earnings extract covers 74 months through March 2023 and therefore contains only three postperiod months. Both outcomes condition on employment. Unemployment duration is described only among persons observed unemployed, and no occupation is imputed to never-employed or out-of-labor-force respondents. These restrictions prevent the outcomes from being misreported as population employment effects.}
+
+ \input{tables/r3_appendix_exit_components.tex}
+
+ \subsection{Selection sensitivity and missing-outcome bounds}
+
+ \text{Table~\ref{tab:app_flow_selection} compares three descriptive linear probability contrasts. Official link weights and origin-weight complete cases use the retained population directly. Observable poststratification instead reweights within fixed observed strata and therefore requires conditional missing-at-random. Factors are neither trimmed nor capped; their maximum is 102.7. Similarity across those three columns cannot rule out selection on unobservables.}
+
+ \input{tables/r3_appendix_flow_selection.tex}
+
+ \text{The last column assigns one common binary missing outcome to each eligible source record and all of its bridge descendants. Its linear interval is sharp under that source-record coupling for the named margin. No conditional-MAR assumption or Lee trimming is used. The analogous log intervals are only conservative marginal outer intervals because bridge coupling need not preserve joint sharpness. These accounting intervals concern missing outcomes, whereas Main Table 5's confidence intervals concern sampling uncertainty.}
+
+ \subsection{Entry denominators and annual timing}
+
+ \text{Table~\ref{tab:app_entry_probabilities} places every observed destination on one retained-nonemployment denominator. The destination probabilities sum to one after including people who remain nonemployed and employed destinations outside the exposure support. The conditional Q1--Q5 allocation used in the regression divides only by observed supported entries and remains a different object; neither is employer hiring.}
+
+ \input{tables/r3_appendix_entry_probabilities.tex}
+
+ \text{The annual exclusion rule drops origins whose twelve-month window straddles January 2023 and is reported with the other primary margins in main Table 5. The exposure-duration sensitivity retains those origins and codes the fraction of the twelve destination months at or after onset. Age and exposure remain defined at the origin. Table~\ref{tab:app_annual_timing} reports only this additional sensitivity, avoiding a second finite-draw interval for the primary target. No paired interval comparing the two rules is available, so their difference is not described as detected or equivalent. October 2025 is absent rather than interpolated, and annual endpoints remain distinct from sums of monthly transitions.}
+
+ \input{tables/r3_appendix_annual_timing.tex}
+
+ \subsection{Why no stock--flow calibration is reported}
+
+ \text{A compatible accounting identity would require entries from nonemployment, exits, switches among all occupations, aging into and out of ages 22--25, corresponding movements for the older denominator, job-to-job moves, and population/sample-composition residuals. The CPS endpoint links do not measure all of these components on a common employer-match risk set, and the BCC hiring and separation objects are not identical to CPS employment transitions. A single implied stock response would therefore be assumption driven. The revision reports the measurable margins and the precise comparability gap instead of manufacturing a calibration.}
diff --git a/paper/appendix/sections/r3_H.bcc_architecture.tex b/paper/appendix/sections/r3_H.bcc_architecture.tex
new file mode 100644
index 0000000..119b9dc
--- /dev/null
+++ b/paper/appendix/sections/r3_H.bcc_architecture.tex
@@ -0,0 +1,87 @@
+ \section{Payroll Bridge and Exposure-Construction Diagnostics} \label{app:bcc_architecture}
+
+ \subsection{BCC version and grouping bridge}
+
+ \text{The comparison uses the August 12, 2026 version of Brynjolfsson--Chandar--Chen and records the accessed paper, dashboard, and workbook in the evidence ledger. Their reported 19-percent young-worker "kept pace" shortfall is a stock-growth counterfactual in ADP data, not a regression coefficient, employment probability, or hiring rate. Their public dashboard forms equal-occupation exposure groups; later regressions employ other and employer controls. Exact dashboard memberships and the treatment of ties are not released.}
+
+ \text{The CPS bridge therefore applies an equal-occupation, tie-preserving top-two-versus-bottom-three approximation on the present support and labels it as such. The pooled estimate is $-0.0720$, with interval $[-0.1223, -0.0218]$. The SOC2-by-calendar-month estimate is $-0.0168$, with interval $[-0.0808, 0.0473]$. Their paired movement is 0.0553, with interval $[-0.0013, 0.1118]$. The intermediate SOC2-by-post specification moves by 0.0573 relative to the pooled model, with interval $[0.0005, 0.1141]$. This post-outcome bridge contrast is detected narrowly, whereas the stronger family-month comparison is not. The archive provides group summaries and a membership file; it cannot provide concordance with undisclosed assignments. Differences in occupation universe, age comparison, denominator, endpoint, firm controls, and employer-matched flows remain.}
+
+ \input{tables/r3_appendix_bcc_bridge.tex}
+

```



```

+\\subsection{Annual ACS reconstruction}
+
+The annual extension uses ordinary one-year ACS PUMS for 2017--2019 and
+2021--2024; no standard 2020 one-year file exists, and no experimental 2020
+estimate is spliced into the panel. Eighty replicate weights per endpoint year
+form the successive-difference-replication intervals in
+Table-\\ref{tab:appendix_acs}. The BCC-labelled rows remain public analogues:
+complete BCC occupation membership is unavailable, so they cannot be exact
+replications.
+
+\\input{tables/r3_appendix_acs.tex}
+
+For the benchmark, changing from all employed to full-time civilian
+\\wage-and-salary workers is not detected under common ACS replicate
+\\perturbations: the paired differences are $-0.0025$
+\\$[-0.0266,0.0216]$ for the equal-occupation BCC analogue and $0.0056$
+\\$[-0.0174,0.0285]$ for fixed primary groups. The fixed-primary versus BCC-analogue
+\\grouping difference is likewise not detected among all employed workers
+\\$[0.00908, \\$[-0.0045,0.0224]$). Among full-time civilian wage-and-salary
+\\workers that grouping difference is narrowly detected $[0.0171$,
+\\$[0.0002,0.0340]$), which reinforces rather than resolves the missing-membership
+\\limitation.
+
+For the annual young-relative regressions, full-calendar family-year estimates
+\\retain negative ACS sampling intervals under the fixed primary exposure groups,
+\\while the broader equal-occupation grouping and every nonreuse-calendar
+\\family-year interval contain zero. The paired ACS sampling intervals for the
+\\pooled-minus-family-year movements exclude zero in all eight reported
+\\definition--population--calendar combinations. This is not universal across
+\\stochastic targets: supplementary occupation- and family-multiplier intervals
+\\are wider, and every family-year coefficient reported in
+Table-\\ref{tab:appendix_acs} includes zero under those shock assignments. The
+\\annual exercise thus shows that large respondent counts sharpen person-sampling
+\\uncertainty but do not create within-family occupational comparisons or
+\\identify an AI effect.
+
+\\subsection{Equal-occupation estimation objective}
+
+The public-dashboard grouping bridge above changes the grouping rule. A separate sensitivity changes the regression objective itself: each occupation receives total weight one, while \\texttt{WTFINL}
+\\still determines the young and older shares within occupation--month cells. Under this objective the pooled Q5-by-post coefficient is $-0.0833$, compared with $-0.1321$ under employment-stock
+\\weighting. Their paired difference is $0.0488$ $[\\$[-0.0604,0.1581]$). The corresponding family-month estimates are $-0.0047$ and $-0.0217$, with a paired difference of $0.0170$ $[\\$
+\\[-0.1587,0.1927]$). The design therefore does not detect an objective-driven difference, but the wider equal-occupation intervals and null-containing paired intervals do not establish
+\\equivalence.
+
+\\input{tables/r3_appendix_equal_objective.tex}
+
+\\subsection{Task-capability weighting}
+
+Write the two Eloundou primitives as $E_1$ (the E1 share) and $E_2$ (the E2 share). The score family $E_1+\\lambda E_2$ is evaluated on a fixed grid from zero to one under common preperiod weights and
+\\common bootstrap draws. The implementation verifies numerically that $\\lambda=0.5$ reproduces raw beta before grouping. Any earlier mismatch between that row and the headline arose from a
+\\different support or treatment contract and is preserved only as implementation history.
+
+Two versions are kept separate. The native-rank analysis recomputes the score normalization and tie-preserving groups at each $\\lambda$, so both scale and membership can change. The fixed-membership
+\\analysis changes only the continuous score representation while holding the beta groups fixed. For every grid point the result files report coefficient, paired difference from beta, interval,
+\\occupation coverage, tail turnover, and correlations with computer use and remotability. A focused 408-occupation run adds both computer use and remotability on common support. None of these
+\\paths measures realized deployment.
+
+\\subsection{Primitive and architecture comparisons}
+
+The joint primitive model reports $E_1$ and $E_2$ in raw share units, in preperiod-standard-deviation units, their covariance, and an illustrative common change. This is a two-regressor
+\\parameterization, not a causal decomposition into technologies. Architecture comparisons likewise report both component coefficients and their paired difference on each declared support. Webb
+\\AI comparisons are shown first on identical support and then on broader native support so that score changes are not confused with population changes. Coordinate rotations that do not change
+\\the fitted predictor are not interpreted as additional scientific results.
+
+Raw AIOE is a standardized source-occupation index: one point is approximately one unweighted standard deviation across the 774 occupations in the source workbook, and the source interquartile
+\\difference is 1.879 points. It is not a percentage. Equal-administrative and source-employment-weighted mappings preserve that source scale through averaging, although their target-support
+\\distributions need not retain unit variance. The direct-ability reconstruction has its own raw scale and is not numerically interchangeable. Categorical AIOE estimates use ranks and tie-
+\\preserving groups; continuous comparisons must state whether their increment is a raw point or a current-support weighted standard deviation.
+
+Table-\\ref{tab:appendix_nonlambda_architecture} runs the five non-\\lambda definitions on the current 113-month outcome contract. Native-support pooled coefficients range from $-0.0991$ for direct-
+\\ability AIOE to $-0.0145$ for the reversed OECD gap. Every native-support alternative-minus-beta paired occupation interval contains zero; the OECD comparison nearly reaches the boundary, with
+\\interval $[-0.0004,0.2378]$. On the common 444-occupation AIOE support, however, family-month differences between administrative/equal-component mapping and direct ability $[0.0688$, $[\\$
+\\[-0.0070,0.1447]$ under occupation clustering but $[0.0057,0.1320]$ under family clustering) and source-employment weighting $[0.0586$, $[0.0111,0.1061]$ under occupation clustering) show that
+\\implementation can matter for the conditioned result. These are paired implementation sensitivities within AIOE, not independent replications or causal technology decompositions.
+
+\\input{tables/r3_appendix_nonlambda_architecture.tex}
+
+\\input{tables/r3_appendix_primitive_ds.tex}
+
+\\input{tables/r3_appendix_architecture_pairs.tex}
+
+\\input{tables/r3_appendix_webb_support.tex}
+
+\\subsection{Later self-reported use}
+
+An external descriptive check merges the frozen beta score to the author-maintained Real-Time Population Survey occupation workbook. The workbook pools four waves from August 2025 through May 2026;
+\\it is not a preperiod measure or a detailed occupation-by-time panel. The exact-code merge retains 121 nonsuppressed occupations from the frozen 468, with all five quintiles and 20 major
+\\families represented. The unweighted exposure--adoption correlation is 0.504 and mean adoption differs by 0.267 between Q5 and Q1, although the quintile profile is not monotone because Q4's
+\\mean exceeds Q5's. After major-family means are removed, 26.2 percent of the exposure sum of squares remains and the residualized slope is 0.302.
+
+The workbook supplies pooled point estimates and respondent counts but no detailed-cell standard errors or replicate weights. Table-\\ref{tab:rps_adoption} therefore reports no p-values or confidence
+\\intervals, and its respondent-count weighting is only a descriptive sensitivity. These results show positive alignment between the capability score and later reported use; they do not
+\\establish adoption timing, explain the CPS coefficient, or identify an employment effect.
+
+\\input{tables/r3_appendix_adoption.tex}
+
+\\subsection{What validation would require}
+
+None of the sampling intervals above treats a score as a noisy observation of a validated true exposure. A useful validation sample would begin at the versioned occupation--task level and oversample
+\\both broad families and tasks on which architectures disagree. It would record separate outcomes for technical capability, realized time savings without complementary software, realized time
+\\savings with software, observed workplace adoption, and task importance. Conflating these labels would reproduce the construct problem the validation is meant to solve.
+
+Labels should come from repeated, blinded, independent expert assessments and, where feasible, audited product-use or task-performance records. Calibration and evaluation occupations should be
+\\separated so that crosswalk and occupation aggregation choices are tested out of sample. The task description, model snapshot, prompt, complementary tools, rater instructions, occupational
+\\bridge, and employment weights all require version records. Sample size cannot be stated as one universal number: it depends on within-occupation task correlation, rater reliability, unequal
+\\task weights, the latent quantity being corrected, and whether the downstream target is a continuous slope or tail assignment. In the absence of these inputs, the revision specifies the
+\\validation design but does not apply a generated-regressor correction.
diff --git a/paper/appendix/sections/r3_I_reproducibility.tex b/paper/appendix/sections/r3_I_reproducibility.tex
new file mode 100644
index 0000000..be9556d
--- /dev/null
+++ b/paper/appendix/sections/r3_I_reproducibility.tex
@@ -0,0 +1,33 @@
+\\section{Reproducibility, Numerical Consistency, and Unresolved Items}\\label{app:reproducibility}
+
+\\subsection{Execution contract}
+
+The revision archive separates restricted inputs from distributable code and derived aggregate outputs. Two commands serve different purposes. The restricted-data orchestrator authenticates the

```

licensed inputs, refuses to overwrite an existing output root, runs the rebuilt baseline before dependent modules, passes named treatment objects downstream, invokes each self-check, and runs the pinned official HonestDID implementation. A separate aggregate-only command regenerates exhibits from committed results, checks numerical and prose consistency, runs repository tests, and builds the LaTeX documents. The latter neither reads CPS microdata nor independently re-estimates the empirical models. Environment records report Python and R package versions, seeds, scheduler resources, source hashes, and git state where captured. Every module declares its treatment contract, calendar, support, objective, inference target, and output paths before interpreting the result.

+
+The retained R3 \texttt{baseline_inventory} directory contains a provisional
+machine-readable allowlist (\nolinkurl{PUBLIC_REPLICATION_MANIFEST.csv}) and a
+written scan contract (\nolinkurl{INPUT_PROVENANCE_RECEIPT.md}).
+They have not yet been updated and applied to a separately staged V3 release
+candidate. Accordingly, neither the current git checkout nor this delivery is
+labeled a sanitized public-release artifact. Before release, a new empty
+stage must be built from the updated positive allowlist and actually checked
+for private absolute roots, credential-token patterns, restricted-data
+formats, symlinks, caches, and direct CPS link-identifier columns, followed by
+manual disclosure, licensing, and PDF-metadata review. Licensed CPS records,
+account metadata, credentials, longitudinal identifiers, and operational path
+history are not eligible for that future stage. The retained input manifest
+records IPUMS extract numbers 9, 10, and 11 and available request, data, DDI,
+and codebook hashes; unavailable API timestamps remain blank rather than
+inferred.
+
+The canonical results ledger is the numerical source of truth. Manuscript tables are generated from or checked against it, and repeated estimates use one declared draw set. If an interval comes from
+a different stochastic target---occupation shocks, broad-family shocks, or household resampling---the table labels that procedure rather than presenting multiple intervals as interchangeable
+versions of one result. The test suite verifies calendar membership, one-sided zeros, route conservation, fixed-support pairing, covariance symmetry and rank, draw hashes, and table/prose
+consistency.
+
+\subsection{Implemented failures}
+
+The failure registry is part of the replication evidence. It includes the literal vintage merge, the original March sample selection, nonconvergence of saturated seasonality variants, and indefinite
+positive-lag joint HAC matrices from the superseded historical-treatment audit. A failure is never converted into a different estimator without an explicit amendment. Successful reruns
+preserve the earlier log and record the replacement's package and solver versions; operational installation history remains outside the scientific exposition.
+
+\subsection{Unresolved limitations}
+
+The analysis cannot observe firm adoption, employer identifiers, worker--firm hiring, confidential CPS design variables, exact BCC dashboard memberships, age-specific occupational route probabilities,
+or a validated latent "true" AI exposure. The public data therefore do not identify an AI causal effect or a complete stock--flow mechanism. The direct within-family tail population is
+small, unrestricted preperiod differences are present, several linked-flow intervals are wide, and January 2025--2026 survey-production changes cannot be undone with published aggregate factors
+. The annual ACS extension omits the unavailable standard 2020 one-year file, combines year-specific sampling components under a block-independence approximation, and cannot recreate BCC's
+undisclosed membership or missing family-by-tail support. These are limits on interpretation, not missing robustness rows that a specification search can solve.
+
+
+Author affiliation, email, acknowledgments, funding, conflicts, data-access wording, and journal-required disclosures remain marked \texttt{[AUTHOR TO COMPLETE]} wherever author-supplied information
+is unavailable. No identifying or disclosure information is inferred from directory names, credentials, or institutional access.
+diff --git a/paper/main/abstract_restat.tex b/paper/main/abstract_restat.tex
+index 5e4b5a0..9c89b2c 100644
+--- a/paper/main/abstract_restat.tex
+++ b/paper/main/abstract_restat.tex
@@ -1,11 @@
-I ask whether detailed occupational AI exposure explains a public-CPS young-worker employment pattern beyond broad occupational composition. The corrected highest-versus-lowest exposure contrast is \$
-\$-0.1346\$ log points. Conditioning on two-digit occupation-family post changes reduces it to \$-0.0315\$ (95-percent interval \$[-0.1676,0.1046]\$), while family-specific monthly paths give \$-0.0317\$
+\$. These models retain 30 percent of baseline target information and change the estimand. The primary 80-percent minimum detectable effect is \$0.1244\$, and paired differences across seven
+exposure alternatives include zero. The CPS association is principally a broad-occupation comparison, not an identified effect of detailed AI susceptibility.
+Using public monthly CPS data, I estimate the post-December-2022 change in the log conditional-mean employment-stock ratio of ages 22--25 to 26--65 in the highest versus lowest GPT-4-exposure quintile
+. The pooled estimate is \$-0.132\$ log points, but only 29 occupations--5.03 percent of preperiod stock--directly support within-family tail comparisons. Broad occupation-family paths reduce
+the estimate to \$-0.0228\$. Annual ACS data improve person-sampling precision without repairing occupational support, and preperiod dynamics reject parallel evolution. The association therefore
+relies predominantly on cross-family comparisons and does not identify a causal AI effect.
+diff --git a/paper/main/abstract_working.tex b/paper/main/abstract_working.tex
+index a77901b..e61bb62 100644
+--- a/paper/main/abstract_working.tex
+++ b/paper/main/abstract_working.tex
@@ -1,11 @@
-I rebuild a widely discussed young-worker employment pattern in public Current Population Survey data and ask whether detailed occupational AI exposure carries information beyond broad occupational
+composition. In the corrected January 2017--July 2026 panel, the post-January-2023 young-to-older employment-stock contrast between the highest and lowest Eloundou exposure quintiles is \$
-\$-0.1346\$ log points. Conditioning on two-digit occupation-family-specific post changes reduces it to \$-0.0315\$ (95-percent interval \$[-0.1676,0.1046]\$); allowing each family its own monthly
+young-relative path gives \$-0.0317\$\$. The reduction is statistically detectable in a paired comparison, but the conditional models retain only 30 percent of baseline target information and
+change the estimand. Food-service exclusions explain little of the reduction. Architecture comparisons are less informative than their point estimates suggest: the baseline design has an 80-
+percent minimum detectable effect of \$0.1244\$ log points, and paired differences across seven alternative constructions all include zero. A public implementation of Brynjolfsson--Chandar--Chen'
+s top-two-versus-bottom-three exposure grouping yields a smaller \$-0.0728\$ CPS contrast. Taxonomy repair and construction choices remain consequential, but the defensible conclusion is narrower
+: the CPS association is principally a broad-occupation comparison, not an identified effect of detailed AI susceptibility.
+Recent evidence associates generative-AI exposure with weaker employment among workers in their early twenties. I reconstruct the public monthly CPS comparison and ask which occupational contrasts
+support it. The estimand is the post-December-2022 change in the log conditional-mean employment-stock ratio of ages 22--25 to 26--65 in the highest versus lowest GPT-4-exposure quintile,
+conditional on intermediate quintiles and preperiod software exposure. Although nominal support contains 468 occupations, only four broad families span both tails; their 29 occupations
+represent 5.03 percent of preperiod stock, and direct-tail information is equivalent to 6.2 occupations. The pooled estimate is \$-0.132\$ log points. Broad occupation-family paths reduce it to \$
-\$-0.0228\$, but the direct-tail interval cannot resolve a within-family gradient. An annual ACS extension narrows person-sampling uncertainty without creating the missing occupational comparisons.
+Computer-use and family conditioning move the coefficient in opposite directions, preperiod dynamics reject parallel evolution, and linked-CPS estimates do not isolate a flow mechanism. The
+public-CPS association is supported predominantly by cross-family comparisons; it is not an identified causal AI effect or a replication of employer hiring results.
+diff --git a/paper/main/preamble.tex b/paper/main/preamble.tex
+index 12483c3..225d433 100644
+--- a/paper/main/preamble.tex
+++ b/paper/main/preamble.tex
@@ -8,5,8,5 @@
\fi
\usepackage{styles/yax-common}
\graphicspath{{figures/}}
\hypersetup{pdftitle={AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment},pdfauthor={Lily Luo}}
\newcommand{\papertitle}{AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment}
\hypersetup{pdftitle={Occupational AI Exposure and Young-Worker Employment: Support and Comparisons in the CPS},pdfauthor={Lily Luo}}
\newcommand{\papertitle}{Occupational AI Exposure and Young-Worker Employment: Support and Comparisons in the CPS}
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+index 097fdfa..6d76fc7 100644
+--- a/paper/main/sections/01_introduction.tex
+++ b/paper/main/sections/01_introduction.tex
@@ -1,28 +1,20 @@
\section{Introduction}\label{sec:introduction}
+Recent evidence has linked occupational exposure to generative artificial intelligence with weaker employment growth among workers in their early twenties. The pattern matters because it bears on
+whether new technology is already changing entry-level work. It is also unusually hard to interpret. Occupational exposure is fixed before the labor outcome is observed, is highly correlated
+with broad occupational structure, and is available in several constructed forms that do not necessarily measure the same object. A negative exposed-versus-unexposed contrast can therefore
+reflect detailed AI susceptibility, broad occupational composition, or both.
+Recent evidence associates generative-AI exposure with weaker employment growth among workers in their early twenties. Interpretation is difficult because exposure is constructed rather than observed
+, high- and low-exposure jobs differ broadly, and the period around ChatGPT's release includes several aggregate disruptions.
+This paper asks a deliberately narrower question: \emph{does a detailed occupational AI score explain the public-CPS early-career employment pattern beyond the evolution of broad occupation families?}
+I aggregate Basic Monthly Current Population Survey records into occupation-by-age-by-month employment stocks from January 2017 through July 2026. The design compares ages 22--25 with ages
+26--65 within occupations and asks whether that relative stock changes after January 2023 across exposure groups. The outcome is a stock, not hiring, and the design is descriptive. It does not
+observe employer adoption and does not turn the release of ChatGPT into an instrument.
+This paper asks which occupational comparisons support the public Current Population Survey (CPS) pattern. I rebuild monthly employment stocks from January 2017 through July 2026 and classify
+occupations with the GPT-4 beta score of \cite{eloundou2024gpts}. The target is the post-December-2022 change in the log conditional-mean employment-stock ratio of ages 22--25 to ages 26--65
+in the highest relative to the lowest exposure quintile, conditional on intermediate quintiles and preperiod Webb software exposure. I examine broad-family paths, direct within-family support,
+preperiod computer use, dynamics, and linked-CPS flows. Exposure measures susceptibility rather than adoption; January 2023 is chronology, not an instrument.
+The corrected-calendar baseline reproduces a sizeable pattern. The post-January-2023 young-relative coefficient for the highest versus lowest employment-weighted quintile of Eloundou et al.'s GPT-4
+beta exposure is \$-0.1346\$ log points, with a canonical 9,999-draw 95-percent wild-score interval of \$[-0.2223,-0.0468]\$. The frozen 108-month chronology gives \$-0.1311\$ and its canonical
+interval \$[-0.2171,-0.0451]\$. These numbers describe the change in a young-to-older employment-stock ratio, not a percentage decline in individual employment probability.
+Direct support is thin. Four families span both exposure tails; their 29 occupations represent 5.03 percent of preperiod stock, and the interval includes large effects of either sign. The
+conditioned comparison otherwise relies on intermediate quintiles and common-coefficient restrictions, so its near-zero estimate is not economic equivalence or evidence of no AI effect.
+The central result changes the interpretation. Adding two-digit-occupation-family by young by post interactions reduces the corrected coefficient to \$-0.0315\$ with interval \$[-0.1676,0.1046]\$.

Allowing every broad family its own young-relative monthly path gives $\beta = -0.0317\%$. The paired baseline-to-conditioned movement is about $\$0.103\%$ log points and its interval excludes zero. Yet this is not an additive decomposition: the conditioning model asks whether detailed quintile status predicts outcomes among the comparisons left after broad-family evolution is absorbed. Only four of 22 broad families contain both Q1 and Q5, and conditional target information falls to 30 percent of the baseline. The evidence says that the baseline is predominantly a comparison across broad occupation families and that the remaining within-family association is weakly measured. It does not assign a causal share to broad composition.

+The reconstructed coefficient is $\beta = -0.132\%$ log points, with 95-percent occupation wild-score interval $[-0.221, -0.044]\%$; it is neither an individual probability nor an employer hiring rate. With two-digit occupation-family young-relative monthly paths, it is $\beta = -0.022\%$ ($[-0.161, 0.117]\%$). Their paired difference is $\$0.110\%$ ($[\$0.011, 0.210]\%$). This distinguishes conditional estimands without making a causal decomposition: family paths may absorb AI-related and non-AI-related evolution. The coefficient movement is therefore a consequence of the support result, not evidence that one known share of an AI effect is explained by composition.

-Nor does a simple food-service story account for the result. Excluding low-exposure food-service occupations attenuates the coefficient by 8.8 percent, to $\beta = -0.1196\%$. Excluding food, personal-care, and protective-service occupations makes it slightly more negative. Jointly removing the five occupations with the largest prior leave-one-out movements yields $\beta = -0.101\%$, while symmetric trimming and continuous down-weighting return estimates close to baseline. Influence is consequential and signed in both directions; ‘‘food-service recovery explains the pattern’’ is not supported.

+Annual ACS data separate person-sampling resolution from occupational support.

+With fixed primary exposure labels, the full-calendar family-year coefficient is negative

+with an ACS replicate interval excluding zero, but occupation- and family-shock

+intervals include zero; the nonreuse-calendar family-year coefficient is also

+undetected. More respondents therefore narrow one uncertainty component

+without creating the missing within-family tail comparisons.

-The second result concerns precision. The realized occupation-clustered standard error is 0.0444, compared with 0.0122 in the planning simulation. The frozen design’s normal-theory minimum detectable effect at 80-percent power is 0.1244 log points, approximately the size of the headline coefficient. Paired minimum detectable differences across alternative exposure constructions range from 0.0609 to 0.1689. Every paired Q5–Q1 interval comparing beta with an alternative contains zero. The data therefore do not detect architecture differences at conventional levels; they also do not establish that the architectures are economically equivalent. This distinction replaces earlier language that treated smaller or insignificant alternative point estimates as evidence that the conclusion did not generalize.

+Adding a pre-2022 O*NET computer-use measure makes the common-support coefficient more negative; one control therefore does not resolve computerization. Unrestricted quarterly coefficients reject preperiod equality, while linked-flow intervals include zero. The data neither isolate an AI counterfactual nor attribute the stock pattern to exit, switching, or entry allocation.

-The third result connects the public-data exercise directly to \cite{brynjolfsson2026canaries}. Implementing their published GPT-4 beta classification---the top two exposure quintiles against the bottom three---inside the same CPS model yields $\beta = -0.0728\%$ (interval $[-0.1240, -0.0216]\%$). This is a reproducible component of their design, not a replication of proprietary ADP outcomes, job-title mapping, firm controls, hiring, or separation. Reapplying the grouping across exposure architectures produces coefficients between approximately $\beta = -0.088\%$ and $\beta = -0.008\%$ on common support; only beta’s difference from the OECD capability measure is detected under that grouping. The contrast itself is part of the empirical object.

-Measurement still matters, but in more specific ways than the earlier manuscript claimed. A literal SOC-2010 exact-code merge into post-2018 data covers only 3.33 percent of computer-and-mathematical employment; the documented taxonomy repair covers 97.7 percent. Most of the associated coefficient change comes from readmitting occupations rather than changing exposure values on fixed support. The original six implementations also span little independent variation: two weighted principal components explain 96.11 percent of their common-support variance. These facts motivate a taxonomy that separates implementation errors, same-construct measurement changes, construct changes, support changes, estimand changes, representation changes, and pure reparameterizations.

-The paper contributes a public-data diagnosis of a prominent descriptive pattern and a disciplined way to audit constructed treatments. Its substantive conclusion is negative but informative: in the CPS, detailed AI quintiles do not carry a precisely estimated association after broad occupational evolution is absorbed. Its measurement conclusion is equally bounded: alternative constructed scores can differ in coverage and classification while available outcomes remain too noisy to distinguish many downstream coefficients. The paper does not estimate a causal AI effect, infer economic equivalence from a confidence interval containing zero, or claim that broad-family conditioning identifies the non-AI counterfactual.

-The remainder of the paper defines the construction layers, documents the corrected data and inferential target, presents the broad-family result, and then evaluates architecture precision, reference groups, influence, population controls, and timing. Mobility and algebraic reparameterizations are demoted to the online appendix because they do not resolve the employment-stock question.

-

-\begin{figure}[htbp]

-\centering

-\includegraphics[width=\textwidth]{figure1_architecture_genealogy_revised.pdf}

-\caption{From exposure construction to an economic statement}\label{fig:genealogy}

-\tabnote{The six initial implementations belong to two selected families. Webb AI and the OECD capability gap change the construct as well as the score. Mapping, support, representation, comparison, and conditioning determine the downstream estimand. Potential exposure is not observed adoption.}

-\end{figure}

+The paper shows that a sizeable public-CPS association is identified predominantly by comparisons spanning broad occupation families, quantifies the resolution of direct and flow evidence, and records calendar and taxonomy failures that silently alter support. This localization tells future adoption or quasi-experimental studies which comparisons they must replace rather than inherit. It narrows rather than overturns the motivating evidence: it neither replicates proprietary payroll outcomes nor identifies AI, computerization, or pandemic recovery as the cause.

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-\section{Constructed Treatments and Related Evidence}\label{sec:literature}

+\section{Related Evidence and the Empirical Object}\label{sec:literature}

-\subsection{A layered empirical object}

+The closest motivating study is the August 2026 revision of \cite{brynjolfsson2026canaries}. Using ADP payroll records through June 2026, it reports weaker employment growth among ages 22--25 in occupations with high GPT-4 beta exposure and studies employer-match hiring and separation. The CPS cannot reproduce its employer panel, standardized-title mapping, firm controls, or worker--firm flows. Because that study also reports public CPS and ACS evidence, novelty here lies in auditing a fixed national-CPS estimand, not discovering the age pattern.

-Write a constructed-treatment analysis as a sequence from an economic construct $\mathcal{C}\$, through a measurement rule $\mathcal{M}\$, a harmonization rule $\mathcal{H}\$, a retained support $\mathcal{S}(\Omega)\$, and a representation $\mathcal{R}\$, to an estimand $\mathcal{Q}\$:$$$$$$

-\[

-\mathcal{C} \xrightarrow{\mathcal{M}} \mathcal{N} \xrightarrow{\mathcal{H}} \mathcal{X} \xrightarrow{\Omega, w} \mathcal{X} \setminus \{\Omega\}

-\mathcal{X} \xrightarrow{\mathcal{R}} \mathcal{Z} \xrightarrow{\mathcal{Q}, P} \theta.

-\]

-Calling every perturbation a different ‘‘measure’’ conceals economically important distinctions. An invalid taxonomy merge is an implementation error. Reweighting the same application--ability links is a same-construct measurement change. Moving from LLM task acceleration to patent overlap or a broad capability gap changes the construct. Expanding mapped occupations changes the target population. Converting a level to quintiles changes the treatment representation. Replacing Q5--Q1 by Q5--Q2 changes the comparison. Rotating a fixed two-regressor column space changes only coordinates.

-Table-\ref{tab:architectures} gives the reporting taxonomy used throughout. A row can carry more than one label: repairing a mapping both corrects an implementation and may readmit occupations. The analysis separates the fixed-support score change from the support expansion rather than calling their sum a ‘‘crosswalk effect.’’

-

-\input{tables/table1_architectures.tex}

-

-Two simple invariance results discipline the exercise. First, for fixed positive weights and a fixed tie rule, a positive affine transformation preserves ranks and weighted-quantile membership. Second, if two regressor matrices are related by a nonsingular coordinate transformation and sample, objective, nuisance terms, and constraints remain fixed, fitted values and transformed contrasts are identical. The $\mathcal{F}/\mathcal{G}$ to $\mathcal{A}/\mathcal{E}$ exercise in the prior draft is therefore a basis change, not additional empirical corroboration; it is retained only in the online appendix.

-

-The task-family construction has an equally useful structure. If $\mathcal{D}_o\$ is the share of tasks directly accelerated by an LLM and $\mathcal{S}_o\$ the share accelerated only with complementary software, then$$

-\begin{equation}

-\mathcal{X}(\alpha_o) = \mathcal{D}_o + \mathcal{Q} \mathcal{X}(\beta_o) = \mathcal{D}_o + \mathcal{T} \mathcal{X}(\gamma_o) = \mathcal{D}_o + \mathcal{S}_o. \label{eq:ds}

-\end{equation}

-For any pair of occupations, rankings along $\mathcal{D} + \lambda \mathcal{S}$ can change only where their affine score paths cross. Weighted-bin assignments are consequently piecewise constant in λ on fixed support. There is no reason, however, for estimated categorical coefficients to be monotone in λ .

-

-\subsection{Exposure architectures}

+Administrative and workplace studies observe adoption or tool access in particular settings \cite{humlum2025still, brynjolfsson2025generative}. Firm-level evidence also distinguishes direct task substitution, reallocation across tasks, and firm-growth effects \cite{hampole2025artificial}. Survey evidence shows that realized generative-AI use varies substantially even among workers doing similar work and that exposure explains only part of adoption \cite{bick2026work}. Occupation scores instead describe tasks a technology might affect. Remote-work evidence supplies another competing interpretation \cite{manuel2026proximity}. None turns a capability score into observed deployment.

-The core audit contains three AIOE implementations and three Eloundou implementations. AIOE links progress in ten AI applications to 52 O*NET abilities and then to occupational ability requirements \cite{felten2018method, felten2021occupational}. Administrative, direct-ability, and source-employment-weighted versions vary reconstruction or aggregation within that system. The Eloundou family changes the technology boundary through equation-\ref{eq:ds} \cite{eloundou2024gpts}. These are six implementations from two families, not six independent measurements of a known latent exposure.

+The primary score is Eloundou et al.’s GPT-4 beta: tasks are labeled by whether an LLM, alone or with complementary software, could reduce completion time at constant quality \cite{eloundou2024gpts}. AIOE maps AI applications to abilities \cite{felten2018method, felten2021occupational}; Webb measures patent--task overlap \cite{webb2020impact}; Dingel--Neiman classifies teleworkability \cite{dingel2020home}. Computer use, routine content, education, and remotability are therefore overlapping occupational characteristics, not clean alternative treatments. The task literature likewise allows technology to substitute for some tasks, complement others, and reallocate work within occupations \cite{autor2003skill, autor2013growth, acemoglu2019automation}. Faster-changing technical careers may also combine early entry advantages with skill obsolescence later in the career \cite{deming2020earnings}, while the employment consequence of automating a task depends on whether the remaining bundle requires more or less expertise \cite{autor2025expertise}. These mechanisms make an ages-22--25 versus older-worker contrast economically relevant but not uniquely diagnostic of entry-level AI substitution.

-The treatment-side dependence is strong. On the exact 444-occupation common support, the first weighted principal component explains 80.04 percent of standardized-score variance and the first two explain 96.11 percent; three explain 99.58 percent. The intersection--union sign calculation for the six coefficients is therefore retained only as a predeclared statement about a dependent set. It is not presented as six independent confirmations, and the failed simultaneous-band criterion remains in the design record.

+This distinction changes the burden of interpretation. Agreement across correlated scores can show that a direction is not unique to one coding rule, but cannot validate a latent exposure construct. Conversely, coefficient differences can arise from task definitions, occupational support, or treatment-group membership. The empirical work below therefore matches supports where possible

and names population changes where it is not.

-Two externally admitted constructions use different primitives. Webb maps patent text to occupational tasks, while the OECD measure compares nine capability domains with occupational requirements \citep{webb2020impact,oecd2026exposure}. They are different economic constructs, not reliability draws for the same LLM variable. Frey--Osborne remains an automation comparator rather than being relabeled as AI. A repository-bounded census records six core implementations from two families, two admitted external implementations from two additional families, one scope failure, and several published candidates whose score artifacts or mappings are not yet locally verified. This is not claimed to be a timeless census of the literature.

-\subsection{Relation to labor-market evidence}

-\citep{brynjolfsson2026canaries} use proprietary ADP payroll records and emphasize employment declines among ages 22--25 in highly exposed jobs, operating largely through hiring. Their public rule places GPT-4 beta occupations in employment-weighted quintiles and contrasts the top two with the bottom three. Section-\ref{sec:results} implements that public classification in CPS. It cannot recover their employer panel, title-to-occupation mapping, firm-time controls, hiring, or separations.

-Other studies answer different questions. \citep{humlum2025still} link reported adoption to Danish administrative outcomes; workplace deployments observe actual tool access rather than occupational potential \citep{brynjolfsson2025generative}. These designs distinguish exposure from use but do not validate one universal occupation score. The broader task literature likewise emphasizes substitution, complementarity, and occupational adjustment rather than a one-dimensional technology dose \citep{autor2003skill,autor2013growth,acemoglu2019automation}.

-This paper is closest in spirit to sensitivity analysis that asks which moments or trends support an empirical result \citep{andrews2017sensitivity,rambachan2023credible} and to work on generated covariates \citep{battaglia2024inference,christensen2026valid,duan2026inference,ludwig2026large}. Conventional occupation-clustered inference here conditions on labels, mapping, and support. The online appendix specifies what a validation sample would need to contain before generated-covariate corrections could address occupational AI exposure itself.

+I distinguish six layers:

+[\

+C\longrightarrow M\longrightarrow H\longrightarrow \Omega\longrightarrow R\longrightarrow Q,

+]\

+the intended construct, measurement rule, occupational harmonization, retained support, representation, and regression estimand. A vintage mismatch is an implementation error; score repair on fixed support changes measurement; occupation readmission changes population; and a different exposure contrast changes the estimand. Common-draw paired inference tests coefficient movement, while each component interval measures its own resolution. Neither a detected movement nor a nondetection assigns a causal share or establishes equivalence. This focus connects the exercise to specification-sensitivity and generated-covariate inference \citep{andrews2017sensitivity,rambachan2023credible,battaglia2024inference,christensen2026valid,duan2026inference,ludwig2026large}.

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-\section{Data, Calendar, and Occupational Harmonization}\label{sec:data}

+\section{Data and Exposure Construction}\label{sec:data}

-\subsection{CPS stocks and the corrected calendar}

+\subsection{CPS stocks and calendar}

-The public-data application uses IPUMS Basic Monthly CPS records from January 2017 through July 2026. Employed respondents with positive final weights are aggregated into occupation-by-age-group-by-month stocks. Ages 22--25 form the young group and ages 26--65 the comparison group. Age is not labor-market experience, and the outcome can change through employment entry, exit, occupational switching, or the older denominator.

+I use IPUMS Basic Monthly CPS data from January 2017 through July 2026. Employed respondents ages 18--65 with positive final weights are aggregated to occupation--month stocks; the regression compares ages 22--25 with ages 26--65. Age is not experience, and older workers need not be untreated.

-The original wide extract accidentally selected the March ASEC sample ('03s') rather than the Basic Monthly sample ('03b') in 2017--2021. ASEC records carry zero Basic Monthly final weights for this purpose. A separately authenticated extract restores those five March samples. October 2025 remains absent because no CPS was collected during the federal shutdown. The combined files contain 9,843,021 records and 114 observed survey months. Omitting December 2022 as a transition leaves 113 months in the substantive static model.

+The corrected source calendar contains 114 of 115 possible months. Five March files in the wide extract were ASEC samples and are replaced by authenticated Basic Monthly files for 2017--2021. October 2025 is absent because no CPS was collected during the funding lapse; it is not interpolated. December 2022 is observed but excluded as the transition month, leaving 113 static months.

-The corrected calendar is now the substantive baseline. Restoring March changes the coefficient from \$-0.1311\$ to \$-0.1346\$, with a 9,999-draw interval \$[-0.2223,-0.0468]\$. The frozen result remains visible as the confirmatory chronology record but no longer defines the default exposition.

+The reconstruction scans 9,843,021 source rows and retains 5,322,047 eligible records before routing. Fractional pre-2020 crosswalk descendants are not distinct respondents. The 468-occupation grid contains 52,884 cells; 51,891 enter the likelihood, including 12,965 valid one-sided zeros, while 993 both-age-zero cells contribute nothing. Young cells are sparse: 26.25 percent have zero respondent-equivalent records and 69.83 percent have fewer than five.

-\subsection{Taxonomy bridge and universe reconciliation}

+\input{tables/r3_table1_data.tex}

-CPS occupations use Census-2010 codes through 2019 and Census-2018 codes thereafter, while source exposures use SOC vintages. Pre-2020 records are expanded along official Census-2010-to-2018 routes and aggregated to the common target taxonomy. In a one-to-many route, official conversion proportions are applied equally to young and older source records. This preserves the source occupation's age ratio across target components by construction; official files do not supply age-specific probabilities.

+\subsection{Harmonization and support}

-The full route-expanded preperiod universe contains 539 target occupations with positive young and older stock. Two counts that appeared to be a five-occupation discrepancy are instead nonnested, architecture-specific supports. The stored 490-occupation universe requires full beta coverage. The 495-occupation continuous-mapping universe requires repaired AIOE and Webb coverage. They overlap in 466 occupations: 24 appear only in beta support and 29 only in AIOE-plus-Webb support. Adding Webb to beta leaves the 468-occupation primary model. The online appendix names every member and reason.

+Pre-2020 Census-2010 occupations are routed to Census-2018 using versioned official shares. Applying the same shares to young and older records preserves source stock and age ratios by construction, but does not validate age-specific routing. The route reaches 99.546 percent of eligible early stock and conserves routed weight to machine precision.

-Within primary support, split sources contribute 14.93 percent of young and 17.75 percent of older weighted stock during 2017--2019. Ninety-nine target occupations can receive a split route and carry 23.38 percent of fitted target information; proportional allocation of information by split-stock share assigns 4.44 percent to split stock. The former is a conservative touched-target diagnostic and the latter is not an identified decomposition.

+Primary support requires finite GPT-4 beta and Webb software scores and positive preperiod young and older stock. It contains 468 occupations and 87.19 percent of candidate-universe stock; a named file records all 73 exclusions. Raw AIOE is a standardized source-occupation index, not a percentage; the source workbook has mean approximately zero, unit sample standard deviation, and an interquartile difference of 1.879 raw points. A literal SOC-2010 AIOE merge into post-2018 employment data covers only 3.33 percent of computer-and-mathematical employment, versus 97.7 percent after version-aware routing. Fixed-support score repair changes little; occupation readmission changes the population. I treat the literal merge as invalid and do not attribute it to another paper without implementation evidence.

-The negative association does not depend on split routes. Restricting the corrected model to 369 targets with no structurally possible split inbound route gives \$-0.1674\$ under fixed labels and \$-0.1471\$ after recomputing the comparison on that support. Aggregating to a stable Census-2010 taxonomy gives \$-0.1522\$. Each changes the population and should not be called a correction of the primary estimate.

+The support file separates missing exposure, missing Webb, and absent young preperiod stock. This matters because a common-support regression estimates a population defined by all required scores, whereas native-support rows can change both the exposure definition and who remains. The appendix's four-step crosswalk table holds support fixed before adding newly reachable occupations, so measurement correction is not conflated with sample composition.

-Age-specific route allocation remains uncertain. Under unrestricted official-route accounting, the 2017--2019 young/older stock ratio can range from .1031 to .1480 in Q1 and from .0790 to .1175 in Q5. These are sharp accounting ranges for stocks, not bounds on the nonlinear regression coefficient. A predeclared tilt that changes young-versus-older allocation odds toward the highest-beta eligible target by factors from .5 to 2 moves the coefficient from \$-0.1292\$ to \$-0.1391\$. Only 6.88 percent of early stock is eligible for that scenario, so it is not a global robustness claim

+\subsection{Exposure groups and production breaks}

-\subsection{A detectable workflow failure}

+The primary exposure is Eloundou et al.'s GPT-4 beta share. CPS stock from January 2017 through November 2022 sets normalization and employment-weighted quintile cuts, with ties retained. The quintiles contain 129, 99, 82, 67, and 91 occupations and about one fifth of preperiod stock each. No postperiod stock enters the rebuilt labels.

-Literal exact-code matching of SOC-2010 AIOE into a post-2018 outcome taxonomy covers just 3.33 percent of computer-and-mathematical employment. The documented bridge reaches 97.7 percent. A machine-readable compatibility gate now rejects a merge when declared source and target vintages are not identical and no versioned route is supplied.

+Employment weighting balances stock rather than occupation counts. Ties are assigned to the lower group at a cut and never split to force exact fifths. Consequently, group membership is a versioned treatment object rather than a percentile transformation that can be recreated from rounded published values.

-The corresponding continuous sequence separates layers: original score/original 410-occupation support gives \$-0.0188\$; repaired score/same support gives \$-0.0192\$; repaired score/expanded 495-occupation support gives \$-0.0315\$; excluding computer/math gives \$-0.0294\$. The invalid exact-code result is not an economically defensible alternative. The small first movement concerns score correction; the larger second movement is population re-admission.

+A historical production treatment instead used the full static panel for weights. On the corrected calendar its coefficient is \$-0.1346\$. Rebuilding from the 71-month preperiod changes nine assignments and moves the coefficient by 0.0024 log points (\$[-0.0013,0.0062]\$). The revised analysis uses only the rebuilt treatment; the historical row is a checkpoint.

-\subsection{Treatment formation}

-

-The primary Eloundou beta score and Webb software control are standardized using preperiod employment weights. Beta is divided into employment-weighted quintiles with ties retained. On the 468-occupation primary support, Q1 contains 133 occupations and 20.02 percent of employment; Q5 contains 91 and 19.53 percent. All support and sensitivity rows state whether cutoffs and scales are frozen or recomputed. Respondent-equivalent counts after route expansion are constructed records, not unique people.

-

-\input{tables/table_sample_flow_revised.tex}

+Official documentation also identifies population-control changes in January 2025 and January 2026, modified November 2025 collection and weighting, and later longitudinal-identifier corrections. These establish survey-production breaks, not their causal effect on the coefficient. I report endpoint and shutdown-month sensitivities, preserve elapsed calendar time in lags, and do not

apply aggregate population adjustments to age-by-occupation cells without the required counterfactual controls.

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diff --git a/paper/main/sections/04_data_design.tex b/paper/main/sections/04_data_design.tex
index c61e12f..2f2d0ef 100644
--- a/paper/main/sections/04_data_design.tex
+++ b/paper/main/sections/04_data_design.tex
@@ -1,41 +1,39 @@
\section{Estimand and Inference}\label{sec:design}

-\subsection{Employment-stock contrast}
-
-Let  $\$N_{\text{ot}}\$$  be the CPS-weighted employment stock in occupation  $\$o\$$ , age group  $\$a\$$ , and month  $\$t\$$ . Let  $\$Y_{\text{a}}\$$  identify ages 22--25,  $\$P_{\text{t}}\$$  identify January 2023 onward,  $\$Q_{\text{q}}\$$  denote exposure quintile  $\$q\$$ , and  $\$W_{\text{o}}\$$  denote standardized Webb software exposure. The maintained mean is
+Let  $\$N_{\text{ot}}\$$  be CPS-weighted employment stock in occupation  $\$o\$$ , age group  $\$a\$$ , and month  $\$t\$$ ;  $\$Y_{\text{a}}\$$  indicate ages 22--25;  $\$P_{\text{t}}\$$  indicate January 2023 onward;  $\$Q_{\text{q}}\$$  indicate rebuilt exposure quintile  $\$q\$$ ; and  $\$W_{\text{o}}\$$  be standardized Webb software exposure. The pooled mean is
\begin{equation}
E[N_{\text{ot}}|X]=\exp\{\left[\gamma_{\text{oa}}+\delta_{\text{ot}}+\lambda_{\text{bat}}\right.
+\sum_{\text{q}=2}^5\beta_{\text{aq}}Q_{\text{q}}Y_{\text{a}}P_{\text{t}}\theta_{\text{W_oY_aP_t}}\right]\}.\label{eq:headline}
\end{equation}
+Let  $\$N_{\text{ot}}\$$  be CPS-weighted employment stock in occupation  $\$o\$$ , age group  $\$a\$$ , and month  $\$t\$$ ;  $\$Y_{\text{a}}\$$  indicate ages 22--25;  $\$P_{\text{t}}\$$  indicate January 2023 onward;  $\$Q_{\text{q}}\$$  indicate rebuilt exposure quintile  $\$q\$$ ; and  $\$W_{\text{o}}\$$  be standardized Webb software exposure. The pooled mean is
\begin{equation}
E[N_{\text{ot}}|X]=\exp\{\left[\gamma_{\text{oa}}+\delta_{\text{ot}}+\lambda_{\text{bat}}\right.
+\sum_{\text{q}=2}^5\beta_{\text{aq}}Q_{\text{q}}Y_{\text{a}}P_{\text{t}}\theta_{\text{W_oY_aP_t}}\right]\}.\label{eq:headline}
\end{equation}
-Occupation-by-age, occupation-by-month, and age-by-month fixed effects absorb time-invariant age composition within occupation, common occupation shocks, and aggregate age-specific shocks. The target
\begin{equation}
\delta_{\text{ot}}
\end{equation}
is the change in the young-to-older employment-stock ratio for Q5 relative to Q1 after January 2023, conditional on Q2--Q4 and Webb software exposure. It is not an individual employment probability, a hiring rate, or a continuous exposure slope.
+The target  $\delta_{\text{ot}}$  is the postperiod change in the young-to-older stock ratio for Q5 relative to Q1, conditional on Q2--Q4 and Webb. It is not an individual probability, continuous slope, or hiring rate.

-With two age groups, conditioning independent Poisson pseudo-likelihood contributions on  $\$T_{\text{ot}}=N_{\text{oyt}}+N_{\text{oot}}\$$  gives
+With two age groups, conditioning on  $\$T_{\text{ot}}=N_{\text{oyt}}+N_{\text{oot}}\$$  removes the occupation--month effect and gives
\begin{equation}
\ell_{\text{ot}}=N_{\text{oyt}}\log p_{\text{ot}}+N_{\text{oot}}\log(1-p_{\text{ot}}),\quad
\operatorname{logit}(p_{\text{ot}})=\log\left(\frac{\mu_{\text{oyt}}}{\mu_{\text{oot}}}\right).\label{eq:conditional}
\end{equation}
-Occupation-month effect cancels. CPS final weights construct the two continuous cell stocks and are not applied a second time to the objective. The grouped-binomial expression is an estimating equation for weighted stocks, not a literal model of independent survey-weighted trials.

-The central conditioning exercise augments equation~\eqref{eq:headline} with SOC2-family-specific  $\$Y_{\text{a}}P_{\text{t}}\$$  terms. A saturated version instead absorbs SOC2-family-specific young-relative effects in every month. These models ask whether detailed exposure groups predict outcomes after broad-family evolution is conditioned out. They do not decompose the baseline coefficient into causal AI and non-AI components, because the conditioning set and therefore the estimand change.
+Final weights construct continuous cell stocks once; equation~\eqref{eq:conditional} is an estimating criterion, not a literal independent-binomial sampling model.

-\subsection{Maintained source of randomness}
+The central alternative gives each SOC two-digit family its own young-relative calendar-month path; a less saturated companion uses family-specific post shifts. Both change the conditional estimand. Neither the residual coefficient nor its movement from equation~\eqref{eq:headline} is an AI remainder or composition share.

-The principal confidence interval treats occupation-level economic score contributions as independent across 468 occupation clusters and permits arbitrary serial dependence within occupation. It conditions on the observed CPS-weighted cells, constructed exposure labels, taxonomy mapping, support, quintile cuts, and fitted specification. Thus it is model-based uncertainty for a stochastic occupational-shock parameter. It is not a complete design-based CPS variance estimator and does not cover exposure-label generation, taxonomy-allocation uncertainty, or every dependence induced by respondents moving across occupations.
+The occupational-conditioning, influence, dynamic, and flow analyses added in this revision were designed after the initial outcome pattern was known. They are post-outcome exploratory analyses, not preregistered confirmatory tests; simultaneous inference does not undo that selection.

-For transparency, let  $\$x_{\text{ot}}\$$  be the target and nuisance regressors after the grouped-binomial fit,  $\$p_{\text{ot}}\$$  the fitted young share, and  $\$h_{\text{ot}}=T_{\text{ot}}p_{\text{ot}}(1-p_{\text{ot}})\$$ . The implementation absorbs the fixed effects from  $\$x_{\text{ot}}\$$  under weights  $\$h_{\text{ot}}\$$ , producing  $\$x_{\text{ot}}\$$ . It forms the information matrix
\begin{equation}
H=\sum_{\text{o,t}}h_{\text{ot}}x_{\text{ot}}x_{\text{ot}}',
\end{equation}
-the occupation score  $\$s_{\text{o}}=\sum_{\text{t}}x_{\text{ot}}(N_{\text{oyt}}-T_{\text{ot}}p_{\text{ot}})\$$ , and the one-step influence  $\$H^{-1}s_{\text{o}}\$$ , with the finite-cluster factor  $\sqrt{G/(G-1)}$ . Each wild-score draw multiplies the entire occupation influence vector by one Rademacher sign. The analytic occupation-cluster standard error studentizes the centered draw; the 95-percent critical value is the empirical .95 quantile of the absolute studentized shifts. Paired comparisons reuse the same signs for both fits, thereby preserving estimated covariance. The procedure does not create pseudo-outcomes or refit each draw.
+\subsection{What the design establishes}

-All central revised intervals use 9,999 multipliers. The sole canonical interval for the frozen  $\delta$ -0.1311 estimate is  $[-0.2171, -0.0451]$ ; previously displayed variants came from different finite sets of 999 multipliers, except a minimum-count row that changed the sample. Pair-specific member intervals remain labeled as such and never replace the canonical interval.
+The design measures the magnitude, support, and specification sensitivity of a public-CPS association. It distinguishes paired coefficients and locates the occupational comparisons carrying the contrast, but cannot allocate movement among AI, computerization, pandemic recovery, or other forces. Attenuation is not a causal share, survival is not an AI remainder, and an interval containing zero is nondetection rather than economic equivalence. The limitations section returns to preperiod evolution, sparse direct support, exposure without adoption, and incomplete flow and survey-design information.

-\subsection{Precision, sparse cells, and rotation sensitivity}
+For shared observations I reuse bootstrap multipliers and form each draw of  $\Delta\widehat{\beta}^{(2)}-\widehat{\beta}^{(1)}$ . A paired interval excluding zero shows statistical distinguishability under that procedure; it does not identify the reason. An interval including zero means nondetection, not economic equivalence. I report  $\widehat{\text{MDE}}_{(80)}\approx\text{se}(1.96+0.84)\text{SE}$  only as a precision diagnostic.

-The realized primary SE is 0.0444, versus 0.0122 in the planning simulation. Its normal-theory 80-percent minimum detectable effect is therefore 0.1244 log points, or 13.25 percent on the relative-stock-ratio scale. The paper uses MDEs to characterize what the design can resolve; it does not substitute an arbitrary smallest effect of interest or interpret a confidence interval containing zero as equivalence.

-Young cells are sparse: the 10th, 50th, and 90th percentiles of respondent-equivalent counts are 0, 2, and 16; 26.3 percent are zero and 69.9 percent are below five. Older-cell percentiles are 4, 25, and 199. One-sided cells contribute valid finite terms to equation~\eqref{eq:conditional}; only 965 occupation-months with both stocks zero are omitted. Conditioning on at least five observed young respondents attenuates the coefficient to about  $\delta$ -0.1098, but this selects on realized outcomes, retains fewer than half of occupation-months, and changes the estimand.
+To describe where a target is learned, let  $\$h_{\text{ot}}=T_{\text{ot}}p_{\text{ot}}(1-p_{\text{ot}})\$$  and let  $\$r_{\text{ot}}\$$  be the target regressor after projection on every nuisance regressor in the fitted-information metric. I report
\begin{equation}
I=\sum_{\text{o,t}}h_{\text{ot}}r_{\text{ot}}^2,\quad
s_{\text{o}}=\frac{\sum_{\text{t}}h_{\text{ot}}r_{\text{ot}}^2}{I},\quad
G_{\text{eff}}=\left(\sum_{\text{o}}s_{\text{o}}^2\right)^{-1}.
\end{equation}
-These are fitted target information and concentration diagnostics, not an effective number of CPS sampling clusters. Family and occupation shares, leave-one-out estimates, and the top-five information share are reported without using influence to select a preferred sample.

-Quarterly aggregation gives  $\delta$ -0.1310 under the frozen chronology and  $\delta$ -0.1345 under the repaired chronology, essentially matching monthly estimates and SEs. To probe CPS rotation and common monthly shocks, a post-outcome score-covariance sensitivity adds contemporaneous and lagged covariance in aggregate month scores while subtracting the occupation-month intersection. At a 12-month Newey--West lag the SE rises to 0.0491 and the normal interval is  $[-0.2273, -0.0349]$ . Occupation clustering already permits arbitrary within-occupation serial dependence; this extra calculation addresses declared cross-occupation time covariance but remains model-based rather than a CPS replicate-weight variance.
+The principal 9,999-draw wild-score interval treats occupations as shock clusters and allows arbitrary serial dependence within occupation. It conditions on weighted cells, exposure labels, mapping, support, and specification. Separate sensitivities use 22 broad-family shocks, household/sample-unit full refits under released weights, and elapsed-calendar time-HAC scores. A sparse-cell and family-shock simulation audits finite-sample behavior. These procedures have different stochastic targets; they are not mechanically combined, and none supplies complete CPS design inference or exposure-label uncertainty.

-Finally, a historical simulation that preserves each donor month's entire contemporaneous cross-occupation residual vector under a sharp global-sign design closes none of the prospective-to-realized precision gap. It rules out that specific implementation as the explanation; it does not validate the planning calculation or identify which unmodeled dependence generated the gap.
+The household exercise keeps every month and fractional route descendant of a positive CPS household identifier together. It still omits unavailable multistage-selection and calibration uncertainty. The time-HAC construction uses elapsed calendar lags, so the missing October 2025 is not treated as adjacency between September and November.
diff --git a/paper/main/sections/05_support.tex b/paper/main/sections/05_support.tex
index 0983e58..828c2b2 100644
--- a/paper/main/sections/05_support.tex
+++ b/paper/main/sections/05_support.tex
@@ -1,41 +1,40 @@
\section{Where the Employment Association Comes From}\label{sec:results}
+\section{Which Occupational Comparisons Support the Pattern?}\label{sec:results}

-\subsection{Baseline and broad occupation families}
+\subsection{Public benchmark reconstruction}\label{subsec:public_benchmark}

```

-The substantive baseline uses the repaired 113-month calendar and frozen beta treatment definition. The Q5--Q1 coefficient is $-\$0.1346$ with occupation-clustered SE 0.0450 and canonical 95-percent interval $[-\$0.2223, -\$0.0468]$. The frozen design record remains $-\$0.1311$ with the canonical interval $[-\$0.2171, -\$0.0451]$. Calendar repair therefore does not explain the association.

+We first reconstruct the public stock comparison before conditioning on broad occupation families. These are independent CPS analogues using equal-occupation +Rule-A beta quintiles on the fixed primary support, not an exact replication of +BOC: exhaustive BOC occupation membership is unavailable. For all employed +workers, the annual 2022--2024 Q5--Q1 growth-factor difference is $-\$0.0346$ $+\$(-0.1455, 0.0763)$; the November 2022--June 2026 contrast is $-\$0.2312$ $+\$(-0.4660, 0.0036)$. The top-two/bottom-three dashboard analogue is a +different grouping, with estimates $-\$0.0269$ $+\$(-0.0972, 0.0435)$ and $-\$0.1007$ $+\$(-0.2380, 0.0367)$, respectively. Its inference is computed +separately; no Q5--Q1 rejection or precision statement transfers to it.

-The direct regression analogue of the earlier within-SOC2 permutation does. Adding broad-family-specific young-relative post terms reduces the coefficient to $-\$0.0315$; allowing each family its own young-relative monthly path gives $-\$0.0317$. Both intervals include zero. On common draws, the SOC2-by-post estimate minus baseline is $+\$0.1031$, with paired interval $[\$0.0035, 0.2026]$; the monthly version gives $+\$0.1028$, with interval $[\$0.0035, 0.2021]$. Thus the movement is detected even though the residual coefficient is not.

+No-control occupation long differences provide another distinct estimand. With +baseline employment-stock weights and HC1 standard errors, the all-employed +Q5--Q1 coefficient is $-\$0.0318$ $+\$(-0.1447, 0.0811)$ over 2022--2024 and $-\$0.2466$ $+\$(-0.4771, -0.0160)$ from November 2022 to June 2026. On the +468-occupation common support, the separately defined no-Webb young-relative +model gives a pooled coefficient of $-\$0.1146$ $+\$(-0.2057, -0.0236)$; allowing +family-specific young-relative calendar-month paths gives $-\$0.0124$ $+\$(-0.1446, 0.1197)$. Family conditioning therefore changes the descriptive +estimand by absorbing cross-family evolution. It does not identify a causal AI +effect, and intervals containing zero establish non-detection rather than +equivalence.

-This is strong evidence about the comparison and weak evidence about a structural mechanism. In point-estimate terms, conditioning absorbs about 76 percent of the baseline magnitude. But the target changes from a cross-economy Q5--Q1 contrast to a coefficient assembled from comparisons remaining within broad families after their differential evolution is removed. Only four of 22 SOC2 families contain both Q1 and Q5, and conditional target information falls from 26.4 million to 8.0 million, about 30 percent of baseline. The conditional coefficient is connected through intermediate quintiles in most families. It should not be described as "the remaining AI effect," and the difference should not be described as the causal contribution of broad composition.

+An independent annual ACS reconstruction gives a different view of sampling +precision. Between 2022 and 2024, the equal-occupation BOC analogue has an +all-employed Q5--Q1 growth-factor difference of $-\$0.0859$ $+\$(-0.1238, -0.0480)$ under ACS successive-difference replication) and a +full-time civilian wage-and-salary difference of $-\$0.0834$ $+\$(-0.1272, -0.0396)$. The corresponding published BOC comparisons are $-\$0.022$ and $-\$0.019$. Because exhaustive BOC occupation membership is not +public, the gap cannot be assigned separately to samples and grouping rules.

-\input{tables/table2_support.tex}

+The annual young-relative ACS extension also remains sensitive to broad-family +conditioning and calendar construction. On the fixed primary grouping and the +full 2017--2019, 2021, 2023--2024 calendar, the all-employed coefficient moves +from $-\$0.1089$ $+\$(-0.1310, -0.0869)$ pooled to $-\$0.0474$ $+\$(-0.0885, -0.0063)$ with family-year effects. On a nonreuse calendar that +drops 2018--2019, the corresponding estimates are $-\$0.0951$ $+\$(-0.1204, -0.0698)$ and $-\$0.0220$ $+\$(-0.0705, 0.0264)$. These intervals +capture ACS person-sampling uncertainty under a year-block approximation. +Occupation- and family-shock intervals are wider and include zero for each of +the four reported family-year variants. The ACS therefore supplies a larger +public-sample analogue, not an exact payroll replication or causal AI design.

-\begin{figure}[htbp]

-\centering

-\includegraphics[width=0.92\textwidth]{figure2_composition_major_revision.pdf}

-\caption[Broad occupational composition and alternative comparisons]{label{fig:composition}}

-\tabnote[All rows are post-outcome exploratory and use 9,999 occupation wild-score draws. SOC2 rows change the conditioning estimand. Food and influence rows retain frozen groups but change support. The BOC row changes the exposure-group contrast. Intervals are therefore informative within rows but the coefficients are not estimates of one invariant parameter.]{}

-\end{figure}

+Table-\ref{tab:occupation} begins with the rebuilt pooled result. The Q5--Q1 coefficient is $-\$0.1321$, with standard error 0.0452 clustered by occupation. Its 95-percent interval is $[-\$0.2206, -\$0.0437]$. Giving every SOC2 two-digit family its own young-relative calendar-month path yields $-\$0.0217$ $+\$(-0.1607, 0.1173)$. On common draws, the conditioned-minus-pooled movement is 0.1104 $+\$([0.0107, 0.2102])$, with paired $\mathbb{P}(\text{MDE}_{80})=0.1454$. Thus the conditional estimands are distinguishable while the residual gradient is imprecise.

-The prior within-SOC2 permutation is consistent with this result but is no longer asked to carry it. Randomly reassigning detailed beta scores within broad families produces a mean coefficient of $-\$0.1097$ against the frozen $-\$0.1311$. Its exchangeability assumption is not established, so the permutation percentile is descriptive. The conditioned regressions state the relevant comparison directly and show the support and information cost.

+ \input{tables/r3_table2_occupation.tex}

-\subsection{Profile, reference tail, and stable classifications}

+The ratio $\$0.1104/0.1321$ is not a causal "share explained." Broad families differ in AI-relevant tasks, computerization, education, work location, and pandemic recovery; family paths absorb all such common evolution. The result says that the pooled estimate relies mainly on comparisons across families, not what would have happened without AI.

-The fitted post coefficients relative to Q1 are $\$0$, $-\$0.0855$, $-\$0.0478$, $-\$0.0970$, and $-\$0.1311$. A three-restriction test does not reject a common coefficient for Q2--Q5 ($\$p=.119$), but cannot establish their equality. The visible Q3--Q2 increase also does not yield a rejection of monotone decline under a least-favorable max- t test ($\$p=.393$). The data therefore neither establish nor reject monotonicity.

+The full Q2--Q5 profile reaches the same conclusion. Pooled coefficients are jointly different from zero but do not establish a monotone dose response. After family-month conditioning, the four coefficients are jointly indistinguishable from zero. Simultaneous intervals and paired movements appear in the online appendix.

-Changing the requested contrast clarifies Q1's role. Q5--Q2 is $-\$0.0456$ with interval $[-\$0.1248, 0.0335]$; Q5--Q4 is $-\$0.0340$ with interval $[-\$0.1110, 0.0429]$; Q5--Q3 is $-\$0.0833$ with interval $[-\$0.1581, -0.0084]$. These are different economic contrasts, not alternative omitted-category parameterizations. Figure-\ref{fig:paths} shows why: both the low-exposure trajectory and the older-worker denominator matter. The fitted contrast is not simply a collapse of young employment in Q5.

+ \subsection{Direct support and restrictions}

-\begin{figure}[htbp]

-\centering

-\includegraphics[width=\textwidth]{figure3_q1_q5_paths.pdf}

-\caption[Q1 and Q5 employment stocks and ratios]{label{fig:paths}}

-\tabnote[Monthly CPS-weighted stocks under frozen beta membership. The upper row indexes to observed 2019 means and the lower row to observed 2022 means. The dashed line marks January 2023. October 2025 is absent because no CPS was collected. The paths are descriptive and do not additively decompose the nonlinear coefficient.]{}

-\end{figure}

+Only SOC27, 29, 31, and 41 contain Q1 and Q5 occupations. Elsewhere the regression links partial segments---for example Q1--Q3 in one family and Q4--Q5 in another---through common quintile coefficients. That is valid regression variation but not repeated side-by-side tail comparisons.

+ \input{tables/r3_table2b_family_quintile_support.tex}

-Only 46 occupations are always Q1 and 18 always Q5 across the six original implementations. These stable tails account for 9.74 percent of common-support employment. Their contrast is $-\$0.2120$ with interval $[-\$0.4131, -0.0109]$, but has only 15.0 effective information occupations and a 46.7-percent top-five information share. The negative estimate survives in this narrow population; it does not show that unstable classification is innocuous in the broader one.

+A direct-tail benchmark uses only Q1 and Q5 occupations in the four spanning families. Its 29 occupations account for 5.03 percent of preperiod stock. The family-month estimate is 0.1494 $+\$(-0.1694, 0.4682)$, so the CPS cannot precisely resolve a direct tail contrast within broad families. This row changes population and support and cannot replace the primary estimand.

-\subsection{Influence and the food-service hypothesis}

+A continuous companion instead subtracts each family's preperiod-weighted beta mean and imposes a common within-family slope. Its family-month coefficient is $-\$0.0025$ per 0.1084 raw beta units $+\$(-0.0240, 0.0191)$. It uses more residual variation but trades the tail contrast for linearity and a common slope. Together, the specifications expose the support--functional-form tradeoff.

-Influence is economically consequential but not one-directional. Jointly deleting the five occupations with the largest absolute prior leave-one-out movements reduces the estimate to $-\$0.1011$. Deleting the top ten or twenty instead makes it more negative, $-\$0.1522$ and $-\$0.1555$, because later occupations have opposing signed influence. A symmetric 2.5-percent trim of each signed-influence tail gives $-\$0.1281$, and Huber-style continuous down-weighting gives $-\$0.1314$. These are outcome-adaptive diagnostics rather than preferred specifications.

+The pre-2020 occupational bridge adds a separate allocation uncertainty. A +jointly feasible odds-tilt grid reallocates the 14.86 percent of supported +early-period stock routed through 31 split source occupations while preserving +source--age--month mass. With exposure groups rebuilt after each allocation, +the pooled coefficient ranges from $-\$0.1446$ to $-\$0.1163$, and all six paired +intervals relative to the no-tilt allocation contain zero. The family-month +estimand is less stable: at the $\$K=0.25$ endpoint it moves from $-\$0.0217$ to $+\$0.0439$, a paired change of $\$0.0656$ $+\$([0.0031, 0.1280])$, and five of six

+nonbaseline paired intervals exclude zero. A wider 25-point feasible grid
+places the pooled coefficient between $-\$0.1599\$$ and $-\$0.0702\$$. These are
+post-outcome mapping sensitivities over finite declared grids, not estimates of
+true age-specific routes, sharp bounds, or identified sets.

-Cleaner occupation-side exclusions do not support the strong claim that food-service recovery explains the headline. Dropping the five Q1 food-service occupations yields $-\$0.1196\$$, an 8.8-percent
reduction in absolute magnitude. Dropping all food-service occupations gives $-\$0.1201\$$. Dropping all food, building-and-grounds, personal-care, and protective-service occupations yields $-\$0.1370\$$; dropping only their Q1 members gives $-\$0.1387\$$. These exercises cannot isolate reopening, immigration, minimum wages, or labor supply, but they reject the simpler assertion that
removing in-person service occupations makes the pattern disappear.

+\subsection{Influence and observed paths}
+
+The online appendix reports nuisance-adjusted information, effective information counts, and family and occupation deletions. Fast-food and counter workers have substantial signed influence because
low-exposure service employment recovered differently after the pandemic, but other occupations push oppositely. Outcome-selected deletion is not a design correction; symmetric trimming and
economically defined service exclusions remain diagnostics.

+
+Figure-\ref{fig:paths} separates young and older stocks for both tails. Exact
+equal-month accounting gives a Q5--Q1 young-stock change of $-\$0.1098\$$ log point
+and an older-stock change of $\$0.0483\$$, so their young-relative difference is
+ $-\$0.1581\$$. The numerator supplies the larger movement, while the denominator
+reinforces it. This identity is not the grouped-binomial coefficient
+ $-\$0.1321\$$, and neither component is an individual probability or a causal
+share. A symmetric family accounting assigns $-\$0.1293\$$ to within-family
+ratio changes and $-\$0.0288\$$ to changing older-family weights; the appendix
+reports the zero-denominator audit. The low-exposure reference trajectory is
+part of every comparison.

+
+\begin{figure}[htbp]
+\centering
+\includegraphics[width=0.94\textwidth]{r3_figure1_q1_q5_paths.pdf}
+\caption{Young and older employment stocks in the exposure tails}\label{fig:paths}
+\tabnote{Monthly CPS-weighted stocks use rebuilt beta labels and are indexed to each series' 2019 mean. The ratio panel divides young by older stock before indexing. The vertical line marks January
2023; December 2022 is the transition month, and October 2025 is absent rather than interpolated. Paths are descriptive.}
+\end{figure}

diff --git a/paper/main/sections/06_stock_results.tex b/paper/main/sections/06_stock_results.tex
index 427faae..5452f98 100644
--- a/paper/main/sections/06_stock_results.tex
+++ b/paper/main/sections/06_stock_results.tex
@@ -1,36 +1,68 @@
-\section{What Exposure Construction Changes---and What the Data Resolve}\label{sec:family}
+\section{Characteristics and Inferential Resolution}\label{sec:characteristics}

-\subsection{Paired architecture precision}
+The leading computerization measure is the May 2020 O*NET rating
+“Interacting With Computers.” On the 455-occupation matched support, its
+preperiod-weighted correlation with beta is 0.797. Table-\ref{tab:characteristics}
+holds that support, treatment, outcome, and calendar fixed while crossing a
+computer-use control with family-specific young-relative monthly paths. The
+pooled coefficient is $-\$0.0957\$$. Adding computer use moves it to $-\$0.1966\$$;
+the paired change is $-\$0.1009\$$ ($[-\$0.1772, -\$0.0245]\$$). Adding family paths
+without computer use instead gives $\$0.0354\$$, while adding both gives $-\$0.0467\$$.
+The combined-minus-baseline change is $\$0.0491\$$ ($[-\$0.0642, \$0.1624]\$$).

-Alternative exposure constructions change occupation ranks, tail membership, and point estimates. The key inferential question is whether the downstream coefficient differences are resolved once both
estimates use common support and common occupation multipliers. Table-\ref{tab:stock} answers no for the original Q5--Q1 comparison. Every beta-minus-alternative paired interval contains zero.
The closest case is beta minus the OECD capability gap, $-\$0.1115\$$ with interval $[-\$0.2257, \$0.0027]\$$; its paired MDE_{80}\\$ is 0.1689. Beta minus Webb AI is $-\$0.0646\$$ with interval $[-\$0.1682, \$0.0389]\$$ and MDE_{80}\\$ 0.1499.

+The computer-use-by-young-by-post coefficient is $\$0.0460\$$ per preperiod-
+weighted standard deviation in the pooled computer model
+($[\$0.0117, \$0.0803]\$$) and $\$0.0487\$$ in the combined model
+($[\$0.0105, \$0.0868]\$$). Computer use reduces retained Q5 target information
+from 69.1 to 29.0 percent in the pooled comparison, a VIF-like ratio of 3.45.
+The positive computer coefficient and more negative exposure coefficient are
+joint-projection facts under collinearity, not evidence that computerization
+raises employment or that the residual isolates AI.

-\input{tables/table3_stock_results.tex}
+On a separate 347-occupation common support, remotability, wages, education
+requirements, and routine/manual content do not individually produce detected
+movements from their common baseline; computer use remains the exception.
+A repaired pandemic-shortfall exercise uses bounded predictions and a stricter
+363-occupation support; total-stock and young-relative conditioning move the
+pooled coefficient by only 0.0004 and 0.0011 log point. Regenerating these
+controls within linked-household draws widens uncertainty but does not resolve
+a conditioning movement. These are exploratory diagnostics, not equivalence
+results.

-The correct conclusion is not that all architectures yield the same economic relationship, nor that a smaller alternative estimate shows the pattern “does not extend.” The design does not detect
these paired differences and is too imprecise to rule out economically important movements in either direction. This is itself a result: the employment application reveals much more about how
treatments are constructed than about which construction has a different population coefficient.

+\input{tables/r3_table3_characteristics.tex}

-\begin{figure}[htbp]
-\centering
-\includegraphics[width=0.90\textwidth]{figure4_architecture_precision.pdf}
-\caption{Paired differences and the resolution of the CPS design}\label{fig:architectureprecision}
-\tabnote{Points are beta-minus-alternative Q5--Q1 coefficient differences on pair-specific common support. Thin lines are 95-percent common-draw paired intervals. Gray bars span plus or minus the
normal-theory MDE at 80-percent power. An interval containing zero is a failure to detect a difference, not evidence of equivalence.}
-\end{figure}

+Broad-family paths and a computer-use control need not move the coefficient
+alike: the former remove all family-common evolution, while the latter partials
+one correlated dimension. The matched four-model comparison shows that their
+projections interact rather than add. Neither is a causal technology
+decomposition.

-The original six-score intersection--union calculation is correspondingly demoted. All six marginal point estimates are negative on their literal intersection, and the maximum one-sided marginal tail
area is .045, while the predeclared simultaneous-band criterion fails. With two principal components explaining 96.11 percent of score variance, this is a dependent-set sign statement, not six
independent validations. Nothing in it overturns the paired-precision finding.

+Industry conditioning is built from occupation--industry--age--month cells rather than dominant-industry labels. Allowing 13 broad industries their own young-relative post shifts moves the stratified
coefficient from $-\$0.138\$$ to $-\$0.099\$$; the paired 0.039 interval $[\$0.008, \$0.070]\$$ excludes zero. It does not measure firm AI adoption. BA and non-BA estimates are $-\$0.138\$$ and $-\$0.123\$$; their
paired difference is $-\$0.015\$$ ($[-\$0.156, \$0.125]\$$), with $\mathrm{MDE}_{80}=0.201\$$. Exact-age estimates vary but simultaneous paired intervals do not isolate a particular age mechanism.

-\subsection{A direct bridge to the motivating result}
+The industry row changes the cell objective as well as the nuisance set, so it is reported separately from occupation-level characteristic controls. Education and exact-age estimates use literal
common supports and common draws; no subgroup is selected by its point estimate. The age coefficients are jointly unequal, but each simultaneous age-minus-pooled interval contains zero.

-The published component of \cite{brynjolfsson2026canaries} that can be reproduced is a different representation: GPT-4 beta exposure divided into employment-weighted quintiles, with Q4--Q5 compared
against Q1--Q3. Implementing that classification in the CPS model gives $-\$0.0728\$$ with interval $[-\$0.1240, -\$0.0216]\$$, roughly half the corrected Q5--Q1 log-point magnitude. The unconditioned
result is $-\$0.0728\$$ as well. Young CPS stock rises 5.78 percent in the bottom-three group and falls 5.03 percent in the top-two group between November 2022 and June 2026, while older stocks
rise by about 0.6 percent in each. These aggregate growth calculations do not inherit a hiring interpretation.

+Changing the older comparison band does not remove the pooled pattern. For ages
+22--25 relative to ages 26--65, 26--30, 31--40, 41--50, and 51--65, the pooled
+coefficients are respectively $-\$0.132\$$, $-\$0.121\$$, $-\$0.160\$$, $-\$0.139\$$, and
+ $-\$0.097\$$. Relative to the 26--65 benchmark, only the 51--65 comparison has a
+paired interval excluding zero: the movement is 0.035
+($[\$0.004, \$0.066]\$$). All five family-month coefficient intervals include zero.

+Holding the older group's preperiod exact-age composition fixed changes the
+pooled coefficient by only 0.0008 ($[-\$0.0011, \$0.0027]\$$). Restricting the young
+numerator to nonenrolled workers changes it from $-\$0.132\$$ to $-\$0.124\$$, a paired

+movement of 0.008 (\$[-0.012,0.029]\$). These are non-detections, not evidence of equivalence, and older workers are comparison groups rather than untreated +controls.

-Reapplying the BCC grouping rule to eight exposure constructions produces native-support coefficients from \$-0.0817\$ for Eloundou broad to \$-0.0096\$ for the OECD capability gap. On literal 426-occupation common support, beta is \$-0.0810\$ and OECD is \$-0.0076\$; their paired difference is \$-0.0733\$ with interval \$[-0.1390,-0.0076]\$. The other six paired differences are not detected. This does not contradict Table-\ref{tab:stock}: grouping changes the estimand and its precision. It illustrates that the economic contrast is part of the constructed empirical object.

+Separate annual profiles show heterogeneous, nonmonotone enrollment and education changes across the five exposure quintiles. The national +ages-22--25 profile, reported without assigning exposure to nonworkers, also +changes in enrollment, education, and weighted population, so these diagnostics +document rather than eliminate cohort composition as an alternative (Online +Appendix D.6).

-\input{tables/table4_reallocation.tex}

+Table-\ref{tab:inference} separates uncertainty targets. Occupation wild scores allow serial dependence within occupation. A 22-family sensitivity permits common broad-family shocks. Household full -refit multipliers preserve repeated observations and co-residence under released weights but are not full CPS design inference. None covers exposure labels or route allocation.

-\subsection{Taxonomy repair and construction layers}

+\input{tables/r3_table4_inference.tex}

-Taxonomy repair is the clearest measurement finding because it separates an invalid workflow from defensible alternatives. Literal matching of SOC-2010 AIOE codes into a post-2018 occupation universe covers only 3.33 percent of computer-and-mathematical employment. The documented route reaches 97.7 percent. In a continuous comparison, the original score on original support gives \$-0.0188\$; the repaired score on the same 410 occupations gives \$-0.01920\$; expanding to 495 repaired-score occupations gives \$-0.03156\$; excluding computer/math from the expanded support gives \$-0.02940\$ \$\$. Most movement comes from changing who enters, not correcting scores among already matched occupations.

- This sequence is not a multiverse of equally defensible estimates. The exact-code merge is an implementation failure. The fixed-support row measures score correction. The expansion changes the target population. The computer/math exclusion asks whether one family drives the expanded-support coefficient. Reporting these layers separately is the portable contribution.

-

-\subsection{Primitive models rather than rotated labels}

-

-The prior draft emphasized ‘‘consensus’’ and ‘‘disagreement’’ coordinates $\$F=(A+E)/2\$$ and $\$G=(A-E)/2\$$. Those coordinates are retained in the online appendix only as a complete record. A representative two-score model is more transparent: administrative AIOE enters at $\$+0.044\$$ and beta at $\$-0.078\$$ when included jointly. In direct task units, $\$D\$$ and $\$S\$$ enter at $\$-0.0323\$$ and $\$-0.0303\$$. These are conditional associations among correlated constructed regressors, not causal technology channels. Transforming $\$F/G\$$ back to $\$A/E\$$ reproduces the same fitted predictor exactly and supplies no additional evidence.

+With 22 family clusters, the paired movement remains detected under Rademacher and Webb weights; the Webb interval is $\$[0.0059,0.2150]\$$. The completed finite-sample audit finds small coefficient bias but target-specific interval performance. In the empirical observed-effect design, occupation-wild coverage is 0.865 for the pooled coefficient, 0.921 for the family-conditioned coefficient, and 0.889 for their paired movement; family clustering is conservative for the pooled and paired targets but covers the family-conditioned target only 0.895 of the time. In the clean zero-target design, rejection rates for the family-conditioned coefficient are 6.1 percent under occupation weights, 8.9 percent under family weights, and 5.6 percent for the cross-fit oracle. No one procedure is therefore validated for all targets and declared designs. In the rebuilt-treatment audit, elapsed-calendar HAC leaves scalar uncertainty similar and none of the 15 five-parameter covariance matrices across three targets and five lags is indefinite. Earlier indefinite matrices belong to the superseded historical-treatment audit and remain labeled in the archive.

diff --git a/paper/main/sections/07_reallocation.tex b/paper/main/sections/07_reallocation.tex
index f0b742d..1d068f7 100644
--- a/paper/main/sections/07_reallocation.tex
+++ b/paper/main/sections/07_reallocation.tex
@@ -1,23 +1,92 @@
-\section{Timing, Weights, and Remaining Interpretations}\label{sec:reallocation}
+\section{Dynamics, Flows, and External Comparison}\label{sec:dynamics}

-\subsection{Pre-AI pseudo-breaks and post-period dynamics}

+\subsection{Dynamics}

-The authenticated extract begins in January 2017, so the literal request for monthly pseudo-breaks beginning in 2015 is infeasible. I estimate every feasible break in the fully pre-AI January 2017--December 2019 window and emphasize 12 dates with at least 12 retained months on each side. Under both pre-AI classifications and the frozen classification, the balanced pseudo-break distributions are centered near zero; none approaches \$-0.131\$. Because the 12 estimates overlap and were designed after the headline was known, the plus-one empirical tail of $\$1/13\$$ is a resolution floor, not a randomization $\$P\$$ -value. Unbalanced early-window extremes are artifacts of having as little as one pre-break month.

+The binary post coefficient is not an event-study design. I estimate quarterly coefficients for all four nonreference quintiles, omitting 2022Q4 (October--November; December is the transition month). The Q5 path has 23 preperiod and 15 postperiod coefficients in pooled and family-month specifications.

-The actual post path remains descriptive rather than a causal event study. A model with separate periods gives \$-0.1356\$ in 2023, \$-0.0783\$ in 2024, and \$-0.1629\$ in 2025--26. The last is distinguishable from 2024 but not from 2023. No detected pretrend and an end-2022 break do not isolate AI from the technology-sector correction, interest rates, return-to-office changes, post-pandemic normalization, or occupational composition.

+Figure-\ref{fig:dynamics} shows the rebuilt Q5 path. Relative to the omitted +2022Q4 coefficient, joint tests reject that all 23 preperiod coefficients are +zero in the pooled ($\$p=.0171\$$) and family-conditioned ($\$p=.0053\$$) models. A +reference-invariant test of equality within the entire preperiod rejects for +the pooled path ($\$p=.0121\$$) but not for the family-conditioned path +($\$p=.0641\$$). The appendix reports fixed pre-pandemic, post-reopening, and +pandemic-exclusion windows. An insignificant anchored linear slope cannot +override the unrestricted pooled test, while the family result illustrates +the importance of the unusually short 2022Q4 reference. Neither pattern +supports a causal January-2023 interpretation.

+When 2020Q2--2020Q4 are excluded, the pooled test no longer rejects at five +percent under either the short-reference null ($\$p=.0772\$$) or the +reference-invariant within-window null ($\$p=.0676\$$). For the family-conditioned +path the corresponding values are $\$p=.0079\$$ and $\$p=.1048\$$, underscoring that +the conclusion depends on the null as well as the calendar window.

-I do not add a national adoption series merely to replace the post indicator. Interacting one national series with fixed occupation exposure would largely reparameterize an already displayed exposure-by-time path. Industry-level BTOS adoption would require an occupation--industry ecological bridge and would still not generate exogenous occupational adoption. Such data could support a separately designed adoption paper, but forcing it into this measurement-centered revision would change both treatment and estimand without resolving the main confound.

+\begin{figure}[htbp]
+\centering
+\includegraphics[width=0.94\textwidth]{r3_figure2_dynamics.pdf}
+\caption{Quarterly Q5--Q1 young-relative employment coefficients}\label{fig:dynamics}
+\tabnote{Coefficients are relative to Q1 and omitted 2022Q4 under rebuilt labels; lines are pointwise 95-percent occupation wild-score intervals. The online appendix reports simultaneous bands and the Q2--Q5 system. The conditioned model gives each SOC2 family its own young-relative monthly path.}
+\end{figure}

-\subsection{Population controls and response composition}

+Three related targets separate a genuine estimand comparison from an event-time +normalization artifact. The static grouped-binomial coefficient $\$S\$$ is + $\$-0.1321\$$ pooled and $\$-0.0217\$$ after family conditioning. The calendar- +weighted post functional relative to omitted 2022Q4, $\$P\$$, is $\$-0.1199\$$ +($\$[-0.2637,0.0239]\$$) and $\$-0.2074\$$ ($\$[-0.4252,0.0103]\$$). By contrast, the +reference-invariant post-minus-pre functional $\$D\$$ is $\$-0.1314\$$ and $\$-0.0217\$$. +The paired $\$D\$$ - $\$S\$$ differences are 0.0007 ($\$p=.454\$$) and less than 0.0001 in +absolute value ($\$p=.995\$$). Thus $\$D\$$ nearly reproduces the static estimates; +the design does not detect the family-model $\$P\$$ - $\$S\$$ difference at five percent +($\$-0.1858\$$, $\$p=.074\$$). What it does detect is that family conditioning changes + $\$P\$$ - $\$S\$$ by $\$-0.1980\$$ ($\$p=.021\$$), while it does not detect a change in $\$D\$$ - $\$S\$$ +($\$-0.0008\$$, $\$p=.693\$$). The short-reference functional therefore responds to +conditioning differently from the reference-invariant comparison. This +numerical reconciliation is not an equivalence claim and does not make any +estimand causal.

-January 2025 introduced unusually large CPS population-control revisions. A literal age-by-occupation counterfactual series ‘‘without’’ those controls cannot be constructed: BLS provides aggregate experimental adjustments, not subgroup weights, and official earlier microdata are not revised. I therefore report what the available data identify rather than rescale cells ad hoc.

+Official \texttt{HonestDiD} 0.2.8 applies smoothness and relative-magnitude restrictions to the full event vector and covariance for $\$P\$$, not for $\$S\$$ or $\$D\$$. Both the full 2017Q1--2022Q3 calibration and a consecutive 2021Q1--2022Q3 post-reopening calibration are reported. Under the latter, the family-conditioned smoothness interval is $\$[-0.772,-0.007]\$$ when differential-trend bias is restricted to be exactly linear ($\$M=0\$$), but expands to $\$[-1.100,0.207]\$$ at the first positive grid value, $\$M=0.005\$$. Both conventional $\$P\$$ intervals already include zero, so no positive zero-exclusion threshold exists. The exercise cannot repair rejected preperiod equality or identify adoption.

-On the repaired calendar, official weights give \$-0.1346\$ and respondent-equivalent cells give \$-0.1348\$. Ending in December 2024 gives \$-0.1135\$ and \$-0.1056\$, respectively. In a joint official-weight model, 2023--24 is \$-0.1108\$ and 2025--26 is \$-0.1664\$; their paired difference is \$-0.0556\$ with interval $\$[-0.1228,0.0116]\$$. Under respondent equivalents the era difference is more negative and detected. The raw official-weight Q5--Q1 log-ratio contrast becomes slightly less negative from December 2024 to January 2025. These checks do not provide a counterfactual weight vintage, but they are inconsistent with revised population weights mechanically creating the full association.

+Onset estimates from November 2022 through June 2023 are similar within each structure, but common timing does not distinguish AI from coincident shocks. Ending in December 2024 gives a pooled coefficient of $\$-0.111\$$ ($\$[-0.201,-0.021]\$$); its paired movement relative to the July-2026 estimate is $\$0.021\$$ ($\$[-0.007,0.049]\$$). The corresponding movements at the September- and December 2025 endpoints are 0.0225 and 0.0212, essentially the same magnitude despite different interval endpoints. The full window remains the canonical target and December 2024 a prominent pre-production-break benchmark; neither is causally clean. Splitting the postperiod instead gives $\$-0.111\$$ in 2023--24 and $\$-0.168\$$ in 2025--26; their paired difference is $\$-0.057\$$ ($\$[-0.125,0.012]\$$), so the official-weight design does not detect a change between eras. Family-conditioned endpoint changes remain near zero, and survey-gap exclusions are small. A coding-stable post-2020 fit gives $\$-0.118\$$ pooled and $\$-0.030\$$ with family-month effects; their paired movements from the full-window models are $\$0.014\$$ ($\$[-0.029,0.057]\$$) and $\$-0.009\$$ ($\$[-0.105,0.088]\$$). Quintile- and occupation-specific month-of-year controls move either coefficient by at most $\$0.0011\$$. All six replacement fits pass the same numerical certification; the earlier failed attempts remain in the failure registry rather than being treated as results.

--'Respondent equivalent' means that cells are reconstructed from observed employed records after route expansion without their CPS final weights. It changes the empirical cell outcome; dividing all weighted stocks by a common constant would not change the maximizer and is not what is done here.

+{subsection(Linked CPS margins)}

-{subsection(Age, entry, and mobility)}

+Table-{ref(tab:flows)} estimates distinct adjacent- and twelve-month margins using validated links and official longitudinal weights. Every primary interval includes zero. Normal-theory MDEs_{80}\$ values range from 0.115 to 0.336 log point, so these are low-resolution nondetections rather than evidence of absent flow responses.

-The pooled older comparison is not the sole source of the pattern. Ages 22--25 versus 26--35 give $\$-0.1307\$$, nearly identical to the frozen pooled result. Comparisons with ages 26--45 and 36--55 are $\$-0.1475\$$ and $\$-0.1464\$$. The contrast against ages 51--65 is smaller, $\$-0.0978\$$, and its paired difference from the pooled comparison is detected. Age 22--25 is an age group, not observed experience; the paper therefore uses "young" rather than treating every respondent as a labor-market entrant.

+{input{tables/r3_table5_flows.tex}}

-The main outcome cannot determine whether hiring, separation, or occupation switching generated the stock association. CPS longitudinal links do allow distinct flow estimands, but their intervals are wide. Entry destination is an allocation conditional on moving from nonemployment to employment, not an employment-finding probability. The employed-to-employed switch frame says still less about hiring. For that reason, the mobility analysis, family-balanced disagreement statistics, rematching benchmark, and flow regressions appear in the online appendix rather than as a main contribution. No immediate-reversal rate is treated as a validated occupation-coding-error rate.

+Employment exit starts from employed origins; reported occupation change requires employment at both endpoints; entry destination conditions on nonemployment-to-employment links and measures allocation across exposure groups. The last is neither a finding probability nor employer hiring. Twelve-month links are endpoints, not sums of monthly transitions. Among fixed-support employed origins, postperiod twelve-month retention ranges from 48.7 to 52.8 percent across young quintiles and from 63.9 to 70.2 percent across older quintiles; preperiod ranges are 50.1--54.0 and 68.8--73.3 percent. Hours ($\$-0.119\$$, $\$[-0.295,0.057]\$$) and conditional weekly earnings ($\$-0.0245\$$, $\$[-0.1024,0.0534]\$$) are also imprecise and condition on employment; the legacy earnings series ends in March 2023 and contains only three postperiod months, while a full-window

+{textrtt{EARNWEEK2}} analysis remains unavailable in the authorized extracts.

+{+}

+Observable poststratification does little to the adjacent employment-exit

+probability contrast: the descriptive Q5--Q1, young--older, post--pre difference

+is 0.0088 under official link weights and 0.0086 after reweighting retained

+links to the eligible-origin distribution. At twelve months it moves from

+0.0166 to 0.0210. This reweighting requires outcome missingness to be

+independent of linkage within the observed strata and includes a maximum

+untrimmed factor of 102.7. Without that assumption, the jointly feasible

+linear accounting interval for twelve-month employment exit is

+ $\$[-1.570,1.575]\$$, leaving the sign of selection bias unrestricted. These are

+descriptive probability contrasts and missing-outcome bounds, not the nonlinear

+regression coefficient and its confidence interval.

+{+}

+The annual primary rule excludes origins whose twelve-month link straddles the

+January 2023 onset. Coding instead the fraction of destination months exposed

+gives an employment-exit coefficient of 0.158 ($\$[-0.070,0.387]\$$), compared

+with the Table-{ref(tab:flows)} primary estimate of 0.123

+($\$[-0.111,0.357]\$$). Both intervals contain zero;

+no paired interval for their difference is available.

+{+}

+All flow exposure is defined at the origin where an origin occupation exists. No current occupation is imputed to a nonemployed respondent. Annual endpoints miss intervening jobs and transitions, and positive longitudinal-weight eligibility changes the population; neither horizon is a mechanical decomposition of the stock coefficient. Link-rate and observable-balance differences diagnose selection but do not determine its direction, and alternative clustering addresses dependence rather than missing outcomes.

+{+}

+{subsection(Payroll bridge and exposure architecture)}

+{+}

+BOC's August 2026 19-percent "kept pace" shortfall is a stock-growth comparison, not this regression coefficient or a hiring rate. Complete public membership and tie rules are unavailable, so I use an equal-occupation, tie-preserving approximation and label it a bridge. Its top-two-versus-bottom-three CPS coefficient is $\$-0.0720\$$ ($\$[-0.1223,-0.0218]\$$); family-month conditioning gives $\$-0.0168\$$ ($\$[-0.0808,0.0473]\$$). Their paired movement is $\$0.0553\$$ ($\$[-0.0013,0.1118]\$$), so the design does not detect a difference. Differences in universe, ages, outcome, endpoints, employer controls, and worker--firm flows remain.

+{+}

+That grouping bridge is distinct from changing the regression objective. Giving each of the 468 occupations equal total weight, while retaining {textrtt{WTFINL}} for within-cell age shares, changes the pooled Q5-by-post coefficient from $\$-0.1321\$$ to $\$-0.0833\$$; the paired difference is $\$0.0488\$$ ($\$[-0.0604,0.1581]\$$). The family-month coefficient changes from $\$-0.0217\$$ to $\$-0.0047\$$, with paired difference $\$0.0170\$$ ($\$[-0.1587,0.1927]\$$). These intervals are nondetections, not evidence that the objectives are equivalent.

+{+}

+The bridge therefore asks whether a closer public grouping changes the CPS stock pattern; it does not calibrate CPS transitions to proprietary employer flows. The required entry, exit, job-to-job, occupation-switching, aging, and denominator quantities are not jointly observed.

+{+}

+Finally, $\$E_1 + \lambda E_2$ varies the weight on tasks requiring complementary software. Here $\$E_1\$$ is the directly accelerable task share and $\$E_2\$$ is the share requiring complementary software. Each row uses rebuilt weights and common draws, and $\lambda = .58$ exactly reproduces beta. Native-tail coefficients range from $\$-0.103\$$ at $\lambda = 0$ to $\$-0.156\$$ at $\lambda = 1$; every comparison with beta includes zero, even though 61.6 percent of stock changes quintile at $\lambda = 0$. This is large classification movement with limited coefficient resolution, not evidence about deployed technology.

+{+}

+Five non- λ definitions give pooled native-support tail coefficients from $\$-0.099\$$ (direct-ability AIOE) to $\$-0.014\$$ (the reversed OECD AI-skills gap). Every alternative-minus-beta paired occupation interval contains zero, although the OECD lower endpoint is $\$-0.0004\$$. On the common AIOE support, some family-conditioned implementation contrasts do exclude zero; the result is therefore sensitive to how one construct is mapped in that specification, without implying that the alternatives measure the same technology.

+{+}

+{input{tables/r3_table6_lambda.tex}}

+diff --git a/paper/main/sections/08_competing_interpretations.tex b/paper/main/sections/08_competing_interpretations.tex

+index 344106a..f5df9ff 100644

+--- a/paper/main/sections/08_competing_interpretations.tex

++++ b/paper/main/sections/08_competing_interpretations.tex

+@@ -1,13 +1,9 @@

+{section(Conclusion)}{label{sec:competing}}

+{section(Conclusion)}{label{sec:conclusion}}

-Public CPS data reproduce a negative young-relative employment-stock association under a widely used occupational AI-exposure construction. After repairing the calendar, the top-versus-bottom beta-quintile contrast is $\$-0.1346\$$ log points. Implementing the public top-two-versus-bottom-three grouping used by {citet{brynjolfsson2026canaries}} gives a smaller but negative $\$-0.0728\$$ contrast. These are descriptive population comparisons, not replications of proprietary hiring results or causal AI estimates.

+Public CPS data contain a sizeable negative association between GPT-4 task exposure and employment for ages 22--25 relative to ages 26--65. The rebuilt Q5--Q1 coefficient is $\$-0.132\$$ log points, but it falls to an imprecise $\$-0.022\$$ when broad occupation families receive their own young-relative monthly paths. Their paired movement is detected; it is a change between conditional estimands, not a causal composition share.

-The decisive fact is where the comparison lies. Broad two-digit occupation-family controls reduce the detailed Q5--Q1 coefficient to about $\$-0.032\$$. The paired movement is detectable, while the residual coefficient is imprecise and draws on only 30 percent of baseline target information. Only four broad families span both exposure tails. Thus, the baseline pattern is principally a comparison across broad occupational families; the design does not precisely isolate a detailed within-family AI gradient.

+Direct within-family tails cover only 29 occupations and are highly imprecise. Annual ACS data reduce person-sampling uncertainty but do not repair that support: the fixed-label full-calendar family-year coefficient is negative under ACS replicate inference, whereas occupation- and family-shock intervals include zero and the nonreuse-calendar family-year estimate is undetected. Preperiod computer use strengthens rather than attenuates the coefficient, showing that computerization is not settled by one control. Unrestricted dynamics reject joint preperiod equality, and linked-CPS flow estimates do not identify a mechanism or replicate employer-match hiring and separation.

-That conclusion should not be embellished into a mechanism the data reject. Low-exposure food-service occupations matter, but removing them attenuates the point estimate by less than one tenth. Broader in-person-service exclusions do not attenuate it. Joint influence exercises show large, oppositely signed occupation contributions rather than one occupation whose removal makes the result vanish. The evidence supports a broad-composition interpretation, not a demonstrated food-service-recovery explanation.

+Later self-reported use is descriptively aligned with the score. An exact-code merge to a public RPS workbook retains 121 nonsuppressed occupations: the unweighted exposure--adoption correlation is 0.504 and mean adoption is 0.267 higher in Q5 than Q1. The profile is not monotone, and only 26.2 percent of exposure variation remains after removing broad-family means. Because the rates pool four 2025--26 waves and lack detailed-cell sampling uncertainty, this check validates neither adoption timing nor an employment mechanism.

-Nor do alternative exposure estimates permit a horse-race winner. All seven paired beta-minus-alternative Q5--Q1 intervals include zero, with minimum detectable differences as large as 0.169 log points. A confidence interval containing zero means that the design does not detect a difference; it does not establish economic equivalence. The original six implementations are also highly dependent, with two components accounting for 96.11 percent of their score variance. Claims of six independent validations or universal architecture robustness are therefore retired.

-{+}

-The measurement work remains useful precisely because it distinguishes objects that should not be pooled. Exact-code matching across incompatible occupational vintages is an implementation error. Repairing scores on fixed support is a measurement comparison. Readmitting occupations changes the population. Switching from Q5--Q1 to top-two-versus-bottom-three changes the estimand. Moving from LLM task acceleration to patent overlap changes the construct. Rotating $\$F/\G into $\$A/\E changes only coordinates. The revised tables report these layers rather than treating every row as interchangeable robustness evidence.

```

-
-The paper's claim is consequently narrower and stronger than before. The early-career CPS pattern is real as a descriptive broad-occupation contrast, but this design cannot attribute it specifically
  to detailed AI exposure. Future causal work needs observed adoption or deployment, a credible source of timing or assignment variation, and outcomes that directly measure hiring and separation.
  Future measurement work needs occupation-task validation labels collected independently of the constructed indices, repeated ratings or audited use logs, and a taxonomy-stable holdout sample.
  Until then, occupational AI coefficients should travel with their construct, mapping, support, comparison, and resolution limits.
+The defensible conclusion is therefore comparison-specific. Mapping, support, representation, and conditioning determine what an exposure coefficient measures. A detected movement does not identify
  why it occurs; nondetection does not establish equivalence. Causal work requires observed adoption or credible assignment and timing variation. This paper instead establishes that the public-
  CPS young-worker pattern is principally a broad occupational comparison and quantifies what direct within-family and flow evidence cannot resolve. Online Appendix I.3 collects the remaining
  data, design, and interpretation limits rather than treating them as additional specification searches.
diff --git a/paper/main/title.tex b/paper/main/title.tex
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@@ -20,10 +20,16 @@
\space{0.15in}
\noindent\textbf{JEL codes:} J23, J24, O33, C81.

\noindent\textbf{Keywords:} artificial intelligence, occupational exposure, occupational composition, early-career employment, constructed treatments.
\noindent\textbf{Keywords:} artificial intelligence, occupational exposure, occupational composition, young-worker employment, Current Population Survey.

\space{0.1in}
\noindent\textbf{Acknowledgments:} [AUTHOR TO COMPLETE].
+
\noindent\textbf{Funding:} [AUTHOR TO COMPLETE].
+
\noindent\textbf{Conflicts of interest:} [AUTHOR TO COMPLETE].
+
\noindent\textbf{Data and code disclosure:} Replication code and nonconfidential derived outputs will accompany the paper; restricted CPS microdata access and journal-specific disclosure language are
  [AUTHOR TO COMPLETE].
\end{minipage}
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\end{titlepage}
diff --git a/paper/revision/README.md b/paper/revision/README.md
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--- a/paper/revision/README.md
+++ b/paper/revision/README.md
@@ -1,25 +1,33 @@
-# Major-advice referee revision
-# V3 substantive referee revision

-Build with:
+This directory contains the editable response materials for the current V3
+revision of *Occupational AI Exposure and Young-Worker Employment: Support and
+Comparisons in the CPS*. It is a working revision package, not a claim that the
+paper is submission-ready or independently replicated.
+
+## Editable deliverables
+
+- manuscript: 'paper/main/working.tex' and its files under 'paper/main/';
+- online appendix: 'paper/appendix/appendix.tex' and its files under
+  'paper/appendix/';
+- responses to Referees A and B: 'paper/revision/referee_response.tex';
+- revision diagnosis and unresolved-work record:
+  'paper/revision/revision_diagnosis.tex';
+- change record: 'paper/revision/source_diff.tex', using the generated
+  'paper/revision/source_diff.txt'.
+
+After the source tree is finalized, the V3 document target is:
+
+'''sh
+make -C paper substantive-revision
+'''
+
+It produces:
+
+- 'paper/build/YAX_REVISED_MANUSCRIPT.pdf';
+- 'paper/build/YAX_FOCUSED_ONLINE_APPENDIX.pdf';
+- 'paper/build/YAX_REFeree_RESPONSE.pdf';
+- 'paper/build/YAX_REVISION_DIAGNOSIS.pdf'; and
+- 'paper/build/YAX_SOURCE_DIFF.pdf'.
+
+Where 'latexmk' is unavailable but Tectonic is installed, the existing
+alternative is:

'''sh
make -C paper major-revision
paper/scripts/build_substantive_revision_tectonic.sh
'''

-On SOC installations without 'latexmk', submit 'paper/scripts/scc_build_major_revision_pdfs.sh' from the repository root.
+Both commands compile documents only. They do not rerun an empirical model.
+PDFs already present under 'paper/build/' are snapshots and must be rebuilt
+after any source change before their hashes or visual checks are treated as
+current.
+
+## Aggregate-only exhibit build
+
+The existing aggregate exhibit command is separate from both TeX compilation
+and protected-data re-estimation:
+
+'''sh
+python3 paper/scripts/build_r3_exhibits.py --only all
+'''
+
+Despite its historical filename, this script reads retained aggregate results,
+including V3 support-inference outputs. It rebuilds three 'r3_' figures and the
+family-support, profile, leave-one-family-out, and endpoint-sensitivity tables.
+It does *not* regenerate every V3 table, rerun CPS estimation, or certify the
+full revision. Other V3 exhibits remain tied to their phase-specific result
+directories and validators under
+'yax/revision/substantive_v3_20260906/'.
+
+## Evidence and completion boundary
+
+The V3 instruction, immutable seed, working status ledger, specifications,
+sanitized receipts, and retained aggregate results are under
+'yax/revision/substantive_v3_20260906/'. Start with:

-The four required outputs are:
+- 'revision_inputs/EXECUTION_PROMPT_V3.md';
+- 'revision_inputs/requirements_seed.json';
+- 'requirements_status.json';
+- 'source_inventory.json'; and
+- 'runs/' plus the relevant gate specification and validator.

-- 'paper/build/YAX_WORKING_PAPER_REVISED.pdf'
-- 'paper/build/YAX_ONLINE_APPENDIX_REVISED.pdf'
-- 'paper/build/YAX_REFeree_RESPONSE.pdf'

```

```
-- 'paper/build/YAX_REVISION_DIAGNOSIS.pdf'
+Those records provide internal traceability. They do not by themselves prove
+an independent replication: licensed CPS microdata and the protected Gate 1
+cell object are not distributed, and an aggregate exhibit rebuild is not a
+fresh microdata re-estimation.

-The second-round outputs preserve those files and use new names:
+The current genuine external-input limits are:

-- 'paper/build/YAX_WORKING_PAPER_MAJOR_REVISION.pdf'
-- 'paper/build/YAX_ONLINE_APPENDIX_MAJOR_REVISION.pdf'
-- 'paper/build/YAX_REFeree_RESPONSE_MAJOR_REVISION.pdf'
-- 'paper/build/YAX_REVISION_DIAGNOSIS_MAJOR_REVISION.pdf'
++ the authorized CPS extracts do not contain 'EARNWEEK2', so the requested
++ full-window weekly-earnings extension requires a new authorized extract;
++ no verified public artifact supplies exhaustive BCC occupation-to-quintile
++ membership, so public-data comparisons are labeled independent
++ reconstructions rather than exact BCC replications; and
++ affiliation, email, acknowledgments, funding, conflicts, repository/archive
++ identifiers, and journal-specific disclosures remain
++ '[AUTHOR TO COMPLETE]'.
```

-The first two are journal-facing. The response and diagnosis preserve completion status, principled non-adoptions, and author-only metadata tasks. The second-round machine results, response matrix, and receipts are under 'yax/revision/referee_round2_20260905/'. This stage addresses the numbered major comments; secondary copyediting is explicitly reserved for the next pass.

+Final delivery also requires a clean rebuild, updated hashes, visual inspection, +status-ledger reconciliation, and the delivery checks described in the V3 +prompt. The presence of a run directory or a polished PDF is not a substitute +for those checks.

```
diff --git a/paper/revision/referee_response.tex b/paper/revision/referee_response.tex
index 2ff8296..aa76fb0 100644
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 \usepackage{styles/yax-common}
 \geometry{margin=1in}
 \hypersetup{pdftitle={Major-Comment Response to Two Referee Reports},pdfauthor={Lily Luo}}
 \hypersetup{pdftitle={Response to Two Referee Reports},pdfauthor={Lily Luo}}
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 \begin{center}
 -{\Large\bfseries Major-Comment Response to Two Referee Reports}\par
 -\vspace{0.4em}
 -{\large AI Exposure or Occupational Composition?\\Constructed Measures and Early-Career Employment}\par
 -\vspace{0.4em}
 +{\Large\bfseries Response to Two Referee Reports}\par
 +\vspace{0.35em}
 +{\large "Occupational AI Exposure and Young-Worker Employment:\\Support and Comparisons in the CPS"}\par
 +\vspace{0.35em}
 Lily Luo \quad September 2026
 \end{center}
```

-(\textbf{Scope of this revision.}) This document responds to every numbered major comment in the two reports supplied on September 5, 2026. In accordance with the staged revision requested by the author, secondary copyediting and presentation comments are catalogued for the next pass rather than represented as complete. All post-outcome analyses are labeled exploratory; the frozen v1.1 design and results remain intact.

+{\section*{Response to the editor}}

-\section*{Revision in one paragraph}

+We thank both referees for identifying the same central problem from different directions: the earlier manuscript reported a sizeable sensitivity to broad-family conditioning, but it did not yet make occupational support and the precise descriptive target the organizing contribution. We have substantially refocused the paper. Its positive result is now that an apparently broad national exposure comparison contains little direct within-family tail information. On the primary 468-occupation support, only four broad families contain both Q1 and Q5 occupations; the resulting 29 direct-tail occupations represent 5.03 percent of preperiod employment stock, and their information is equivalent to 6.2 occupations. The paper treats the pooled-to-conditioned coefficient movement as a consequence of that support geometry, not as a causal "share explained" or evidence against an AI effect.

-We accepted the central diagnosis and rewrote the paper around the result the CPS can establish. The corrected Q5--Q1 beta coefficient is $-\$.1346\$$, but it falls to $-\$.0315\$$ after broad-family-specific post changes are absorbed and to $-\$.0317\$$ after family-specific monthly paths are absorbed. The paired movement is detectable, although the conditional estimand changes and retains only 30 percent of baseline target information. Clean service exclusions do not support a food-service mechanism. We now report a primary MDES_{80} of $$.1244$ and paired MDEs for every architecture comparison; all seven paired Q5--Q1 intervals include zero. The title, abstract, introduction, results, and conclusion no longer claim general architecture heterogeneity or AI-specific displacement. The revised contribution is a composition-centered empirical finding joined to a framework that separates implementation error, measurement, construct, support, estimand, representation, and reparameterization changes.

+The revision adds the analyses required to make that claim auditable. It places the family-by-quintile support matrix and information diagnostics in the main paper; reports 89 supported family-specific edges and a broader 490-occupation support check; gives exact young/older and within-family/family-weight stock accounting; crosses computer-use and family conditioning on common support; separates the static coefficient from reference-dependent and reference-invariant dynamic functionals; completes a target-specific finite-sample audit; adds age-band, fixed-age-composition, enrollment, cohort-composition, flow-selection, ACS, public-benchmark, and later-adoption checks; and resolves the earlier numerical-existence question for all eleven declared models. The manuscript also distinguishes results that are informative under one sampling target from those that are robust across occupation, family, household, and ACS-replicate perturbations.

-\section*{Response to Referee 1}

+Several requests required modification rather than nominal compliance. We do not promote December 2024 to the primary endpoint after observing the outcomes; it is instead a prominent pre-2025-production-break benchmark beside the unchanged full-window target. We do not manufacture an Oster-style scalar bound for a nonlinear grouped-binomial coefficient when the required selection relation is not identified. Exact BCC occupation memberships and tie handling are not public, so our BCC exercises remain labeled public-data analogues rather than replications. Full-window weekly earnings remain unavailable because the authorized extracts omit \texttt{EARNWEEK2}; the short legacy series is retained only as ancillary evidence. These non-adoptions and input constraints are stated below rather than described as completed work.

-\subsection*{M1. Precision and comparative language}

+{\section*{Response to Referee A: major comments}}

-\emph{Response.} Accepted. The main text now reports the realized primary MDES_{80} of $$.1244$ log points next to the $-\$.1311\$$ frozen estimate. A consolidated paired table reports MDEs from $$.0609$ to $$.1689$ for beta against the three AIOE implementations, alpha, broad, Webb AI, and OECD. Every paired Q5--Q1 interval contains zero. Throughout, we now write "the design does not detect a difference" rather than "the conclusion does not generalize," and never infer equivalence.

+{\subsection*{3.1. Family conditioning, support, and the order of the argument}}

-We also diagnosed the prospective-to-realized SE gap. A sharp historical simulation preserves each donor month's full contemporaneous cross-occupation residual vector with a global sign. Its mean SE remains $$.01217$, so it closes none of the gap to $$.04441$. We report this as a rejection of one proposed mechanism, not a decomposition or validation of the planning simulation. See Sections 1, 4, and 6; Main Table 3 and Figure 4; Appendix D and K.

+{\emph{Response: accepted.}} The abstract, introduction, and first results section now lead with support and information rather than the near-zero conditioned coefficient. The main paper displays the family-by-quintile matrix and reports that only four of 22 families span Q1 and Q5, that the 29 direct-tail occupations account for 5.03 percent of preperiod stock, that family-month conditioning retains 29.7 percent of pooled Q5 target information, and that the direct-tail effective count is 6.2 occupations with a 77.8-percent top-five information share. The direct-tail interval, $[-0.169, 0.468] \$$, is presented before interpreting the conditioned coefficient.

-\subsection*{M2. Effective clustering and CPS rotation}

+The introduction now states explicitly that family paths cannot distinguish an AI effect operating at the broad-family level from unrelated family-level evolution. The movement from $-\$.0.1321\$$ to $-\$.0.0217\$$ is therefore a comparison of conditional estimands and a corollary of the support geometry. It is not a causal decomposition, and the residual coefficient is not evidence that AI has no effect.

-\emph{Response.} Addressed within the available survey design. Young-cell respondent equivalents have $p_{10}/\text{median}/p_{90}$ of 0/2/16; 26.3 percent are zero and 69.9 percent below five. Quarterly aggregation gives $-\$.13103\$$ on the frozen and $-\$.13454\$$ on the repaired calendar, nearly identical to monthly estimates and SEs. A cross-occupation time-score covariance sensitivity increases the SE to $$.04909$ at a 12-month Nevev--West lag, with normal interval $[-.2273, -.0349] \$$.

+{\subsection*{3.2. Pandemic quarters and preperiod diagnostics}}

-The main text now explains that occupation clustering already permits arbitrary serial dependence within occupation but is not a complete CPS design-based variance. The additional HAC targets common monthly and nearby-month score covariance across occupations. The extract begins in 2017, so literal 2015--19 pseudo-breaks are unavailable. We run all feasible 2017--19 dates and emphasize 12 balanced dates; none approaches $-\$.1311\$$. Their $1/13\$$ empirical tail is explicitly a resolution floor in a dependent, post-outcome diagnostic, not a randomization $\$ \$$ -value. See Section 4 and Appendix D.

+{\emph{Response: accepted, with the nulls kept distinct.}} The appendix now reports fixed tests for 2017Q1--2019Q4, 2021Q1--2022Q3, and the full preperiod excluding 2020Q2--2020Q4, under both the short-reference null and a reference-invariant within-window equality null. Over the full preperiod, the reference-invariant test rejects for the pooled model ($\$p=.0121\$$) but not for the family-

conditioned model (\$p=.0641\$). After excluding 2020Q2--2020Q4, neither within-window equality test rejects at five percent (\$p=.0676\$ pooled and \$p=.1048\$ conditioned). Relative to the short 2022Q4 reference, the corresponding pandemic-exclusion values are \$p=.0772\$ and \$p=.0079\$. The difference is reported as a property of the null and the unusually short reference period, not as a favorable re-selection of the window.

-\subsection*{M3. Regression analogue of the permutation, monotonicity, and stable tails}
+Leave-one-quarter diagnostics identify 2019Q2 as producing the largest reduction in the pooled reference-relative Wald statistic and 2018Q4 in the conditioned statistic. Thus the revised paper does not say that rejection is confined to pandemic quarters. It says that the full pooled within-window rejection no longer clears five percent when those quarters are excluded, while the reference-relative family rejection persists. HonestDID is now run on the full event vector and covariance for the explicitly named post functional \$P\$, with both the full calibration and the consecutive 2021Q1--2022Q3 calibration. Its conventional intervals already include zero, and the first positive smoothness allowance also includes zero; the exercise is not presented as a robustness statement for the static coefficient.

-\emph{Response.} This result now determines the paper. On the corrected calendar, SOC2-by-young-by-post terms reduce \$-\$.1346\$ to \$-\$.0315\$; SOC2-by-young-by-month paths give \$-\$.0317\$. Paired baseline-to-conditioned movements are approximately .103 and their intervals exclude zero. We do not call this a causal decomposition: conditioning changes the estimand, only four of 22 families span Q1 and Q5, and target information falls to 30 percent.

+\subsection*{3.3. Young numerator and older denominator}

-We add a three-restriction test of a common Q2--Q5 coefficient (\$p=.119\$) and a least-favorable max-\$t\$ monotonicity assessment (\$p=.393\$). Neither equality nor monotonicity is established. The 46 always-Q1 and 18 always-Q5 occupations yield \$-\$.2120\$ with interval \$[-.4131, -.0109]\$, but only 15.0 effective information occupations and a 46.7-percent top-five share. See Section 5, Main Table 2 and Figure 2, and Appendix C/E.

+\emph{Response: addressed through exact stock accounting rather than unmatched regressions.} Under the saturated occupation--month nuisance structure, separate single-age regressions do not reproduce the ratio target: the corresponding single-age target is absorbed, while an additive occupation-plus-month regression would define a different model. We therefore report an exact, model-free equal-month accounting on the same Q1 and Q5 stocks. The Q5--Q1 young-stock change is \$-\$.0.1098\$ log point and the older-stock change is \$0.0483\$, so their young-relative difference is \$-\$.0.1581\$. The numerator supplies the larger movement and the denominator reinforces it. A second exact Shapley accounting assigns \$-\$.0.1293\$ to within-family ratio changes and \$-\$.0.0288\$ to changing older-family weights, with a zero boundary term after auditing all absent cells.

-\subsection*{M4. Influence concentration and service recovery}
+These identities are reported in the main paper and appendix, but are not forced to equal the nonlinear grouped-binomial coefficient \$-\$.0.1321\$. We label them descriptive stock accounting, not separate causal regressions or percentage shares.

-\emph{Response.} Influence is now a headline diagnostic rather than summarized by the absence of sign reversal. Deleting the five occupations with largest absolute prior LOCO movements gives \$-\$.1011\$. Deleting ten or twenty gives \$-\$.1522\$ and \$-\$.1555\$, showing offsetting signed influence. A symmetric signed-tail trim gives \$-\$.1281\$ and continuous Huber down-weighting \$-\$.1314\$. These data-adaptive rows are labeled diagnostics, not preferred estimators.

+\subsection*{3.4. Finite-sample stress results}

-We replaced the coarse industry proxy with occupation-major-group exclusions. Dropping Q1 food-service occupations gives \$-\$.1196\$, an 8.8-percent attenuation. Dropping broader in-person services gives \$-\$.1370\$ or \$-\$.1387\$. We therefore reject the proposed strong food-service mechanism while accepting that the Q1 trajectory and individual occupations matter. See Section 5 and Appendix E/J.

+\emph{Response: accepted.} We withdrew the earlier 26.7- and 11.3-percent figures rather than defending them. The replacement audit has eleven declared designs, 799--1,599 retained outer replications per design, 9,999 common multiplier draws per joint refit, independently computed pseudo-targets, and separate occupation, family, and cross-fit-oracle procedures. In the empirical observed-effect design, occupation-wild coverage is 0.865, 0.921, and 0.889 for the pooled, family-conditioned, and paired targets; family-Rademacher coverage is 0.990, 0.895, and 0.992. In the clean zero-target design, rejection of the family-conditioned target is 6.1 percent under occupation weights, 8.9 percent under family weights, and 5.6 percent for the cross-fit oracle.

-\subsection*{M5. CPS population controls}
+The simulation also isolates why a structural-null calibration can remain far from a zero pooled projection: removing the structural exposure coefficient while retaining the fitted family-month surface leaves a pooled pseudo-target of \$-\$.0.11035\$. The revision therefore does not offer one omnibus ‘size-corrected’ interval. It reports target-specific performance and states the stronger limitation: no feasible cluster procedure is validated for every target and calibrated design, and rejection of the pooled projection alone is not a test of a structural exposure effect.

-\emph{Response.} The respondent-equivalent result is promoted and defined operationally as cells reconstructed from observed records without final CPS weights. It is not a common rescaling of weighted stocks. On the repaired calendar it is \$-\$.1348\$, versus \$-\$.1346\$ under official weights. Ending in December 2024 gives \$-\$.1056\$ and \$-\$.1135\$, respectively.

+\subsection*{3.5. Computer use, collinearity, and sensitivity}

-BLS does not publish age-by-occupation bridge weights for the January 2025 revision; its experimental factors are aggregate. We therefore do not fabricate a counterfactual micro-weight series. The raw official-weight contrast becomes slightly less negative between December 2024 and January 2025, and the official 2025--26 era is not distinguishable from 2023--24. The documentation, bounded inference, and refusal are now explicit. See Section 7 and Appendix F.

+\emph{Response: accepted in substance; the proposed scalar bound was not imposed.} On a fixed 455-occupation support, the paper now reports the four matched models: baseline, computer use, family-month paths, and both together, all with common draws. Computer use moves the exposure coefficient from \$-\$.0.0957\$ to \$-\$.0.1966\$; family paths alone give \$0.0354\$; both give \$-\$.0.0467\$. The computer-use coefficient itself is \$0.0460\$ per preperiod-weighted standard deviation in the pooled computer model and \$0.0487\$ in the combined model, with both intervals reported. The beta--computer-use correlation is 0.797 on this support, retained Q5 information falls from 69.1 to 29.0 percent, and the VIF-like ratio is 3.45.

-\subsection*{M6. Selected architecture space}
+The text treats the positive computer coefficient and the more negative exposure coefficient as joint-projection factors under collinearity, not as evidence that computerization raises employment or that the residual isolates AI. We added an explicit applicability ruling for Oster, Cinelli--Hazlett, and DiGert--Masten--Poirier style analyses: none maps automatically to this nonlinear grouped-binomial target with generated, highly collinear treatments and changing nuisance spaces. Rather than choose an arbitrary proportional-selection parameter after outcomes were known, we report the matched projection matrix and information loss and leave a scalar confounder bound unreported.

-\emph{Response.} The revision includes a repository-bounded candidate census with source-integrity, construct, mapping/support, and representation status. It records six selected implementations from two core families, two operationally admitted external implementations from two other families, one scope failure, and multiple pending or locally unverified candidates. It does not claim a global count.

+\subsection*{3.6. Enrollment and cohort composition}

-The weighted eigenvalue spectrum confirms the referee's concern: PC1 explains 80.04 percent and the first two components 96.11 percent of common-support score variance. The IUT is demoted to a predeclared marginal sign statement over a dependent set; the failed simultaneous band remains visible. Firm- and demand-side measures are not forced into occupation CPS without a defensible unit bridge. See Section 2 and Appendix A.

+\emph{Response: accepted.} Restricting the young numerator to nonenrolled workers moves the pooled coefficient from \$-\$.0.1321\$ to \$-\$.0.1239\$, with paired movement \$0.0082\$ and interval \$[-.0.0125, 0.0289]\$. On the common ages 22--54 question universe, restricting both groups to nonenrolled workers produces a similarly small paired movement. The appendix reports annual enrollment and BA-or-higher shares by quintile, exact-age composition, approximate birth-year composition, and a national ages 22--25 population profile that assigns no exposure to nonworkers. National enrollment changes from 0.2336 to 0.2211, BA-or-higher from 0.2941 to 0.3121, and average monthly weighted population from 16.94 to 17.60 million between the defined pre- and postperiods. These results document cohort composition as a live alternative; they do not turn enrollment into a causal control.

-\subsection*{M7. Inconsistent confidence intervals}
+\subsection*{3.7. Endpoints and survey-production changes}

-\emph{Response.} Corrected. The only primary frozen interval is the 9,999-draw \$[-.217075, -.045073]\$. We trace the earlier variants to different finite 999-multiplier sets; the minimum-count row additionally changed support. Pair-specific member intervals are labeled and are not presented as primary. A consistency audit checks the canonical coefficient and interval. See Section 4 and Appendix D.

+\emph{Response: accepted on magnitudes; the proposed change of primary endpoint was not adopted.} The endpoint discussion now compares movements rather than sorting nearly identical estimates by whether one interval endpoint crosses zero. Relative to July 2026, the pooled movements through December 2024, September 2025, and December 2025 are 0.0211, 0.0225, and 0.0212 log point. Their magnitudes are essentially the same even though their interval endpoints differ.

-\subsection*{M8. Relationship to Brynjolfsson, Chandar, and Chen}
+We retain the full available window as the canonical target because December 2024 would be selected after observing that it preserves a negative detected coefficient. December 2024 is reported prominently as the pre-2025-production-break benchmark: its pooled coefficient is \$-\$.0.1110\$ with interval \$[-.0.2008, -.0.0213]\$, and its movement from the full-window estimate has interval \$[-.0.0073, 0.0495]\$. The paper separately reports official-weight and respondent-equivalent era comparisons, exclusions around the October 2025 gap, and the January 2025 and January 2026 control changes. It does not claim that aggregate control factors can reconstruct unavailable age-by-occupation counterfactual weights.

-\emph{Response.} We implement the reproducible public component of BCC: GPT-4 beta with top two employment-weighted quintiles against the bottom three. In CPS this gives \$-\$.0728\$ with interval \$[-.1240, -.0216]\$, compared with the corrected YAX Q5--Q1 coefficient \$-\$.1346\$. Applying that grouping across eight constructions yields native-support estimates from \$-\$.0817\$ to \$-\$.0096\$. On literal common support, beta differs detectably from OECD under this grouping but not from the other six alternatives.

+\subsection*{3.8. Crosswalk allocation uncertainty}

-We label this a bridge, not a replication. The ADP outcome, employer panel, job-title mapping, firm controls, hiring, and separation margins are proprietary or outside CPS. See Sections 2 and 6 and Appendix E.

+\emph{Response: accepted as a finite sensitivity, not as a claimed bound.} The current-contract analysis reallocates the 14.86 percent of supported early stock that passes through 31 one-to-many source occupations. A seven-point jointly feasible odds-tilt grid preserves source--age--month mass and official structural zeros, and recomputes exposure groups after each allocation. Pooled coefficients range from \$-\$.0.1446\$ to \$-\$.0.1163\$, with all six paired intervals relative to no tilt containing zero. The family-month coefficient is less stable: at \$K=0.25\$ it moves from \$-\$.0.0217\$ to \$0.0439\$, a paired change of \$0.0656\$ with interval \$[0.0031, 0.1280]\$. A wider 25-point feasible grid puts the pooled coefficient between \$-\$.0.1599\$ and \$-\$.0.0702\$.

-\subsection*{M9. January 2023 and adoption}
+Because the public files do not reveal true age-specific route probabilities, these are declared feasible-grid sensitivities rather than sharp bounds or an identified set. The superseded historical-contract tilt is no longer substituted for the rebuilt-contract question.

-\emph{Response.} We supplement the binary coefficient with the existing event-time profile, balanced pre-AI pseudo-breaks, and joint post-era coefficients. We do not introduce a new adoption series in this revision. A national adoption series interacted with fixed exposure would largely reparameterize exposure-by-time dynamics already shown. BTOS would require an occupation--industry ecological bridge and still would not identify exogenous adoption.

+\subsection*{3.9. Link attrition and flow selection}

-This is a principled scope decision, not a claim that adoption is irrelevant. The binary term is now called a chronology benchmark, not a causal shock. A separate adoption design remains future work.

See Section 7 and Appendix F.

\emph{Response: accepted within observable limits.} The revision reports link retention by age, quintile, period, and horizon, plus observable differences between retained and eligible-unlinked origins. It adds origin-observable poststratification and missing-outcome accounting intervals. For example, the descriptive twelve-month exit contrast moves from 0.0166 under official link weights to 0.0210 after poststratification, but the maximum untrimmed factor is 102.7 and the procedure requires conditional missing at random. Without that assumption, the jointly feasible linear accounting interval is $[-1.570, 1.575]$.

\subsection*{M10. F/G reparameterization}

+These results show why clustering cannot cure selection and why the direction of unobserved attrition bias is not identified. We do not assert attenuation toward zero or use the poststratified result as a correction. Flow estimates remain low-resolution descriptions on their own linked risk sets.

\emph{Response.} Accepted. F/G is removed from the main results and retained in Appendix I as a coordinate transformation. The main text leads with the representative AIOE-plus-beta and direct $\$D, \S models. The task-family centroid's effective raw-unit weights are reported as $\$4.9057D + 1.8104\S , so the embedded $\$S$ -to- $\$D$ weight is .3690. We no longer treat a confidence interval crossing zero after component deletion as evidence of a detected coefficient change.

\subsection*{3.10. Comparison with BCC public evidence}

\subsection*{M11. Mobility, entry, and uncertainty}

\emph{Response: modified because exact membership is unavailable.} We added a public benchmark section before the paper's main conditional models. It distinguishes BCC's ADP stock-growth shortfall from CPS regression coefficients, and distinguishes equal-occupation public-dashboard grouping from employment-stock weighting. Exact dashboard memberships and tie handling are not released, so our top-two/bottom-three rule is explicitly an approximation, not a replication.

\emph{Response.} The employed-to-employed switch analysis is removed from the paper's main contribution. The appendix now separates the entrant-destination allocation among people observed entering employment from any employment-finding probability. Its beta $Q5 \rightarrow Q1$ coefficient is $\$-.0898$ with interval $[-.2787, .1011]$ and $\$p=.383$. Immediate reversals are not treated as a validated coding-error rate, so no purported attenuation correction is applied; the appendix also reconciles the prior 9.865-percent statistic with the present sample's 10.336-percent raw conditional rate.

+In the CPS bridge, the approximation gives $\$-0.0720$ pooled and $\$-0.0168$ with family-month conditioning; the paired movement is 0.0553 with interval $[-0.0013, 0.1118]$. The less saturated family-post movement is narrowly detected, while the stronger family-month movement is not. A separate all-employed public benchmark and an annual ACS reconstruction compare the published horizons and populations. The ACS BCC analogues differ from the published values, but without exhaustive public membership the gap cannot be allocated among sample, grouping, and estimator. The revised paper therefore promotes the bridge while preserving this identification limit.

-The revised mobility audit reports a 20.13-percent family-balanced pairwise disagreement rate and proves that beta adds no new zero-threshold conflict when its task-family endpoints agree. On identical 98.305-percent represented support, realized conflict is 53.2819 percent and the benchmark is 52.3227 percent, a 0.9592-point gap. The declared household-cluster SE is 0.1979 point for the realized rate and 0.0776 point for the gap, versus a 0.00172-point Monte Carlo SE for the benchmark mean. The all-support accounting bound for the gap is $[0.1521, 1.8469]$ points. A second valid repair rule changes the benchmark by only $\$-0.00157$ point, but neither rule is a canonical uniform draw over feasible derangements. The main text draws no hiring conclusion from switches. See Appendix G/H.

\subsection*{3.11. ACS and adoption data}

\subsection*{M12. Generated-covariate validation}

\emph{Response: accepted through two scoped extensions.} The annual ACS extension uses the 2017--2019 and 2021--2024 one-year PUMS files and all 80 replicate weights per endpoint year; no nonexistent standard 2020 one-year file is imputed. It materially narrows person-sampling uncertainty, yet the occupation- and family-shock intervals for every family-year variant include zero and the direct tail support does not expand. This distinguishes person-sampling resolution from occupational comparison support.

\emph{Response.} Appendix A now specifies a validation design: occupation--task units stratified by family and architecture disagreement; separate labels for capability, realized time savings with and without software, adoption, and task importance; blinded independent experts plus audited use logs where feasible; repeated labels; held-out occupations; and versioned task/model/mapping records. We provide a Fisher--S2s power formula and an illustrative independent-unit calculation while emphasizing that clustering, rater noise, sampling weights, and target correction determine the actual sample size. We do not invent one universal number.

+For realized use, we merge the fixed exposure score to the public Real-Time Population Survey detailed-occupation workbook. The exact-code match retains 121 nonsuppressed occupations, with an unweighted exposure--adoption correlation of 0.504 and a $Q5 \rightarrow Q1$ mean-adoption difference of 0.267. The workbook pools four 2025--26 waves and lacks detailed-cell standard errors and adoption timing, so we report no causal interaction or p-values. We did not force the industry--state BTOS measure onto occupation-level CPS outcomes through an ecological bridge. The extension establishes descriptive alignment between exposure and later use, not treatment assignment.

\section*{Response to Referee 2}

\section*{Response to Referee A: minor comments}

\subsection*{M1. Framework separating measurement from the question}

\begin{enumerate}[leftmargin=1.8em]

+item The correct count is 6,188,956 route-expanded descendants. The erroneous 36,188,956 count is removed; 5,322,047 eligible source records and the distinction between fractional descendants and persons are stated beside it.

+item "Chapter" has been replaced by "paper."

+item The industry comparison is labeled as an industry-cell baseline with a different objective and is separated from occupation-level controls.

+item The family-path figure now uses a symmetric-log vertical scale, keeps literal zero-stock months visible, and states the linear and logarithmic regions in its note.

+item Every computer-use correlation and estimate names its support. The matched four-model main comparison uses 455 occupations; the broader characteristic table separately identifies its 347-occupation common support.

+item Principal MDE quantities are consolidated in one appendix precision table with their scales and target types. Main-text repetitions are limited to cases where resolution is itself the result.

+item The introduction now states first what the paper establishes---the support geometry and the comparisons it localizes---and then states the causal and equivalence limits. Repeated implementation caveats have been moved to the appendix and archive.

+item The literature discussion now engages Bick, Blandin, and Deming; Hampole, Papanikolaou, Schmidt, and Seegmiller; Deming and Noray; and Autor and Thompson, and explains how adoption, firm investment, technological obsolescence, and expertise bear on the descriptive target.

+item The abstract defines the log conditional-mean young-to-older employment-stock ratio before reporting coefficient values and distinguishes it from an employment probability.

+item The conclusion now signposts Online Appendix I.3's consolidated data, design, and interpretation limits.

\end{enumerate}

\emph{Response.} Accepted and used to reorganize the paper. Main Table 1 distinguishes implementation repair, same-construct measurement, construct change, population/support change, estimand change, representation change, and pure reparameterization. The paper states positive-affine rank/bin invariance, piecewise-constant bin membership along $\$D + \lambda \S , and full-rank coordinate invariance. Every empirical row now identifies which layer changed. See Section 2 and Appendix A.

\section*{Response to Referee B: major comments}

\subsection*{M2. Degree of architecture heterogeneity}

\subsection*{1. Benchmark and contribution}

\emph{Response.} Accepted. The revision separates point-estimate sign, a dependent-set sign test, and paired coefficient differences. Main Table 3 contains direct paired differences, intervals, and MDEs; all $Q5 \rightarrow Q1$ intervals contain zero. Native-support and BCC-grouping estimates are shown separately rather than pooled into one common-sample experiment. We no longer infer heterogeneity from individual significance patterns.

\emph{Response: accepted.} The revised contribution is not that a literal merge error proves a published finding wrong. The introduction explicitly declines that attribution without another study's implementation evidence. Instead, the contribution is that a national occupational AI-exposure comparison can contain much less direct within-family information than its nominal occupation count suggests. The paper defines a public benchmark and separately records differences in population, outcome, grouping, weighting, endpoint, nuisance structure, and available employer controls. The title and interpretation no longer claim to adjudicate "AI versus composition." They describe support and comparisons and preserve the observational equivalence between family-level AI effects and family-level confounding.

\subsection*{M3. Construct validity of external architectures}

\subsection*{2. From coefficient attenuation to explicit accounting}

\emph{Response.} The architecture matrix now records technology scope, primitive, label generation, publication and occupational-information vintages, horizon where verified, mapping, and the downstream hypothesis. OECD is described as a broader capability/demand and horizon construct, not a noisy LLM reliability measure. Webb AI and Webb software are distinguished. The external-admission rule is preserved as an operational screen, while the distinct-cut condition is explicitly representation-dependent rather than construct validity. The paper also reports unconditioned and Webb-conditioned BCC beta results, which are nearly identical for that binary grouping without making a general computerization claim. See Section 2/6 and Appendix A/E.

\emph{Response: accepted.} The revision distinguishes three objects: employment reallocations across families, within-family young-to-older ratio changes, and movements in a fitted coefficient after family paths are added. Exact log-stock accounting gives a young $Q5 \rightarrow Q1$ change of $\$-0.1098$, an older change of $\$0.0483$, and a relative change of $\$-0.1581$. A symmetric Shapley decomposition of the same family stocks assigns $\$-0.1293$ to within-family ratio changes and $\$-0.0288$ to changes in older-family weights. It audits 36 structurally absent family-tail-period cells and obtains a zero boundary term. This accounting is deliberately kept distinct from the nonlinear regression coefficient.

\subsection*{M4. Reference group and broad occupational variation}

+We also added the requested matched-support four-model comparison. On the same 455 occupations and common draws, the baseline, computer-use, family-month, and combined coefficients are $\$-0.0957$, $\$-0.1968$, $\$0.0354$, and $\$-0.0467$. Thus family and computer-use conditioning are different projections whose effects interact; no one residual is labeled an AI component.

\emph{Response.} This is now the substantive center of the paper. We replace "reference dependence" with exact descriptions of changed contrasts, retain the $Q1/Q5$ stock paths, add profile/equality/monotonicity tests, and estimate both SOC2-by-post and SOC2-by-month conditioned models with information and support diagnostics. The regression results, not the assumption-dependent permutation percentile, determine the framing. See Section 5 and Appendix C/E.

\subsection*{3. Direct overlap and common-coefficient restrictions}

\subsection*{M5. Corrected pipeline and bridge uncertainty}

\emph{Response: accepted.} The family support matrix is now in the main paper, alongside direct-tail coverage, information retention, effective occupation count, and concentration. The text explains how intermediate quintiles and a common profile connect families that do not span $Q1$ and $Q5$, and notes that computer and mathematical occupations contribute only a small share of the conditioned target's information.

\emph{Response.} The repaired 113-month coefficient is now the substantive baseline; the frozen result is a chronology benchmark. We add a fail-closed taxonomy-compatibility gate, one-to-one-target estimates, split-source stock and information shares, age-allocation accounting ranges, and predeclared tilt scenarios. We do not claim the common-proportion bridge resolves age-specific allocation.

+The appendix now retains all 89 supported family-specific pairwise edge functionals. Their joint covariance has rank 50, so we do not report an invalid 89-degree-of-freedom Wald test or select a

discovery list after inspecting pointwise results; the associated rank-50 coefficient system is reported with rank-aware inference. Relaxing Webb availability expands support from 468 to 490 occupations and direct tails from 29 to 62 occupations and from 5.03 to 12.61 percent of stock. SOC43 becomes a fifth spanning family, but overlap remains confined to five of 23 families. The pooled coefficient changes little. This shows that Webb availability contributes to, but does not create, the support limitation.

-The 490/495 discrepancy is fully reconciled. The raw balanced universe is 539. The 490 support requires beta; the 495 support requires repaired AIOE plus Webb. They overlap in 466, with 24 beta-only and 29 AIOE-plus-Webb-only occupations. See Section 3, the sample-flow table, and Appendix B.

+\\subsection*{4. Static and dynamic reconciliation}

-\\subsection*{M6. Source of randomness and bootstrap mechanics}

+\\emph{Response: accepted.} We now report three targets with fixed weights and covariance-preserving comparisons. The static grouped-binomial target $\$\\$$ is $\$-0.1321\$$ pooled and $\$-0.0217\$$ conditioned. The post functional relative to the short 2022Q4 reference, $\$\\$, is $\$-0.1199\$$ and $\$-0.2074\$$. The reference-invariant post-minus-pre functional $\$D\$$ is $\$-0.1314\$$ and $\$-0.0217\$$. The paired $\$D-\$\\$$ differences are 0.0007 ($\$p=.454\$$) and less than 0.0001 ($\$p=.995\$$), so the design does not detect differences between the static and reference-invariant dynamic targets. By contrast, family conditioning changes $\$P-\$\\$$ by $\$-0.1980\$$ ($\$p=.021\$$). This quantifies how the short reference period, rather than a contradiction in the post-minus-pre comparison, produces the apparent reversal. HonestDID is explicitly limited to $\$P\$$ and is not used to make claims about $\$\\$$ or $\$D\$$.$

-\\emph{Response.} Section 4 now states the target as a stochastic occupation-shock parameter conditional on realized CPS cells and constructed treatment. It lists household survey design, label generation, and mapping allocation as uncovered uncertainty. The nuisance-adjusted score, fitted information, cluster aggregation, finite-cluster factor, Rademacher multipliers, studentizer, critical value, and common-draw paired construction are described explicitly. The failed pseudo-outcome refit is correctly treated as evidence that that construction was inappropriate, neither validation nor invalidation of the score bootstrap. Draws increase from 999 to 9,999 for central revised inference; IUT and simultaneous-band claims remain distinct.

+\\subsection*{5. Inference, numerical existence, and stable-taxonomy checks}

-\\subsection*{M7. Primitive units and F/G}

+\\emph{Response: accepted.} The finite-sample audit now evaluates occupation-Rademacher, occupation-Webb, family-Rademacher, family-Webb, and cross-fit full-refit oracle procedures for the pooled, family-conditioned, and paired targets under empirically motivated and adverse designs. It reports target-specific coverage rather than treating one stress number as the CPS sampling law. Household resampling remains explicitly a companion for repeated sampled units and released weights, not independent validation of the saturated family-month parameter. Elapsed-calendar HAC is also kept as a distinct stochastic target.

-\\emph{Response.} Accepted. Direct $\$D,\$\\$$ and AIOE-plus-beta models are presented before the appendix rotation. Effective raw-unit centroid weights are reported, and leave-one-component changes are described as changes in implicit weights. F/G and A/E are explicitly the same fitted column space. See Section 6 and Appendix I.

+The appendix now states existence results directly. All eleven declared grouped-binomial models pass rank, recession-direction, independent trust-path versus zero-start Newton/IRLS agreement, fitted-mean, gradient, Hessian, and two-sided profile checks at unchanged thresholds. Post-2020 fits profile one boundary nuisance group spanning 77 positive-total rows; occupation-seasonality fits profile 110 boundary groups spanning 770 rows. The groups are handled on the extended-likelihood face rather than deleted, no direction can move the reported treatment target to infinity, and every focal target is finite. L-BFGS-B remains a disclosed failed-score diagnostic, not the corroborating solver.

-\\subsection*{M8. Mobility scale and benchmark}

+The coding-stable post-2020 pooled and family-month estimates are $\$-0.1181\$$ and $\$-0.0304\$$; their paired movements from the full-window fits have intervals $\$[-0.0288,0.0569]\$$ and $\$[-0.1053,0.0878]\$$. Occupation-specific seasonality fits also pass and move the targets by at most 0.0011. Earlier failed attempts remain in the registry but are no longer the only evidence on these questions.

-\\emph{Response.} Mobility is demoted and its claims narrowed. Thresholds are decision rules in standardized or rank units, not welfare magnitudes. The family-balanced matrix prevents multiplicity of near-duplicate measures from defining the result. The exact identity $\$\\Delta I(\\lambda) = \\Delta D + \\lambda \\Delta \\Delta$ shows that beta cannot introduce a new zero-threshold sign conflict when alpha and broad agree. Benchmark and realized conflict are recomputed on identical represented support; omitted support is bounded. Household-linked sampling uncertainty is separated from Monte Carlo error, and at least two valid no-self repair distributions are compared. See Appendix H.

+\\subsection*{6. Age comparison and links to flows}

-\\subsection*{M9. Verifiable chronology without revision-history prose}

+\\emph{Response: accepted, with remaining data limits explicit.} The paper now compares ages 22--25 with 26--30, 31--40, 41--50, 51--65, and the original 26--65 group on fixed occupational support. Pooled coefficients remain negative, and all family-month coefficient intervals contain zero. Holding the older group's exact-age composition fixed moves the pooled target by only 0.0008 with interval $\$[-0.0011,0.0027]\$$. These are descriptive denominator checks; no older band is called untreated.

-\\emph{Response.} Revision-history phrases are removed from the main scientific narrative. The appendix preserves the dated protocol, protected-outcome definition, frozen and confirmatory tags, information set, hypothesis-to-test mapping, scripts, samples, seeds, and output hashes. We do not imply that repository hashes constitute external prospective registration. A result manifest links each headline number to its script and machine-readable output. See Appendix L and the reproducibility package.

+The flow appendix reports weighted baseline transitions, retention by age, quintile, period, and horizon, observable differences between linked and eligible-unlinked origins, poststratification, and missing-outcome accounting intervals. Entry destinations are also placed on one common nonemployment denominator, including continued nonemployment and destinations outside the support, while the conditional Q1--Q5 allocation remains separately labeled. All primary flow intervals include zero, so the paper shortens their interpretation and does not infer an absent mechanism. A complete stock--flow calibration remains infeasible because aging, all job-to-job moves, unsupported destinations, denominator flows, and employer matches are not jointly observed on one risk set.

-\\section*{Claims retired}

+The short weekly-earnings series through March 2023 is retained only as an ancillary, employment-conditioned result. A full-window earnings analysis is not claimed: \\texttt{EARNWEEK2} is absent from the authorized extracts.

-The revised paper does not claim a causal AI employment effect, six independent validations, architecture equivalence, a monotone exposure response, a food-service mechanism, a CPS replication of proprietary ADP hiring results, survey-design-complete inference, or a coding-error correction inferred from occupation reversals. It also does not treat an invalid taxonomy merge as a defensible robustness specification.

+\\section*{Response to Referee B: specific presentation and reproducibility comments}

-\\section*{Next revision pass}

+\\begin{enumerate}[\\leftmargin=1.8em]

+\\item \\textbf{Dependent variable.} Main Table 2 now says that the grouped-binomial criterion is estimated on the two age-specific weighted stocks and that coefficients parameterize log ratios of conditional means. No observed log ratio is constructed, and one-sided zero cells remain.

+\\item \\textbf{Exposure-group reconstruction.} Characteristic tables state whether groups are rebuilt on a support or fixed from the primary universe. The central computer/family comparison uses fixed assignments, one 455-occupation support, one calendar, and common draws. Maximal-support and 347-occupation common-support rows are separated, with their different baselines explained in the notes.

+\\item \\textbf{Qualifications and implementation history.} The main paper is reorganized around support, accounting, and a smaller set of economic targets. Repeated caveats, installation history, superseded numerical results, hashes, and failure receipts are moved to the online appendix or archive. Nondetection-versus-equivalence and association-versus-causation rules are stated once in the design section and applied consistently.

+\\item \\textbf{Reproducibility and submission fields.} Central exhibits are tied to named treatment, population, estimator, and inference objects, with a machine-readable results ledger and aggregate-only exhibit checks. This is an auditable package, not a claim that the referees independently replicated it. Affiliation, email, acknowledgments, funding, conflicts, and journal disclosures remain visibly marked \\texttt{\\{AUTHOR TO COMPLETE\\}}; none has been invented.

+\\end{enumerate}

-The two reports' secondary comments are preserved in the machine-readable response matrix. The next pass will focus on copyediting, citation alignment, remaining table conventions, and author-supplied affiliation/disclosure fields. None of those items will be marked complete merely because the major empirical revision is complete.

+\\section*{Claims retired and unresolved inputs}

+

+The paper no longer claims a causal AI effect, a causal composition share, economic equivalence, a monotone dose response, a food-service mechanism, a validated latent exposure measure, complete survey-design inference, or replication of proprietary employer hiring and separation results. It does not infer that a literal vintage merge was used by another study without implementation evidence.

+

+Two supplied-data limitations remain. Exact BOC public-dashboard occupation membership and tie handling are unavailable, so the CPS and ACS BOC exercises remain labeled approximations. The authorized full-window extracts omit \\texttt{EARNWEEK2}, so the legacy earnings sample ending in March 2023 is not presented as a substitute for a full-window earnings analysis. Author identity and journal disclosure fields remain marked for completion by the author.

\\end{document}

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+Evidence-led framing decision, completed work, and unresolved limits\\par
+September 2026
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-\\section*{Decision}

+\\section*{Framing decision}

-The paper should be a \\textbf{composition-centered measurement paper}, titled \\emph{AI Exposure or Occupational Composition? Constructed Measures and Early-Career Employment}. The strongest defensible substantive result is that the public-CPS young-relative association is largely a comparison across broad occupation families. The strongest methodological result is that implementation repair , same-construct measurement, construct choice, support, estimand, representation, and coordinate changes have to be audited separately.

+The defensible paper is a \\textbf{comparison-centered empirical audit}, not an AI-displacement design and not a generic ‘‘measurement choices matter’’ essay. Its question is: what occupational comparisons support the negative association between GPT-4 task exposure and young-relative employment stock in the public CPS?

-The earlier thesis---that robustness is generally statement-specific---was true but too generic. The stronger revised thesis is empirical and falsifiable: after correcting the data, a $-\$.1346\%$ Q5--Q1 association falls to about $-\$.032\%$ when broad-family evolution is absorbed, while the design is too imprecise to resolve most paired architecture differences. The chapter is not a causal AI-displacement paper.

+The answer is specific. The reconstructed pooled Q5--Q1 comparison is $-\$.0.132\%$ log point, with 95-percent occupation wild-score interval $[-\$0.2206, -\$0.0437]\%$. Giving every broad occupation family its own young-relative calendar-month path yields $-\$.0.0217\%$, with interval $[-\$0.1607, 0.1173]\%$. Their paired difference is 0.1104 , with interval $[0.0107, 0.2102]\%$. The movement is detected under the declared occupation-shock procedure, but it compares different conditional estimands and is not a causal share.

-\\section*{Evidence that forced the decision}

+Direct support is the binding economic qualification and the paper’s organizing contribution. Only four of 22 broad families contain both exposure tails; their 29 tail occupations account for 5.03 percent of preperiod stock, and the direct-tail comparison has an effective information count of 6.2 occupations. Other families connect the coefficient through intermediate quintiles and common-coefficient restrictions. The article should therefore say that the pooled pattern is principally a comparison across broad occupational structure, not that ‘‘composition explains’’ a known fraction of an AI effect.

-\\begin{enumerate}[\\leftmargin=1.6em]

- \\item \\textbf{Broad-family condition.} SOC2-by-post and SOC2-by-month specifications yield $-\$.0315\%$ and $-\$.0317\%$, versus $-\$.1346\%$ at baseline. Paired movements of approximately .103 have intervals excluding zero.
- \\item \\textbf{Support cost.} Only four of 22 SOC2 families contain both Q1 and Q5, and conditioned target information is 30 percent of baseline. The residual detailed-exposure coefficient is weakly measured and is not a causal remainder.
- \\item \\textbf{Food-service mechanism rejected.} Removing Q1 food-service occupations attenuates the coefficient only 8.8 percent. Broader in-person-service exclusions make it slightly more negative.
- \\item \\textbf{Architecture heterogeneity unresolved.} The primary MDEs, $\{.80\}$ is .1244. All seven paired beta-minus-alternative Q5--Q1 intervals include zero; paired MDEs reach .1689. Point-estimate differences cannot carry claims of detected heterogeneity or equivalence.
- \\item \\textbf{Direct public BOC bridge.} Their top-two-versus-bottom-three beta grouping produces a smaller $-\$.0728\%$ CPS coefficient. This anchors the audit to the motivating classification while preserving the boundary around proprietary hiring evidence.
- \\item \\textbf{Construction dependence.} Two components explain 96.11 percent of the six core score implementations. The prior six-measure sign exercise is a dependent-set statement, not six independent validations.
- \\item \\textbf{Data corrections do not explain the negative association.} March repair moves $-\$.1311\%$ to $-\$.1346\%$. One-to-one-target and stable-taxonomy estimates remain negative. Population-control and respondent-equivalent diagnostics do not reveal a mechanical January-2025 explanation.
- \\item \\textbf{Mobility is diagnostic rather than mechanistic.} A family-balanced pairwise statistic is 20.13 percent, while 53.28 percent is a six-way ‘‘any conflict’’ event. The aligned-support realized-minus-rematching gap is 0.959 point with a declared household-cluster sampling SE of 0.0776 point, but the algorithm-defined benchmark and switch-only sample do not identify hiring or coding error.

+\\section*{Evidence that determines the interpretation}

+\\begin{enumerate}[\\leftmargin=1.8em]

- +\\item \\textbf{Corrected baseline.} Five wrongly selected March ASEC samples are replaced by Basic Monthly files; October 2025 is absent; December 2022 is the transition month. All 71-month preperiod weights, normalizations, cuts, ties, and memberships are rebuilt. Nine occupations change quintile, but reconstructed-minus-historical treatment moves the coefficient only 0.00245, with interval spanning zero.
- +\\item \\textbf{Support geometry, not attenuation alone, carries the contribution.} The pooled-to-SOC2-month movement is 0.1104 and statistically distinguishable from zero on common draws, while the conditioned coefficient and direct-tail benchmark are imprecise. A fully interacted system yields 89 supported family-specific edge functionals, but their covariance has rank 50; it is not converted into an invalid 89-degree-of-freedom test or a selected discovery list. Relaxing Webb availability expands the beta-valid support from 468 to 490 occupations and direct-tail stock from 5.03 to 12.61 percent, yet only five of 23 families then span both tails. Thin direct overlap is not solely a Webb-support artifact.
- +\\item \\textbf{Exact stock accounting separates economic objects.} Equal-month accounting gives a Q5--Q1 young-stock change of $-\$.0.1098\%$, an older-stock change of $\$.0.0483\%$, and hence a young-relative change of $-\$.0.1581\%$. An exact log-Shapley accounting assigns $-\$.0.1293\%$ to within-family ratio changes and $-\$.0.0288\%$ to changes in older-family weights; audited boundary mass is zero. These identities describe aggregate stocks. They neither decompose the nonlinear $-\$.0.132\%$ coefficient nor identify causal shares, and no separate young/older regressions are claimed.
- +\\item \\textbf{Computer use and family paths are distinct conditioning dimensions.} On the same 455-occupation support with fixed exposure assignments, the pooled baseline, computer-only, family-month-only, and combined coefficients are $-\$.0.0957\%$, $-\$.0.1966\%$, $\$.0.0353\%$, and $-\$.0.0466\%$. The computer-use coefficients are positive in both augmented models, while adding computer use sharply reduces retained target information. Thus computer use acts as a suppressor in this projection; it does not purify the remaining coefficient as AI. The repaired pandemic-shortfall exercise uses bounded predictions, a strict 363-occupation population, and linked-household regeneration; its conditioning-movement intervals include zero. These are correlated diagnostics, not causal horse races.
- +\\item \\textbf{Industry, education, cohort, and enrollment checks clarify the target.} Conditioning occupation--industry microcells on 13 broad-industry post shifts moves $-\$.0.1376\%$ to $-\$.0.0987\%$; the paired movement is detected while the residual remains negative. BA and non-BA coefficients are not statistically distinguishable. Narrower older comparison bands produce different point estimates, but fixing the older group’s exact-age composition changes the pooled coefficient by only 0.0008. Restricting the young numerator to nonenrolled workers moves it by 0.0082, with an interval containing zero. Nationally, enrollment among ages 22--25 falls from 0.2336 to 0.2211 while the BA-or-higher share rises from 0.2941 to 0.3121. These profiles document composition and selection; they do not identify cohort, experience, or enrollment mechanisms.
- +\\item \\textbf{Preperiod conclusions depend on both the null and the window.} Against the short 2022Q4 reference, the full 23-coefficient preperiod null rejects in both structures. The reference-invariant full-window equality test rejects for the pooled path ($-\$.0.0121\%$) but not the family path ($-\$.0.0641\%$). Excluding 2020Q2--2020Q4 removes the pooled five-percent rejection under both nulls ($-\$.0077\%$ and $-\$.0676\%$) for the family path the short-reference null rejects but within-window equality does not ($-\$.0079\%$ and $-\$.1048\%$). A linear-slope nondetection cannot validate parallel evolution. January 2023 remains descriptive chronology, not causal timing.
- +\\item \\textbf{Static and dynamic targets are reconciled rather than conflated.} The static coefficients $\$\\$$ are $-\$.0.132\%$ pooled and $-\$.0.0217\%$ conditioned. The short-reference post functionals $\$P\\$$ are $-\$.0.1199\%$ and $-\$.0.2074\%$, while reference-invariant post-minus-pre functionals $\$D\\$$ are $-\$.0.1314\%$ and $-\$.0.0217\%$. Paired $\$D\\$$ differences are not detected. Within the family model, $\$P\\$$ $-\$.0.1858\%$ has $-\$.0.074\%$; what is detected is the change in $\$P\\$$ across conditioning structures ($-\$.0.1980\%$, $-\$.0.021\%$), not the corresponding change in $\$D\\$$ ($-\$.0.693\%$). HonestDiD applies to $\$P\\$$, whose conventional intervals include zero, not to $\$S\\$$ or $\$D\\$$.
- +\\item \\textbf{Endpoint extensions do not identify intensification.} Relative to July 2026, the pooled movements at the December 2024, September 2025, and December 2025 endpoints are approximately 0.021--0.023 log point and similar in magnitude; the December 2024 paired interval includes zero. The full window remains canonical and December 2024 is a prominent pre-2025 benchmark, not an outcome-selected replacement. Family-conditioned endpoint movements remain near zero.
- +\\item \\textbf{Flows do not resolve the mechanism.} Corrected adjacent and twelve-month CPS links give intervals containing zero for employment exit, reported occupation change, and exposure allocation conditional on entry. Annual young-worker retention is substantially lower than older-worker retention, and observable reweighting does not resolve missing-outcome selection. Hours and conditional earnings are imprecise. None of these targets is an employer hiring rate or a mechanical decomposition of the stock coefficient.
- +\\item \\textbf{Public benchmarks and ACS sharpen comparability without creating a replication.} An equal-occupation approximation to BOC’s top-two-versus-bottom-three grouping gives $-\$.0.0720\%$ in CPS and $-\$.0.0168\%$ after family-month conditioning; their paired movement is not detected at five percent. Annual ACS reconstructions provide much tighter person-sampling intervals, but family-year results remain sensitive to calendar construction and broader shock uncertainty. Exact dashboard membership, firm controls, and proprietary worker--firm flows cannot be reproduced.
- +\\item \\textbf{The estimation objective is separately audited.} Giving each occupation total weight one, while retaining final-weight shares within occupation--month cells, changes the pooled coefficient from $-\$.0.1321\%$ to $-\$.0.0833\%$. The paired difference is $\$.0.0488\%$ ($[-\$0.0604, 0.1581]\%$), so this design does not detect a difference and does not establish equivalence.
- +\\item \\textbf{Architecture and mapping evidence are paired on the current contract.} Five non- $\\lambda$ definitions give pooled native-support coefficients from $-\$.0.0991\%$ to $-\$.0.0145\%$, and every alternative-minus-beta paired occupation interval contains zero. On common AIOE support, some family-conditioned implementation contrasts exclude zero. A declared bridge-allocation grid leaves the pooled coefficient negative but moves some family-conditioned comparisons across zero. The paper therefore treats construct choice and mapping implementation as real sensitivities without calling different constructs replications or finite grids identified sets.
- +\\item \\textbf{Later reported use is aligned but not identifying.} A public RPS detailed-occupation workbook retains 121 nonsuppressed occupations. The unweighted correlation of beta exposure with pooled 2025--26 adoption is 0.504 and the Q5-minus-Q1 adoption difference is 0.267, but Q4 exceeds Q5 and only 26.2 percent of exposure variation remains after major-family means. The workbook provides neither detailed-cell sampling uncertainty nor adoption timing, so this is descriptive construct validation rather than a mechanism test.
- +\\item \\textbf{Mapping matters for population, not just score values.} A literal SOC-vintage mismatch retains 3.33 percent of computer-and-mathematical employment; documented routing restores 97.7 percent. Fixed-support score repair changes little, while readmission changes the population. The invalid merge is an implementation failure, not a robustness specification attributed to another paper.
- +\\item \\textbf{Finite numerical targets are certified.} All 11 declared grouped-binomial specifications pass treatment-rank, recession-direction, independent same-objective solver, fitted-mean, normalized-gradient, fitted-Hessian, and two-sided profile checks at unchanged thresholds. Boundary nuisance groups in the post-2020 and occupation-seasonality models are profiled on the extended-likelihood face rather than silently deleted. No audited recession direction moves a reported target, and every focal target is finite. This resolves numerical existence and separation; it does not certify the sampling model.
- +\\item \\textbf{Inference remains conditional.} Occupation wild scores, 22-family dependence weights, household/sample-unit full refits, elapsed-calendar HAC, and the completed 11-design finite-sample audit answer different stochastic questions. In the empirical observed-effect design, neither occupation nor family clustering attains nominal coverage for every target; the clean zero-target family-month rejection rates are 6.1 percent, 8.9 percent, and 5.6 percent for occupation weights, family weights, and the cross-fit oracle. Public Basic Monthly files do not permit complete CPS design-based inference, and these variances cannot be added without a valid decomposition.

-\\end{enumerate}

-\\section*{What the revised paper may say}

+\\section*{What the paper may and may not claim}

+The paper may say that public CPS data contain a sizeable negative young-relative employment-stock association under GPT-4 beta exposure; that its national Q5--Q1 interpretation relies heavily on between-family structure and common restrictions because direct within-family tail support is thin; that broad-family conditioning produces a detected coefficient movement; that exact stock accounting attributes the descriptive aggregate change to both the young numerator and older denominator; that a matched computer-use control strengthens rather than attenuates the coefficient; that preperiod evidence does not validate a causal January-2023 break; and that available flow and adoption evidence does not identify a mechanism.

+It may not say that AI caused an individual employment decline, that occupational composition causally explains a percentage of an AI effect, that the residual coefficient is an AI effect, that computerization, reopening, enrollment, or cohort supply is the cause, that nondetection means economic equivalence, that the numerical audit supplies complete CPS-design inference, or that CPS reproduces ADP employer hiring and separation results.

-It may say that public CPS data reproduce a negative young-relative association under beta exposure and the public BOC grouping; that broad-family conditioning absorbs most of the point-estimate magnitude; that the movement is detectable but changes the estimand; that food-service exclusions do not explain the result; that mapping errors can remove nearly all computer/math coverage; and that available precision does not resolve most architecture differences.

+\\section*{Current revision state}

-It may not say that AI caused employment loss, that 76 percent is a causal share, that the remaining $-\$.032\%$ is an AI effect, that food-service recovery is the mechanism, that intervals containing

zero establish equivalence, that six measures independently validate the finding, or that CPS reproduces ADP hiring and separation evidence.

+The scientific working tree contains the rewritten main manuscript and online appendix; fixed-target stock accounting; support, computerization, cohort, flow, ACS, public-benchmark, mapping, architecture, adoption, dynamic-reconciliation, and inference results; machine-readable registries; module-level code and tests; source and environment manifests; covariance and influence outputs; figure and table generators; failure logs; and numerical-consistency audits. Restricted microdata, credentials, and direct identifiers are excluded.

-\section{Major-comment completion}

+The scientific appendix no longer contains the mobility-rematching benchmark or F/G coordinate machinery. Historical artifacts remain available in the replication archive. Revision chronology is described honestly as internal git ordering rather than external prospective registration. Final delivery still requires synchronization of the point-by-point response, diagnosis, source diff, comment matrix, status ledger, and exhibit hashes with the settled manuscript, followed by clean compilation and visual inspection of every PDF.

-Both reports' numbered major comments have been translated into an exact response matrix. The major empirical program now covers composition, profile and monotonicity, stable tails, joint influence, service exclusions, canonical intervals, realized MDEs, quarterly aggregation, time-score covariance, pre-AI pseudo-breaks, population controls, the BCC grouping, architecture dependence, corrected mapping, age-allocation uncertainty, and mobility/rematching uncertainty. The manuscript and appendix have been reorganized around the economic question rather than revision chronology.

+\section{Unresolved items}

-Two requests receive principled non-adoptions. No counterfactual January-2025 age-by-occupation weights are created because official subgroup bridge factors do not exist. No national adoption or industry BTOS series is appended merely to reparameterize event time or introduce an unvalidated occupation--industry bridge. Both limits are explicit.

+The following limits remain after feasible execution:

-\section{Remaining work before submission}

+\begin{itemize}[leftmargin=1.8em]

+\item observed firm adoption, employer identities, employer-match hiring/separation, and detailed occupation-by-time adoption are unavailable; the RPS extension supplies only a later pooled occupation classification;

+\item public files do not supply all confidential survey-design variables or counterfactual age-by-occupation weights for population-control revisions;

+\item exhaustive BCC dashboard occupation memberships and tie rules are unavailable, so the public bridge and ACS exercises remain approximations rather than exact reproductions;

+\item adjacent occupational coding vintages do not validate age-specific route shares;

+\item no external gold-standard label identifies a latent true occupational AI exposure;

+\item several direct-tail, flow, subgroup, and conditional estimates remain imprecise;

+\item the authorized CPS extracts omit \texttt{EARNWEEK2}, so weekly-earnings coverage after March 2023 cannot be extended without a new extract;

+\item affiliation, email, acknowledgments, funding, conflicts, and required journal disclosures require author input and remain \texttt{[AUTHOR TO COMPLETE]}.

+\item submission files still require final cross-document consistency checks, bibliography/cross-reference validation, compilation, page-by-page visual quality assurance, refreshed manifests and ledgers, and a clean source diff; these are delivery tasks, not completed scientific analyses.

+\end{itemize}

-This is the \texttt{major-advice revision}, not the final copyedit. The next pass should reconcile all secondary comments line by line, finalize citations and table formatting, and ask the author to complete affiliation, email, acknowledgments, funding, conflicts, and data-use disclosures. The paper should also receive an independent econometric review of the maintained occupation-shock inference before being sent to a top journal. These open items do not alter the framing decision.

+These are not reasons to add unrestricted specification searches or manufacture unavailable inputs. They define the boundary of the paper's contribution. Before submission, the author should complete the metadata and disclosure fields and obtain econometric judgment on the maintained shock-cluster interpretation; the present public-data audits do not establish a uniquely correct design-based variance.

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+The canonical editable files and generated PDFs accompany this document.
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+Definition & Population & Calendar/model & Estimate [ACS CI] & Family-year [ACS CI] & Family-year occupation CI & Family-year family CI \\\br/>+\midrule
+\multicolumn{7}{l}{\textit{Panel A. 2022--2024 Q5--Q1 growth-factor benchmark, ages 22--25}} \\\br/>+BCC published & All employed & 2022--2024 & \$\\$-0.022\$; [-0.055,0.011] & \$\\$-0.0474\$; [-0.0885,-0.0063] & \$\\$-0.1316,0.0368\$ & \$\\$-0.1242,0.0294\$ \\\br/>+ACS analogue, equal occupation & All employed & 2022--2024 & \$\\$-0.0859\$; [-0.1238,-0.0480] & \$\\$-0.0220\$; [-0.0705,0.0264] & \$\\$-0.1121,0.0680\$ & \$\\$-0.1013,0.0572\$ \\\br/>+Aligned annual CPS analogue & All employed & 2022--2024 & \$\\$-0.0346\$; [-0.1455,0.0763] & \$\\$-0.0691\$; [-0.1193,-0.0188] & \$\\$-0.1554,0.0173\$ & \$\\$-0.1777,0.0395\$ \\\br/>+BCC published & Full-time civilian wage/salary & 2022--2024 & \$\\$-0.019\$; [-0.059,0.020] & \$\\$-0.0467\$; [-0.1057,0.0122] & \$\\$-0.1317,0.0383\$ & \$\\$-0.1480,0.0546\$ \\\br/>+ACS analogue, equal occupation & Full-time civilian wage/salary & 2022--2024 & \$\\$-0.0834\$; [-0.1272,-0.0396] & \$\\$-0.0292\$; [-0.0790,0.0205] & \$\\$-0.1112,0.0527\$ & \$\\$-0.1307,0.0722\$ \\\br/>+Aligned annual CPS analogue & Full-time civilian wage/salary & 2022--2024 & \$\\$-0.0306\$; [-0.1366,0.0755] & \$\\$-0.0291\$; [-0.0875,0.0293] & \$\\$-0.1118,0.0535\$ & \$\\$-0.1294,0.0712\$ \\\br/>+\addlinespace
+\multicolumn{7}{l}{\textit{Panel B. Annual young-relative Q5-by-post regressions, ages 22--25 versus 26--65}} \\\br/>+Fixed primary groups & All employed & Full & \$\\$-0.1089\$; [-0.1310,-0.0869] & \$\\$-0.0474\$; [-0.0885,-0.0063] & \$\\$-0.1316,0.0368\$ & \$\\$-0.1242,0.0294\$ \\\br/>+Fixed primary groups & All employed & Nonreuse & \$\\$-0.0951\$; [-0.1204,-0.0698] & \$\\$-0.0220\$; [-0.0705,0.0264] & \$\\$-0.1121,0.0680\$ & \$\\$-0.1013,0.0572\$ \\\br/>+Fixed primary groups & Full-time civilian wage/salary & Full & \$\\$-0.1196\$; [-0.1456,-0.0936] & \$\\$-0.0691\$; [-0.1193,-0.0188] & \$\\$-0.1554,0.0173\$ & \$\\$-0.1777,0.0395\$ \\\br/>+Fixed primary groups & Full-time civilian wage/salary & Nonreuse & \$\\$-0.1063\$; [-0.1361,-0.0764] & \$\\$-0.0467\$; [-0.1057,0.0122] & \$\\$-0.1317,0.0383\$ & \$\\$-0.1480,0.0546\$ \\\br/>+Broader equal-occupation groups & All employed & Full & \$\\$-0.0830\$; [-0.1053,-0.0607] & \$\\$-0.0151\$; [-0.0559,0.0257] & \$\\$-0.1000,0.0698\$ & \$\\$-0.0942,0.0639\$ \\\br/>+Broader equal-occupation groups & All employed & Nonreuse & \$\\$-0.0723\$; [-0.0979,-0.0467] & \$\\$-0.0056\$; [-0.0535,0.0424] & \$\\$-0.0918,0.0806\$ & \$\\$-0.0927,0.0815\$ \\\br/>+Broader equal-occupation groups & Full-time civilian wage/salary & Full & \$\\$-0.0933\$; [-0.1203,-0.0664] & \$\\$-0.0292\$; [-0.0790,0.0205] & \$\\$-0.1112,0.0527\$ & \$\\$-0.1307,0.0722\$ \\\br/>+Broader equal-occupation groups & Full-time civilian wage/salary & Nonreuse & \$\\$-0.0828\$; [-0.1137,-0.0520] & \$\\$-0.0291\$; [-0.0875,0.0293] & \$\\$-0.1118,0.0535\$ & \$\\$-0.1294,0.0712\$ \\\br/>+\bottomrule
+\end{tabular}}
+\end{table}

+\tabnote{Panel A reports growth-factor differences without an older comparison group. The public analogues do not recover BCC's undisclosed complete occupation membership. Panel B reports log coefficients. Full uses 2017--2019 and 2021 as preperiod years and 2023--2024 as postperiod years; nonreuse uses 2017 and 2021 as preperiod years. Standard 2020 one-year ACS PUMS are omitted, and 2022 is the transition year. ACS intervals use successive-difference replication with 80 replicate weights per perturbed endpoint year and a year-block independence approximation. Occupation and family intervals instead assign shocks at those cluster levels; the procedures answer different stochastic questions and are not combined. All extensions are post-outcome descriptive analyses.}

+\end{table}


```

diff --git a/paper/tables/r3_appendix_adoption.tex b/paper/tables/r3_appendix_adoption.tex
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+++ b/paper/tables/r3_appendix_adoption.tex
@@ -0,0 +1,19 @@
\begin{table}[htbp]
\centering
\caption{Later self-reported GenAI use by frozen beta exposure}\label{tab:rps_adoption}
\begin{tabular}{lcc}
\toprule
+ & Equal occupation & Respondent-count sensitivity \\
\midrule
+Pearson correlation & 0.504 & 0.647 \\
+Spearman correlation & 0.494 & --- \\
+Adoption-on-beta slope & 0.493 & 0.695 \\
+Mean adoption, Q1 & 0.263 & 0.250 \\
+Mean adoption, Q5 & 0.530 & 0.583 \\
+Q5 minus Q1 & 0.267 & 0.333 \\
+Within-family exposure SS share & 0.262 & 0.191 \\
+Within-family slope & 0.302 & 0.465 \\
\bottomrule
\end{tabular}
\end{table}
\tabnote{The exact-code merge retains 121 nonsuppressed occupations from the frozen 468; all five quintiles and 20 SOC major families are represented. The RPS workbook pools four waves from August 2025 through May 2026 and suppresses rates based on fewer than 20 respondents. The second column weights occupations by the workbook's pooled unweighted respondent count, which is neither a survey weight nor a verified inverse-variance weight. Detailed-cell standard errors and replicate weights are unavailable, so the table reports no inference. The quantities are descriptive validation of score alignment, not an adoption treatment or an explanation of the CPS coefficient.}
\end{table}
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\begin{table}[htbp]
\centering
\caption{Twelve-month exposure-duration sensitivity}\label{tab:app_annual_timing}
\begin{tabular}{lr}
\toprule
+Margin & Exposure-duration coding \\
\midrule
+Employment exit & $0.158$; [-0.070, 0.387] \\
+Unemployment entry & $0.186$; [-0.337, 0.708] \\
+Labor-force exit & $0.144$; [-0.084, 0.372] \\
+Reported occupational outflow & $-0.008$; [-0.082, 0.067] \\
\bottomrule
\end{tabular}
\end{table}
\tabnote{Entries are Q5--Q1, young--older onset coefficients under exposure-duration coding, which codes the share of the twelve destination months at or after onset (102 months). The baseline exclusion rule, which drops origins whose twelve-month link straddles January 2023, is the primary specification and is reported in main Table 5 (89 months). This column is not paired with that table. Intervals use 9,999 occupation wild-score draws.}
\end{table}
diff --git a/paper/tables/r3_appendix_architecture_pairs.tex b/paper/tables/r3_appendix_architecture_pairs.tex
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\begin{table}[htbp]
\centering
\caption{Post-outcome exploratory task-weight architectures on common support}\label{tab:appendix_architecture_pairs}
\small
\setlength{\tabcolsep}{4pt}
\resizebox{\textwidth}{\height}{
\begin{tabular}{llrrrr}
\toprule
+Architecture pair & Alternative coefficient & Beta coefficient & Alternative $-\beta$ & beta [95\% CI] & Paired $\mathbf{MDE}_{(80)}$ \\
\midrule
+$E_{1\text{ versus }E_{1+0.5E_2}}$ & $-0.103$ & $-0.132$ & $0.029$ & [-0.044, 0.103] & 0.106 \\
+$E_{1+0.25E_2}$ versus $E_{1+0.5E_2}$ & $-0.101$ & $-0.132$ & $0.031$ & [-0.005, 0.068] & 0.055 \\
+$E_{1+0.75E_2}$ versus $E_{1+0.5E_2}$ & $-0.143$ & $-0.132$ & $-0.011$ & [-0.039, 0.017] & 0.041 \\
+$E_{1+E_2}$ versus $E_{1+0.5E_2}$ & $-0.156$ & $-0.132$ & $-0.024$ & [-0.064, 0.017] & 0.058 \\
\bottomrule
\end{tabular}
}
\end{table}
\tabnote{All entries are post-outcome exploratory Q5--Q1 coefficients estimated on the identical 468-occupation support. Beta is $E_{1+0.5E_2}$. At each alternative weight, normalization, cutoffs, and tie-preserving quintile memberships are recomputed, so a paired movement combines score and membership changes rather than holding the treatment groups fixed. The intervals use 9,999 common occupation multiplier draws, preserving covariance within each pair. Every paired interval contains zero; this means that the design does not detect a difference, not that the architectures are economically equivalent.}
\end{table}
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\centering
\caption{Post-outcome exploratory BCC-style grouping bridge}\label{tab:appendix_bcc_bridge}
\scriptsize
\setlength{\tabcolsep}{3pt}
\resizebox{\textwidth}{\height}{
\begin{tabular}{llrrrr}
\toprule
+Grouping rule & Conditioning structure & Coefficient [95\% CI] & Change from baseline [95\% CI] & Paired $\mathbf{MDE}_{(80)}$ \\
\midrule
+\multicolumn{5}{l}{\textit{Panel A. Public-dashboard equal-occupation approximation}} \\
+Equal occupation & Occupation and calendar-month FE & $-0.072$ & [-0.123, -0.021] & --- \\
+Equal occupation & SOC2 by young by post & $-0.0148$ & [-0.0792, 0.0497] & $0.0573$; [0.0005, 0.1141] & 0.083 \\
+Equal occupation & SOC2 by young by calendar month & $-0.0168$ & [-0.0808, 0.0473] & $0.0553$; [-0.0013, 0.1118] & 0.082 \\
\addlinespace
+\multicolumn{5}{l}{\textit{Panel B. Historical employment-weighted approximation}} \\
+Employment weighted & Occupation and calendar-month FE & $-0.075$ & [-0.125, -0.025] & --- \\
+Employment weighted & SOC2 by young by post & $-0.027$ & [-0.095, 0.041] & $0.048$; [-0.011, 0.108] & 0.086 \\
+Employment weighted & SOC2 by young by calendar month & $-0.027$ & [-0.093, 0.040] & $0.048$; [-0.011, 0.107] & 0.086 \\
\bottomrule
\end{tabular}
}
\end{table}
\tabnote{All rows are post-outcome exploratory CPS bridges, not replications of BCC's proprietary ADP analysis. Each row compares GPT-4-beta Q4--Q5 with Q1--Q3 on the same 468 occupations, includes the standardized historical Web-Software control, and uses 9,999 common occupation-level Rademacher multipliers. Paired changes compare each SOC2-conditioned row with its own unconditioned baseline. Panel A follows the public dashboard's equal-occupation description as closely as public information permits. Panel B uses earlier employment-weighted cuts formed from 108 months of young-plus-older CPS stock, including postperiod months, and is only a grouping-rule diagnostic. Exact BCC occupation membership, universe, cutoff and tie algorithm are not public, so concordance is unverified. The BCC and CPS objects also differ in ages, denominator, endpoint, employer controls, and observed worker--firm flows. Null-containing paired intervals are nondetections, not equivalence.}
\end{table}
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\centering
\caption{Broader support for CPS bridges}\label{tab:appendix_broader_support}
\scriptsize
\setlength{\tabcolsep}{3pt}
\resizebox{\textwidth}{\height}{
\begin{tabular}{llrrrr}
\toprule
+Grouping rule & Conditioning structure & Coefficient [95\% CI] & Change from baseline [95\% CI] & Paired $\mathbf{MDE}_{(80)}$ \\
\midrule
+\multicolumn{5}{l}{\textit{Panel A. Public-dashboard equal-occupation approximation}} \\
+Equal occupation & Occupation and calendar-month FE & $-0.072$ & [-0.123, -0.021] & --- \\
+Equal occupation & SOC2 by young by post & $-0.0148$ & [-0.0792, 0.0497] & $0.0573$; [0.0005, 0.1141] & 0.083 \\
+Equal occupation & SOC2 by young by calendar month & $-0.0168$ & [-0.0808, 0.0473] & $0.0553$; [-0.0013, 0.1118] & 0.082 \\
\addlinespace
+\multicolumn{5}{l}{\textit{Panel B. Historical employment-weighted approximation}} \\
+Employment weighted & Occupation and calendar-month FE & $-0.075$ & [-0.125, -0.025] & --- \\
+Employment weighted & SOC2 by young by post & $-0.027$ & [-0.095, 0.041] & $0.048$; [-0.011, 0.108] & 0.086 \\
+Employment weighted & SOC2 by young by calendar month & $-0.027$ & [-0.093, 0.040] & $0.048$; [-0.011, 0.107] & 0.086 \\
\bottomrule
\end{tabular}
}
\end{table}
\tabnote{All rows are post-outcome exploratory CPS bridges, not replications of BCC's proprietary ADP analysis. Each row compares GPT-4-beta Q4--Q5 with Q1--Q3 on the same 468 occupations, includes the standardized historical Web-Software control, and uses 9,999 common occupation-level Rademacher multipliers. Paired changes compare each SOC2-conditioned row with its own unconditioned baseline. Panel A follows the public dashboard's equal-occupation description as closely as public information permits. Panel B uses earlier employment-weighted cuts formed from 108 months of young-plus-older CPS stock, including postperiod months, and is only a grouping-rule diagnostic. Exact BCC occupation membership, universe, cutoff and tie algorithm are not public, so concordance is unverified. The BCC and CPS objects also differ in ages, denominator, endpoint, employer controls, and observed worker--firm flows. Null-containing paired intervals are nondetections, not equivalence.}
\end{table}

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```

\begin{table}[htbp]
\centering
\caption{Supported tails when Webb availability is relaxed}\label{tab:app_broader_support}
\resizebox{\textwidth}{!}{%
\begin{tabular}{lrrrrrr}
\toprule
Population and classification & Occupations & Families & Tail-spanning families & Tail occupations & Tail stock & (\%) & Pooled coefficient \\
\midrule
Primary support, with Webb & 468 & 22 & 4 & 29 & 5.03 & \& \$-0.1321\$ \\
Primary support, Webb omitted & 468 & 22 & 4 & 29 & 5.03 & \& \$-0.1338\$ \\
Broader beta-valid support, primary cuts & 490 & 23 & 5 & 62 & 12.61 & \& \$-0.1344\$ \\
Broader beta-valid support, recomputed cuts & 490 & 23 & 5 & 62 & 12.61 & \& \$-0.1348\$ \\
\bottomrule
\end{tabular}}
\begin{tabnote}[The primary support requires both beta exposure and Webb software exposure; omitting Webb from the regression on that support does not change membership. The broader support requires beta exposure but not Webb and adds 22 occupations, equal to 1.88 percent of broader-support preperiod stock. With the primary cuts or recomputed cuts it adds office and administrative support (SOC43) as the fifth family containing both Q1 and Q5. ‘Tail occupations’ and ‘tail stock’ count Q1 and Q5 occupations within tail-spanning families. The coefficient change from 468 to 490 occupations changes the population and is descriptive, not a paired test. On fixed support, the Webb-omitted minus Webb-included coefficient difference is \&-0.0017\$, with paired interval \&-0.0102,0.0069\$, nondetection is not equivalence.}
\end{tabnote}
\end{table}

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\begin{table}[htbp]
\centering
\caption{Post-outcome exploratory characteristic-conditioning estimates}\label{tab:appendix_characteristic_estimates}
\scriptsize
\setlength{\tabcolsep}{3pt}
\resizebox{\textwidth}{!}{%
\begin{tabular}{lrrrrrr}
\toprule
Conditioning block & Occupations & Baseline coefficient & Augmented coefficient & [95\% CI] & Paired change & [95\% CI] & Paired \mathrm{MDE}_{.80}\$ \\
\midrule
\multicolumn{8}{l}{\textit{Panel A. One-at-a-time maximal-support comparisons}} \\
Computer use & 455 & \& \$-0.096\$ & \& \$-0.197\$ & [-0.316,-0.077]\$ & \& \$-0.101\$ & [-0.177,-0.024]\$ & \& 0.108 \\
Remotability & 408 & \& \$-0.115\$ & \& \$-0.122\$ & [-0.222,-0.022]\$ & \& \$-0.007\$ & [-0.048,0.033]\$ & \& 0.058 \\
Preperiod wage & 457 & \& \$-0.144\$ & \& \$-0.161\$ & [-0.248,-0.075]\$ & \& \$-0.017\$ & [-0.044,0.009]\$ & \& 0.038 \\
Education requirement & 404 & \& \$-0.100\$ & \& \$-0.105\$ & [-0.203,-0.008]\$ & \& \$-0.005\$ & [-0.027,0.016]\$ & \& 0.031 \\
Routine-task intensity & 449 & \& \$-0.137\$ & \& \$-0.129\$ & [-0.221,-0.036]\$ & \& \$0.008\$ & [-0.011,0.027]\$ & \& 0.027 \\
Manual/physical content & 455 & \& \$-0.140\$ & \& \$-0.142\$ & [-0.289,0.004]\$ & \& \$-0.002\$ & [-0.092,0.089]\$ & \& 0.130 \\
SOC2 by young by post & 468 & \& \$-0.132\$ & \& \$-0.022\$ & [-0.162,0.119]\$ & \& \$0.111\$ & [0.008,0.213]\$ & \& 0.146 \\
\addlinespace
\multicolumn{8}{l}{\textit{Panel B. One-at-a-time and joint comparisons on the 347-occupation common support}} \\
Computer use & 347 & \& \$-0.124\$ & \& \$-0.248\$ & [-0.391,-0.106]\$ & \& \$-0.125\$ & [-0.209,-0.040]\$ & \& 0.124 \\
Remotability & 347 & \& \$-0.124\$ & \& \$-0.137\$ & [-0.239,-0.034]\$ & \& \$-0.013\$ & [-0.052,0.026]\$ & \& 0.057 \\
Preperiod wage & 347 & \& \$-0.124\$ & \& \$-0.135\$ & [-0.235,-0.036]\$ & \& \$-0.012\$ & [-0.029,0.006]\$ & \& 0.025 \\
Education requirement & 347 & \& \$-0.124\$ & \& \$-0.134\$ & [-0.235,-0.032]\$ & \& \$-0.010\$ & [-0.027,0.006]\$ & \& 0.024 \\
Routine-task intensity & 347 & \& \$-0.124\$ & \& \$-0.112\$ & [-0.220,-0.003]\$ & \& \$0.012\$ & [-0.024,0.048]\$ & \& 0.051 \\
Manual/physical content & 347 & \& \$-0.124\$ & \& \$-0.122\$ & [-0.255,0.010]\$ & \& \$0.001\$ & [-0.079,0.082]\$ & \& 0.117 \\
All six characteristics & 347 & \& \$-0.124\$ & \& \$-0.233\$ & [-0.388,-0.078]\$ & \& \$-0.110\$ & [-0.221,0.002]\$ & \& 0.164 \\
\bottomrule
\end{tabular}}
\begin{tabnote}[All entries are post-outcome exploratory and were designed after the initial outcome pattern was known. Each baseline--augmented pair uses identical occupations, the canonical rebuilt corrected-preperiod beta groups and Webb normalization, and 9,999 common occupation-level Rademacher multipliers. Panel A uses each characteristic's maximal finite subset; Panel B uses the current 347-occupation common support, which covers 76.78 percent of primary-support January 2017--November 2022 treatment-construction stock. For comparison, the unaugmented coefficient on all 468 occupations is \&-0.132\$, while the unaugmented coefficient after imposing the 347-occupation common support is \&-0.124\$; that support-only movement is kept separate from every paired conditioning change. Rows from the superseded unbounded pandemic-shortfall construction are omitted; the corrected shortfall analysis uses a separate 363-occupation population and appears in Table~\ref{tab:appendix_shortfall}. A confidence interval containing zero is a nondetection, not evidence of equivalence; conditioning does not identify a causal AI component.]
\end{tabnote}
\end{table}

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\begin{table}[htbp]
\centering
\caption{Post-outcome exploratory support and target-information diagnostics}\label{tab:appendix_characteristic_information}
\scriptsize
\setlength{\tabcolsep}{3pt}
\resizebox{\textwidth}{!}{%
\begin{tabular}{lrrrrrr}
\toprule
Conditioning block & Employment coverage & (\%) & Information retained & (\%) & Target VIF-like ratio & Effective occupation count & Top-five information share & (\%) \\
\midrule
\multicolumn{8}{l}{\textit{Panel A. One-at-a-time maximal-support comparisons}} \\
Computer use & 96.75 & 29.02 & 3.45 & 52.58 & 22.34 \\
Remotability & 88.58 & 52.66 & 1.90 & 29.25 & 30.65 \\
Preperiod wage & 97.93 & 59.45 & 1.68 & 37.86 & 26.32 \\
Education requirement & 84.54 & 63.60 & 1.57 & 27.48 & 31.24 \\
Routine-task intensity & 98.45 & 65.19 & 1.53 & 43.26 & 24.59 \\
Manual/physical content & 96.85 & 30.61 & 3.27 & 51.05 & 22.34 \\
SOC2 by young by post & 100.00 & 20.81 & 4.81 & 53.06 & 21.87 \\
\addlinespace
\multicolumn{8}{l}{\textit{Panel B. Cumulative comparisons on the 341-occupation common support}} \\
Computer use & 76.77 & 29.33 & 3.41 & 48.88 & 21.98 \\
\&\& remotability & 76.77 & 28.46 & 3.51 & 44.97 & 24.72 \\
\&\& wage, education, routine, and manual & 76.77 & 21.08 & 4.74 & 58.00 & 20.28 \\
\bottomrule
\end{tabular}}
\begin{tabnote}[All diagnostics are post-outcome exploratory. Employment coverage is the share of primary-support January 2017--November 2022 corrected treatment-construction stock retained. ‘Information retained’ is nuisance-adjusted target information divided by fixed-effect-adjusted raw target information; its reciprocal is the reported target VIF-like ratio. These are design-information diagnostics, not an estimate of latent exposure reliability. The effective occupation count is the inverse concentration of occupation-level conditional target-information shares, and the final column is the share supplied by the five largest occupations. In the 341-occupation common-support sample, the 2017--2019-employment-weighted correlations of beta exposure with computer use, remotability, wage, education requirement, routine-task intensity, and manual/physical content are 0.807, 0.562, 0.429, 0.421, 0.197, and \&-0.748\$, respectively. Rows based on the superseded unbounded pandemic-shortfall construction are omitted. Substantial information loss makes residual coefficients less precise and does not purify an AI treatment.]
\end{tabnote}
\end{table}

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\begin{table}[htbp]
\centering
\caption{Older comparison bands and enrollment restrictions}\label{tab:app_cohort_enrollment}
\resizebox{\textwidth}{!}{%
\begin{tabular}{lrr}
\toprule
Comparison & Pooled coefficient and 95\% interval & Family-month coefficient and 95\% interval \\
\midrule
Ages 22--25 versus 26--65 & \& \$-0.1321\$ & [-0.2211,-0.0431]\$ & \& \$-0.0217\$ & [-0.1607,0.1174]\$ \\
Ages 22--25 versus 26--30 & \& \$-0.1205\$ & [-0.1881,-0.0530]\$ & \& \$-0.1093\$ & [-0.2328,0.0141]\$ \\
Ages 22--25 versus 31--40 & \& \$-0.1598\$ & [-0.2455,-0.0742]\$ & \& \$-0.0247\$ & [-0.1709,0.1216]\$ \\
Ages 22--25 versus 41--50 & \& \$-0.1392\$ & [-0.2344,-0.0440]\$ & \& \$-0.0022\$ & [-0.1429,0.1385]\$
\end{tabular}}

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+Agex 22--25 versus 51--65 & $-0.0971\;[-0.2037,0.0094]$ & $-0.0021\;[-0.1627,0.1584]$ \\\
+Older exact-age composition fixed & $-0.1313\;[-0.2202,-0.0425]$ & $-0.0218\;[-0.1597,0.1161]$ \\\
+Young numerators nonenrolled & $-0.1239\;[-0.2154,-0.0324]$ & $-0.0285\;[-0.1666,0.1097]$ \\\
+Both groups nonenrolled, ages 22--54 & $-0.1354\;[-0.2243,-0.0465]$ & $-0.0356\;[-0.1704,0.0991]$ \\\
+bottomrule
+\end{tabular}}
+\tabnote{Intervals use 9,999 occupation multipliers. The fixed-composition row holds the 26--65 exact-age shares at their preperiod values. The young-numerator row retains all employed ages 26--65
in the denominator. The last row restricts both groups to $\mathrm{SCHLCOLE}=5$ on the common ages 22--54 question universe. Older workers are comparison groups, not untreated controls; an
interval containing zero indicates non-detection rather than equivalence.}
+\end{table}
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+++ b/paper/tables/r3_appendix_direct_tail_occupations.tex
@@ -0,0 +1,49 @@
+\begin{center}
+\scriptsize
+\begin{longtable}{@{}r c r p{0.47\textwidth}} r@{}{}
+\caption{Occupations supplying direct within-family tail support}\label{tab:app_direct_tail_occupations}\\
+\toprule
+SOC2 & Tail & \shortstack{Census\\occupation} & Occupation & \shortstack{Preperiod stock\\(million person-months)} \\\
+\midrule
+\endfirsthead
+\multicolumn{5}{l}{\footnotesize Table \thetable{} continued}\\
+\toprule
+SOC2 & Tail & \shortstack{Census\\occupation} & Occupation & \shortstack{Preperiod stock\\(million person-months)} \\\
+\midrule
+\endhead
+\midrule\multicolumn{5}{r}{\footnotesize Continued on next page}\\
+\endfoot
+\bottomrule
+\endlastfoot
+27 & Q1 & 2721 & Athletes and sports competitors & 1.83 \\\
+27 & Q1 & 2740 & Dancers and choreographers & 0.94 \\\
+27 & Q5 & 2850 & Writers and authors & 14.91 \\\
+27 & Q5 & 2825 & Public relations specialists & 8.61 \\\
+27 & Q5 & 2830 & Editors & 8.44 \\\
+27 & Q5 & 2861 & Interpreters and translators & 6.14 \\\
+27 & Q5 & 2810 & News analysts, reporters, and journalists & 5.41 \\\
+27 & Q5 & 2840 & Technical writers & 4.38 \\\
+27 & Q5 & 2751 & Music directors and composers & 2.74 \\\
+27 & Q5 & 2805 & Broadcast announcers and radio disc jockeys & 2.19 \\\
+27 & Q5 & 2862 & Court reporters and simultaneous captioners & 2.10 \\\
+29 & Q1 & 3423 & Surgical technologists & 6.77 \\\
+29 & Q1 & 3402 & Paramedics & 6.47 \\\
+29 & Q1 & 3422 & Psychiatric technicians & 3.59 \\\
+29 & Q5 & 3515 & Medical records specialists & 11.77 \\\
+29 & Q5 & 3030 & Dietitians and nutritionists & 7.71 \\\
+31 & Q1 & 3603 & Nursing assistants & 86.89 \\\
+31 & Q1 & 3601 & Home health aides & 31.36 \\\
+31 & Q1 & 3630 & Massage therapists & 10.70 \\\
+31 & Q1 & 3605 & Orderlies and psychiatric aides & 3.48 \\\
+31 & Q5 & 3646 & Medical transcriptionists & 2.62 \\\
+41 & Q1 & 4900 & Models, demonstrators, and product promoters & 2.18 \\\
+41 & Q5 & 4850 & Sales representatives, wholesale and manufacturing & 81.01 \\\
+41 & Q5 & 4810 & Insurance sales agents & 38.93 \\\
+41 & Q5 & 4840 & Sales representatives of services, except advertising, insurance, financial services, and travel & 36.99 \\\
+41 & Q5 & 4820 & Securities, commodities, and financial services sales agents & 15.73 \\\
+41 & Q5 & 4800 & Advertising sales agents & 11.43 \\\
+41 & Q5 & 4830 & Travel agents & 4.26 \\\
+41 & Q5 & 4940 & Telemarketers & 2.87 \\\
+\end{longtable}
+\tabnote{This is the complete 29-occupation membership of the changed-population direct-tail exercise, ordered by family, tail, and stock. Preperiod stock is the cumulative CPS-weighted employed-
person stock across the 71 observed months from January 2017 through November 2022: for each occupation, the producer sums the monthly young (ages 22--25) and older (ages 26--65) employment
stocks. The displayed unit is one million employed-person months, not average-month employment. The list makes the direct comparisons auditable; it is not a ranking by estimated effect or
influence.}
+\end{center}
diff --git a/paper/tables/r3_appendix_endpoint_sensitivity.tex b/paper/tables/r3_appendix_endpoint_sensitivity.tex
new file mode 100644
index 0000000..ccc3380
--- /dev/null
+++ b/paper/tables/r3_appendix_endpoint_sensitivity.tex
@@ -0,0 +1,27 @@
+\begin{table}[htbp]
+\centering
+\caption{Post-outcome exploratory endpoint sensitivity}\label{tab:app_endpoints}
+\scriptsize
+\setlength{\tabcolsep}{3pt}
+\resizebox{\textwidth}{!}{%
+\begin{tabular}{lrrrr}
+\toprule
+Endpoint & Months & Coefficient [95\% CI] & Endpoint & $-$ full [paired 95\% CI] & Paired $\mathrm{MDE}_{.80}$ \\\
+\midrule
+\multicolumn{6}{l}{\textit{Pooled}} \\\
+Through December 2024 & 95 & $-$0.1110\;[-0.2008,-0.0213]$ & $0.0211\;[-0.0073,0.0495]$ & $0.0407$ \\\
+Through September 2025 & 104 & $-$0.1096\;[-0.1978,-0.0214]$ & $0.0225\;[0.0053,0.0397]$ & $0.0245$ \\\
+Through December 2025 (actual gap) & 106 & $-$0.1109\;[-0.1979,-0.0239]$ & $0.0212\;[0.0057,0.0367]$ & $0.0221$ \\\
+Through July 2026, excluding September and November 2025 & 111 & $-$0.1337\;[-0.2236,-0.0438]$ & $0.0016\;[-0.0075,0.0043]$ & $0.0084$ \\\
+Through July 2026 (reference) & 113 & $-$0.1321\;[-0.2206,-0.0437]$ & Reference & --- \\\
+\addlinespace
+\multicolumn{6}{l}{\textit{SOC2 by calendar month}} \\\
+Through December 2024 & 95 & $-$0.0120\;[-0.1657,0.1417]$ & $0.0097\;[-0.0526,0.0719]$ & $0.0897$ \\\
+Through September 2025 & 104 & $-$0.0219\;[-0.1616,0.1178]$ & $0.0002\;[-0.0345,0.0341]$ & $0.0491$ \\\
+Through December 2025 (actual gap) & 106 & $-$0.0170\;[-0.1555,0.1215]$ & $0.0047\;[-0.0248,0.0342]$ & $0.0422$ \\\
+Through July 2026, excluding September and November 2025 & 111 & $-$0.0252\;[-0.1660,0.1155]$ & $0.0035\;[-0.0131,0.0060]$ & $0.0137$ \\\
+Through July 2026 (reference) & 113 & $-$0.0217\;[-0.1607,0.1173]$ & Reference & --- \\\
+\bottomrule
+\end{tabular}}
+\end{table}
+\tabnote{All rows use the rebuilt corrected-preperiod treatment and the same occupation support. The December 2024 row is the requested pre-2025 endpoint; the December 2025 row retains the actual
missing October survey month. The shutdown-month row excludes September and November 2025 and does not interpolate October. Nonreference endpoint intervals and paired differences use the
common endpoint-grid draws. To avoid printing a second seed-dependent interval for an identical target, each full-window reference uses its declared canonical single-target interval. The MDE
is the two-sided five-percent, 80-percent normal-theory precision diagnostic, not an economic-equivalence threshold.}
+\end{table}
diff --git a/paper/tables/r3_appendix_entry_probabilities.tex b/paper/tables/r3_appendix_entry_probabilities.tex
new file mode 100644
index 0000000..51502a3
--- /dev/null
+++ b/paper/tables/r3_appendix_entry_probabilities.tex
@@ -0,0 +1,22 @@
+\begin{table}[htbp]
+\centering
+\caption{Employment-entry destinations on a common risk set}\label{tab:app_entry_probabilities}
+\scriptsize
+\setlength{\tabcolsep}{2pt}
+\resizebox{\textwidth}{!}{%
+\begin{tabular}{llllllllll}
+\toprule

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+Horizon & Age & Period & Remain nonemployed & Q1 & Q2 & Q3 & Q4 & Q5 & Outside support & Any entry \\
+\\midrule
+Adjacent & 26--65 & Pre & .9154 & .0221 & .0180 & .0117 & .0087 & .0113 & .0128 & .0846 \\
+Adjacent & 26--65 & Post & .9179 & .0210 & .0175 & .0115 & .0091 & .0105 & .0125 & .0821 \\
+Adjacent & 22--25 & Pre & .8491 & .0361 & .0398 & .0191 & .0129 & .0211 & .0219 & .1509 \\
+Adjacent & 22--25 & Post & .8626 & .0361 & .0352 & .0162 & .0117 & .0193 & .0190 & .1374 \\
+Twelve month & 26--65 & Pre & .8169 & .0439 & .0396 & .0252 & .0213 & .0263 & .0268 & .1831 \\
+Twelve month & 26--65 & Post & .8276 & .0399 & .0385 & .0230 & .0215 & .0246 & .0250 & .1724 \\
+Twelve month & 22--25 & Pre & .6408 & .0772 & .0885 & .0492 & .0390 & .0589 & .0465 & .3592 \\
+Twelve month & 22--25 & Post & .6611 & .0695 & .0843 & .0555 & .0351 & .0530 & .0415 & .3389 \\
+\\bottomrule
+\\end{table}
+\\tabnote{The denominator is all retained nonemployed origins, weighted with the horizon-specific official link weight. Q1--Q5 are employed destinations on the retained exposure support. Outside support records an employed endpoint with a valid occupation outside that support. Any entry equals one minus remaining nonemployment and reconciles to the sum of all observed employed destinations.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_equal_objective.tex b/paper/tables/r3_appendix_equal_objective.tex
new file mode 100644
index 0000000..59393c8
--- /dev/null
+++ b/paper/tables/r3_appendix_equal_objective.tex
@@ -0,0 +1,18 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Equal-occupation objective sensitivity}\\label{tab:appendix_equal_objective}
+\\small
+\\setlength{\\tabcolsep}{4pt}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{llrr}
+\\toprule
+Conditioning structure & Objective & Q5 & \\times post [95\\% CI] & Equal occupation & $-\\$ stock weighted [95\\% CI] \\
+\\midrule
+Occupation and calendar-month FE & Employment-stock weighted & $-0.132$; [-0.220,-0.045] & $ & --- \\
+Occupation and calendar-month FE & Equal occupation & $-0.083$; [-0.206,0.040] & $ & $0.049$; [-0.060,0.158] \\
+SOC2 family by calendar month & Employment-stock weighted & $-0.022$; [-0.159,0.115] & $ & --- \\
+SOC2 family by calendar month & Equal occupation & $-0.005$; [-0.203,0.193] & $ & $0.017$; [-0.159,0.193] \\
+\\bottomrule
+\\end{tabular}}
+\\tabnote{This sensitivity changes the estimation objective: every occupation receives total weight one, while \\texttt{WTFINL} continues to determine young and older shares within occupation--month cells. Treatment membership, 468-occupation support, calendar, and nuisance structures are unchanged. Intervals use 9,999 common occupation-level Rademacher score multipliers; paired differences are formed before applying the common draws. A paired interval containing zero means that this design does not detect a difference, not that the objectives are economically equivalent.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_exit_components.tex b/paper/tables/r3_appendix_exit_components.tex
new file mode 100644
index 0000000..16803fb
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+++ b/paper/tables/r3_appendix_exit_components.tex
@@ -0,0 +1,16 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Employment-exit components in linked CPS records}\\label{tab:app_exit_components}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{llrrrr}
+\\toprule
+Endpoint & Component & Raw events & Coefficient & 95\\% interval \\
+\\midrule
+Adjacent month & Unemployment entry & 32,488 & $0.055$ & $[-0.272,0.381]$ \\
+Adjacent month & Labor-force exit & 74,480 & $0.135$ & $[-0.064,0.333]$ \\
+Twelve month & Unemployment at endpoint & 27,330 & $0.246$ & $[-0.295,0.787]$ \\
+Twelve month & Out of labor force at endpoint & 78,773 & $0.078$ & $[-0.181,0.337]$ \\
+\\bottomrule
+\\end{tabular}}
+\\tabnote{Rows use official longitudinal weights and the same young-relative Q5--Q1 log-rate-ratio contrast as the core flow table. Counts are repeated link events rather than unique persons. Unemployment and out-of-labor-force events exhaust observed employment exits, but their nonlinear coefficients do not add. Every interval includes zero.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_family_information_precision.tex b/paper/tables/r3_appendix_family_information_precision.tex
new file mode 100644
index 0000000..e80665d
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+++ b/paper/tables/r3_appendix_family_information_precision.tex
@@ -0,0 +1,16 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Family-support information and precision diagnostics}\\label{tab:app_family_information}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{llrrrrrr}
+\\toprule
+Target and structure & Occ. & $I$ (millions) & $I/I_\\{\\mathrm{raw}\\}$ & $I/I_\\{\\mathrm{parent}\\}$ & Effective occ. & Top-five share & Normal MDE$\\{_{80}\\}$ \\
+\\midrule
+Q5--Q1, pooled & 468 & 26.289 & 0.378 & 1.000 & 43.2 & 0.245 & 0.1266 \\
+Q5--Q1, SOC2 & $\\times$ month & 468 & 7.803 & 0.112 & 0.297 & 53.3 & 0.218 & 0.1998 \\
+Direct-tail Q5--Q1, SOC2 & $\\times$ month & 29 & 0.669 & 0.080 & 0.308 & 6.2 & 0.778 & 0.4576 \\
+Continuous within-SOC2 slope, SOC2 & $\\times$ month & 468 & 261.278 & 0.583 & 0.965 & 48.0 & 0.249 & 0.0313 \\
+\\bottomrule
+\\end{tabular}}
+\\tabnote{$I$ is nuisance-adjusted Fisher information after projecting the target score on the model's nuisance score; $I_\\{\\mathrm{raw}\\}$ is the raw weighted treatment sum of squares. "Parent" is the corresponding pooled model on the same target and population, so the direct-tail and continuous rows each use their own pooled parent. Because target scales and populations differ, absolute $I$ is not comparable across all rows. Effective occupations are the inverse Herfindahl count of occupation-level information contributions; the top-five share is the sum of the five largest contributions. Fitted information and its effective count are design-concentration diagnostics: neither replaces the nominal occupation-cluster count nor measures first-stage CPS sampling precision. MDE$\\{_{80}\\}=2.80\\widehat{\\text{SE}}$ uses the occupation-cluster standard error and is a normal-theory precision diagnostic, not a rejection result. The direct-tail MDE is 0.4576 log point; the continuous MDE is 0.0313 per one within-family standard deviation (0.1084 raw beta units).}
+\\end{table}
diff --git a/paper/tables/r3_appendix_family_support.tex b/paper/tables/r3_appendix_family_support.tex
new file mode 100644
index 0000000..f70f0af
--- /dev/null
+++ b/paper/tables/r3_appendix_family_support.tex
@@ -0,0 +1,42 @@
+\\begin{center}
+\\footnotesize
+\\begin{longtable}{@{}r p{0.25\\textwidth} r r l c r@{}}
+\\caption{Broad-family exposure support}\\label{tab:app_family_support}\\
+\\toprule
+SOC2 & Family & Occs. & Stock (\\%) & Beta range & Quintiles & Q1/Q5 & Q5 info. (\\%) \\
+\\midrule
+\\endfirsthead
+\\multicolumn{8}{l}{\\footnotesize \\itshape Table \\thetable{} continued}\\
+\\toprule
+SOC2 & Family & Occs. & Stock (\\%) & Beta range & Quintiles & Q1/Q5 & Q5 info. (\\%) \\
+\\midrule
+\\endhead
+\\midrule\\multicolumn{8}{r}{\\footnotesize Continued on next page}\\
+\\endfoot
+\\bottomrule
+\\endlastfoot
+11 & Management & 29 & 13.56 & 0.367--0.586 & 3|4|5 & No & 3.04 \\
+13 & Business and Financial Operations & 28 & 6.22 & 0.405--0.636 & 3|4|5 & No & 5.99 \\

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+15 & Computer and Mathematical & 15 & 4.31 & 0.500--0.950 & 4|5 & No & 0.28 \\
+17 & Architecture and Engineering & 19 & 2.28 & 0.406--0.727 & 3|4|5 & No & 1.87 \\
+19 & Life, Physical, and Social Science & 21 & 1.25 & 0.344--0.723 & 3|4|5 & No & 0.54 \\
+21 & Community and Social Service & 12 & 1.31 & 0.111--0.439 & 1|2|3 & No & 0.20 \\
+23 & Legal & 4 & 1.34 & 0.425--0.571 & 3|4|5 & No & 0.28 \\
+25 & Educational Instruction and Library & 7 & 4.53 & 0.229--0.567 & 2|3|5 & No & 3.20 \\
+27 & Arts, Design, Entertainment, Sports, and Media & 25 & 1.80 & 0.000--0.958 & 1|2|3|4|5 & Yes & 4.86 \\
+29 & Healthcare Practitioners and Technical & 37 & 6.38 & 0.067--0.618 & 1|2|3|4|5 & Yes & 3.61 \\
+31 & Healthcare Support & 14 & 3.66 & 0.074--0.875 & 1|2|3|5 & Yes & 7.26 \\
+33 & Protective Service & 17 & 2.28 & 0.036--0.415 & 1|2|3 & No & 1.42 \\
+35 & Food Preparation and Serving Related & 10 & 3.27 & 0.000--0.511 & 1|2|4 & No & 10.84 \\
+37 & Building and Grounds Cleaning and Maintenance & 6 & 2.51 & 0.000--0.355 & 1|2|3 & No & 1.29 \\
+39 & Personal Care and Service & 17 & 2.51 & 0.015--0.400 & 1|2|3 & No & 2.97 \\
+41 & Sales and Related & 17 & 10.06 & 0.133--0.806 & 1|2|3|4|5 & Yes & 17.40 \\
+43 & Office and Administrative Support & 47 & 11.21 & 0.216--0.975 & 2|3|4|5 & No & 18.92 \\
+45 & Farming, Fishing, and Forestry & 5 & 0.15 & 0.043--0.292 & 1|2 & No & 0.14 \\
+47 & Construction and Extraction & 30 & 6.14 & 0.000--0.417 & 1|2|3 & No & 1.97 \\
+49 & Installation, Maintenance, and Repair & 31 & 3.56 & 0.000--0.410 & 1|2|3 & No & 2.64 \\
+51 & Production & 52 & 4.14 & 0.000--0.438 & 1|2|3 & No & 5.67 \\
+53 & Transportation and Material Moving & 25 & 7.52 & 0.031--0.413 & 1|2|3 & No & 5.59 \\
+\\end{longtable}
+\\tabnote{Stock is each family's share of preperiod employment on the 468-occupation rebuilt support. Q5 information is the family's share of nuisance-adjusted fitted information for the family-
conditioned Q5 target; it is neither a sampling-cluster share nor pre-outcome residual-exposure support. Beta ranges and memberships use the corrected preperiod treatment contract. Only Q1/Q5
rows enter the changed-population direct-tail exercise.}
+\\end{center}
diff --git a/paper/tables/r3_appendix_flow_retention.tex b/paper/tables/r3_appendix_flow_retention.tex
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--- /dev/null
+++ b/paper/tables/r3_appendix_flow_retention.tex
@@ -0,0 +1,15 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Official-weight link retention across exposure quintiles}\\label{tab:app_flow_retention}
+\\begin{tabular}{llrr}
+\\toprule
+Horizon & Origin age & Preperiod range & Postperiod range \\
+\\midrule
+Adjacent month & Ages 22--25 & 87.6--88.9\\% & 85.3--86.9\\% \\
+Adjacent month & Ages 26--65 & 90.5--92.4\\% & 87.6--90.5\\% \\
+Twelve month & Ages 22--25 & 50.1--54.0\\% & 48.7--52.8\\% \\
+Twelve month & Ages 26--65 & 68.8--73.3\\% & 63.9--70.2\\% \\
+\\bottomrule
+\\end{tabular}
+\\tabnote{Each entry is the minimum--maximum across fixed beta quintiles among employed origins on the 468-occupation analysis support. Numerators require a validated endpoint and a positive horizon-
specific longitudinal weight; denominators use origin \\texttt{WTFINL}. Rotation ineligibility and a target month outside the extract are recorded separately from eligible origins without a
validated endpoint.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_flow_selection.tex b/paper/tables/r3_appendix_flow_selection.tex
new file mode 100644
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+++ b/paper/tables/r3_appendix_flow_selection.tex
@@ -0,0 +1,20 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Linked-flow selection sensitivities}\\label{tab:app_flow_selection}
+\\scriptsize
+\\setlength{\\tabcolsep}{3pt}
+\\resizebox{\\textwidth}{!}{\\%
+\\begin{tabular}{llrrrr}
+\\toprule
+Horizon & Margin & Official link & Origin-weight complete case & Observable poststratification & Linear accounting interval \\
+\\midrule
+Adjacent & Employment exit & 0.0088 & 0.0088 & 0.0086 & $[-0.441,0.463]$ \\
+Adjacent & Unemployment entry & 0.0037 & 0.0039 & 0.0038 & $[-0.445,0.459]$ \\
+Adjacent & Labor-force exit & 0.0051 & 0.0050 & 0.0048 & $[-0.444,0.460]$ \\
+Twelve month & Employment exit & 0.0166 & 0.0167 & 0.0210 & $[-1.570,1.575]$ \\
+Twelve month & Unemployment entry & 0.0119 & 0.0128 & 0.0130 & $[-1.573,1.572]$ \\
+Twelve month & Labor-force exit & 0.0047 & 0.0039 & 0.0080 & $[-1.576,1.569]$ \\
+\\bottomrule
+\\end{tabular}
+\\tabnote{The first three numerical columns are descriptive linear-probability Q5--Q1, young--older, post--pre differences, not nonlinear regression coefficients. Observable poststratification assumes
outcome missingness is independent of linkage within fixed observed strata. Accounting intervals impose no conditional-MAR assumption or Lee trimming and may exceed the probability-contrast
parameter space because they bound the linear regression contrast jointly over source records.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_heterogeneity.tex b/paper/tables/r3_appendix_heterogeneity.tex
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index 0000000..7691df8
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+++ b/paper/tables/r3_appendix_heterogeneity.tex
@@ -0,0 +1,20 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Industry, education, and exact-age comparisons}\\label{tab:app_heterogeneity}
+\\resizebox{\\textwidth}{!}{\\%
+\\begin{tabular}{llrrrr}
+\\toprule
+Model & Occupations & Coefficient & Pointwise 95\\% interval & Simultaneous 95\\% interval \\
+\\midrule
+Occupation--industry-cell baseline & 468 & $-0.138$ & $[-0.228,-0.048]$ & --- \\
+Industry-conditioned & 468 & $-0.099$ & $[-0.188,-0.009]$ & --- \\
+BA or higher & 429 & $-0.138$ & $[-0.275,-0.001]$ & $[-0.293,0.017]$ \\
+No BA & 429 & $-0.123$ & $[-0.231,-0.014]$ & $[-0.243,-0.002]$ \\
+Age 22 & 439 & $-0.093$ & $[-0.212,0.026]$ & $[-0.239,0.053]$ \\
+Age 23 & 439 & $-0.140$ & $[-0.252,-0.028]$ & $[-0.280,-0.0003]$ \\
+Age 24 & 439 & $-0.222$ & $[-0.358,-0.086]$ & $[-0.394,-0.050]$ \\
+Age 25 & 439 & $-0.082$ & $[-0.188,0.024]$ & $[-0.212,0.047]$ \\
+\\bottomrule
+\\end{tabular}
+\\tabnote{Both industry rows use occupation--broad-industry--age--month weighted-stock cells. The baseline includes common Q2--Q5 and Webb interactions; the conditioned row additionally includes 13
broad-industry-specific young-by-post shifts. Education and age collections use their own common occupational support and 9,999 common occupation multipliers. The four-age common-score
equality test gives $\\chi^2(3)=8.21$, $p=.0419$, but every simultaneous age-minus-pooled interval includes zero; no individual age departure is promoted.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_honestdid.tex b/paper/tables/r3_appendix_honestdid.tex
new file mode 100644
index 0000000..bc06ef8
--- /dev/null
+++ b/paper/tables/r3_appendix_honestdid.tex
@@ -0,0 +1,13 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Official HonestDiD sensitivity intervals}\\label{tab:app_honestdid}
+\\small
+\\begin{tabular}{llrrr}
+\\toprule
+Structure & Calibration & Restriction & Lower & Upper \\
+\\midrule

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+Pooled & 2017Q1--2022Q3 & Conventional &  $\beta = -0.264$  &  $\beta = 0.024$  \\
+ & & Smoothness,  $\beta = 0$  &  $\beta = -0.140$  &  $\beta = 0.016$  \\
+ & & Smoothness,  $\beta = 0.005$  &  $\beta = -0.482$  &  $\beta = 0.516$  \\
+ & & Relative magnitude,  $\beta = 0$  &  $\beta = -0.260$  &  $\beta = 0.019$  \\
+ & & Relative magnitude,  $\beta = 0$  &  $\beta = -1.023$  &  $\beta = 0.789$  \\
+Pooled & 2021Q1--2022Q3 & Conventional &  $\beta = -0.264$  &  $\beta = 0.024$  \\
+ & & Smoothness,  $\beta = 0$  &  $\beta = -0.230$  &  $\beta = 0.204$  \\
+ & & Smoothness,  $\beta = 0.005$  &  $\beta = -0.540$  &  $\beta = 0.495$  \\
+ & & Relative magnitude,  $\beta = 0$  &  $\beta = -0.260$  &  $\beta = 0.019$  \\
+ & & Relative magnitude,  $\beta = 0$  &  $\beta = -0.985$  &  $\beta = 0.742$  \\
+addlinespace
+SOC2 monthly paths & 2017Q1--2022Q3 & Conventional &  $\beta = -0.425$  &  $\beta = 0.010$  \\
+ & & Smoothness,  $\beta = 0$  &  $\beta = -0.351$  &  $\beta = 0.007$  \\
+ & & Smoothness,  $\beta = 0.005$  &  $\beta = -0.869$  &  $\beta = 0.352$  \\
+ & & Relative magnitude,  $\beta = 0$  &  $\beta = -0.420$  &  $\beta = 0.007$  \\
+ & & Relative magnitude,  $\beta = 0$  &  $\beta = -2.222$  &  $\beta = 1.884$  \\
+SOC2 monthly paths & 2021Q1--2022Q3 & Conventional &  $\beta = -0.425$  &  $\beta = 0.010$  \\
+ & & Smoothness,  $\beta = 0$  &  $\beta = -0.772$  &  $\beta = 0.007$  \\
+ & & Smoothness,  $\beta = 0.005$  &  $\beta = -1.100$  &  $\beta = 0.207$  \\
+ & & Relative magnitude,  $\beta = 0$  &  $\beta = -0.420$  &  $\beta = 0.007$  \\
+ & & Relative magnitude,  $\beta = 0$  &  $\beta = -2.222$  &  $\beta = 1.884$  \\
+\\bottomrule
+\\end{tabular}
+\\tabnote{The target is the observed-month-weighted average of the 15 post-quarter Q5--Q1 coefficients relative to 2022Q4. Intervals use the full occupation-cluster covariance and official HonestDiD
0.2.8. Smoothness  $\beta$  is in log points per quarter squared;  $\beta$  bounds post changes relative to the largest signed preperiod change. Both the full 2017Q1--2022Q3 calibration and the
consecutive post-reopening 2021Q1--2022Q3 calibration are fixed and reported. The recent-window family-path interval excludes zero only at the exact-linear-bias restriction;  $\beta = 0$  and includes
zero at  $\beta = 0.005$ . Because each conventional interval includes zero, positive zero-exclusion robustness is undefined, not an omitted threshold.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_lofo.tex b/paper/tables/r3_appendix_lofo.tex
new file mode 100644
index 0000000..7e277e1
--- /dev/null
+++ b/paper/tables/r3_appendix_lofo.tex
@@ -0,0 +1,20 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Largest leave-one-family-out movements for the conditioned Q5 target}\\label{tab:app_lofo}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{lrrrr}
+\\toprule
+Omitted SOC2 & Family & Preperiod stock & (\\%) & Re-estimated coefficient & Movement & [95\\% paired CI] \\
+\\midrule
+43 & Office and Administrative Support & 11.21 & & -0.0614 & & [-0.1131,0.0337] \\
+35 & Food Preparation and Serving Related & 3.27 & & 0.0174 & & [-0.0292,0.1074] \\
+31 & Healthcare Support & 3.66 & & -0.0534 & & [-0.0772,0.0139] \\
+13 & Business and Financial Operations & 6.22 & & 0.0099 & & [-0.0006,0.0638] \\
+39 & Personal Care and Service & 2.51 & & -0.0457 & & [-0.0496,0.0015] \\
+25 & Educational Instruction and Library & 4.53 & & -0.0351 & & [-0.0383,0.0115] \\
+17 & Architecture and Engineering & 2.28 & & -0.0339 & & [-0.0330,0.0085] \\
+33 & Protective Service & 2.28 & & -0.0096 & & [-0.0050,0.0291] \\
+\\bottomrule
+\\end{tabular}}
+\\end{table}
+\\tabnote{The parent SOC2-by-calendar-month coefficient is  $\beta = 0.0217$ . Families are ranked by the absolute coefficient movement, so this is an outcome-informed influence diagnostic rather than a
preferred deletion rule. Quintiles and the continuous centering scale remain fixed; all 22 deletions are available in the machine-readable file.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_mapping_tilts.tex b/paper/tables/r3_appendix_mapping_tilts.tex
new file mode 100644
index 0000000..ad2aa4
--- /dev/null
+++ b/paper/tables/r3_appendix_mapping_tilts.tex
@@ -0,0 +1,16 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Current-contract bridge-allocation sensitivity}\\label{tab:app_mapping_tilts}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{llrrr}
+\\toprule
+Model & Exposure labels & Coefficient range & Paired intervals excluding zero \\
+\\midrule
+Pooled & Fixed at  $\beta = 1$  &  $\beta = [-0.1433, -0.1167]$  &  $\beta = 0$  of 6 \\
+Pooled & Rebuilt at each  $\beta$  &  $\beta = [-0.1446, -0.1163]$  &  $\beta = 0$  of 6 \\
+Family by month & Fixed at  $\beta = 1$  &  $\beta = [-0.0748, 0.0409]$  &  $\beta = 0$  of 6 \\
+Family by month & Rebuilt at each  $\beta$  &  $\beta = [-0.0783, 0.0439]$  &  $\beta = 5$  of 6 \\
+\\bottomrule
+\\end{tabular}}
+\\end{table}
+\\tabnote{The grid is  $\beta \in \{0.25, 0.5, 2/3, 1, 1.5, 2, 4\}$ , where  $\beta$  is the relative young-versus-older high/low allocation odds for jointly feasible splits. Ranges summarize coefficients, not confidence
intervals. The last column counts occupation-clustered paired intervals for the six nonbaseline points relative to  $\beta = 1$ . All rows are post-outcome mapping sensitivities; the grid is neither
an identified set nor a sharp bound.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_nonlambda_architecture.tex b/paper/tables/r3_appendix_nonlambda_architecture.tex
new file mode 100644
index 0000000..b028017
--- /dev/null
+++ b/paper/tables/r3_appendix_nonlambda_architecture.tex
@@ -0,0 +1,19 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Non- $\lambda$  exposure definitions on the current CPS contract}\\label{tab:appendix_nonlambda_architecture}
+\\scriptsize
+\\setlength{\\tabcolsep}{3pt}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{lrrrrrr}
+\\toprule
+Exposure definition & Occupations & Pooled coefficient & [95\\% CI] & Alternative  $\beta$ - $\beta$  & [95\\% CI] & Family-month coefficient & [95\\% CI] & Alternative  $\beta$ - $\beta$  & [95\\% CI] \\
+\\midrule
+AIOE, administrative/equal component & 466 &  $\beta = -0.072$  & [-0.143, -0.001] &  $\beta = 0.064$  & [-0.025, 0.152] &  $\beta = 0.065$  & [-0.050, 0.181] &  $\beta = 0.095$  & [-0.063, 0.253] \\
+AIOE, direct ability & 455 &  $\beta = -0.099$  & [-0.174, -0.024] &  $\beta = 0.027$  & [-0.059, 0.113] &  $\beta = 0.024$  & [-0.084, 0.132] &  $\beta = 0.011$  & [-0.143, 0.165] \\
+AIOE, source-employment weighted & 456 &  $\beta = -0.072$  & [-0.145, 0.000] &  $\beta = 0.068$  & [-0.020, 0.157] &  $\beta = 0.071$  & [-0.044, 0.186] &  $\beta = 0.114$  & [-0.047, 0.274] \\
+Webb artificial intelligence & 443 &  $\beta = -0.068$  & [-0.160, 0.023] &  $\beta = 0.067$  & [-0.036, 0.171] &  $\beta = -0.079$  & [-0.177, 0.019] &  $\beta = -0.036$  & [-0.203, 0.131] \\
+OECD reversed AI-skills gap & 448 &  $\beta = -0.014$  & [-0.096, 0.067] &  $\beta = 0.119$  & [-0.0004, 0.238] &  $\beta = 0.069$  & [-0.033, 0.171] &  $\beta = 0.065$  & [-0.125, 0.255] \\
+\\bottomrule
+\\end{tabular}}
+\\end{table}
+\\tabnote{Rows use each alternative's native finite-score support. The outcome cells, age contrast, calendar, estimator, Webb-software control, and preperiod weighting rule are fixed. Each beta
anchor is rebuilt on exactly the alternative row's support. Coefficients compare Q5 with Q1; intervals and paired differences use 9,999 common occupation-level Rademacher score multipliers. A
null-containing paired interval is a nondetection, not equivalence. Continuous-score estimates and raw-to-weighted-SD scaling are preserved in the public result files.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_pairwise_aggregates.tex b/paper/tables/r3_appendix_pairwise_aggregates.tex
new file mode 100644
index 0000000..b42f9cf9
--- /dev/null
+++ b/paper/tables/r3_appendix_pairwise_aggregates.tex
@@ -0,0 +1,12 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Aggregates of supported within-family quintile contrasts}\\label{tab:app_pairwise_aggregates}
+\\begin{tabular}{lrrrr}
+\\toprule
+Contrast & Eligible families & Estimate & Standard error & Normal  $\beta$ -value \\
+\\bottomrule
+\\end{tabular}

```

```

+\\midrule
+Q2--Q1 & 14 & $-0.0095$ & 0.0368 & .796 \\
+Q3--Q1 & 12 & $\\phantom{-}0.0132$ & 0.0541 & .807 \\
+Q4--Q1 & 4 & $-0.0618$ & 0.0577 & .284 \\
+Q5--Q1 & 4 & $\\phantom{-}0.1020$ & 0.0781 & .192 \\
+Q3--Q2 & 14 & $\\phantom{-}0.0702$ & 0.0348 & .044 \\
+Q4--Q2 & 5 & $-0.0033$ & 0.0354 & .926 \\
+Q5--Q2 & 6 & $\\phantom{-}0.1177$ & 0.0498 & .018 \\
+Q4--Q3 & 9 & $-0.0841$ & 0.0458 & .066 \\
+Q5--Q3 & 11 & $-0.0446$ & 0.0965 & .644 \\
+Q5--Q4 & 10 & $-0.1082$ & 0.0485 & .026 \\
+\\bottomrule
+\\end{tabular}
+\\tabnote{For each pair, only families containing both listed quintiles enter. Family-specific edge coefficients are aggregated using fixed pair-endpoint preperiod employment-stock weights, and the
sequence. Normal $p$-values are descriptive and were not used to select rows; all 89 underlying supported family-specific edges and their simultaneous intervals remain machine readable.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_population_eras.tex b/paper/tables/r3_appendix_population_eras.tex
new file mode 100644
index 0000000..00a63c2
--- /dev/null
+++ b/paper/tables/r3_appendix_population_eras.tex
@@ -0,0 +1,16 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Post-outcome exploratory comparison of early and later postperiods}\\label{tab:appendix_population_eras}
+\\small
+\\setlength{\\tabcolsep}{4pt}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{lrrrr}
+\\toprule
+Cell construction & 2023--2024 coefficient [95\\% CI] & 2025--2026 coefficient [95\\% CI] & Later minus early [95\\% CI] & Paired $\\mathrm{MDE}_{.80}$ \\
+\\midrule
+Official final-weight stocks & $-0.111$; [-0.197, -0.024] & $-0.166$; [-0.270, -0.062] & $-0.056$; [-0.123, 0.012] & $0.097$ \\
+Respondent-equivalent routed counts & $-0.103$; [-0.193, -0.012] & $-0.183$; [-0.277, -0.089] & $-0.080$; [-0.139, -0.021] & $0.085$ \\
+\\bottomrule
+\\end{tabular}}
+\\tabnote{This timing comparison was produced after outcomes were opened and is post-outcome exploratory. Both rows use the corrected 113-month calendar, the same 468 occupations, a joint two-era
specification, and 9,999 common occupation multipliers within each paired later-minus-early contrast. Official final-weight stocks are the paper's employment-stock object. Respondent-
equivalent values are fractional routed-record counts after occupational crosswalking; they are neither distinct survey respondents nor a recovered counterfactual CPS weight vintage. The
official-weight interval does not detect an era difference, whereas the respondent-equivalent interval excludes zero. That divergence does not identify response composition or show that
population-control revisions caused the later estimate.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_precision_summary.tex b/paper/tables/r3_appendix_precision_summary.tex
new file mode 100644
index 0000000..fb0883c
--- /dev/null
+++ b/paper/tables/r3_appendix_precision_summary.tex
@@ -0,0 +1,29 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Consolidated normal-theory precision diagnostics}\\label{tab:appendix_precision_summary}
+\\scriptsize
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{lll r}
+\\toprule
+Analysis & Quantity & Scale & $\\mathrm{MDE}_{.80}$ \\
+\\midrule
+Core pooled & Q5--Q1 coefficient & Log young-to-older stock ratio & 0.1266 \\
+Family-month & Q5--Q1 coefficient & Log young-to-older stock ratio & 0.1998 \\
+Family-month vs. pooled & Paired coefficient difference & Log young-to-older stock ratio & 0.1454 \\
+Direct tail within family & Q5--Q1 coefficient & Log young-to-older stock ratio & 0.4576 \\
+Continuous within family & Exposure slope & One within-family exposure SD & 0.0313 \\
+Computer-use conditioning & Paired coefficient difference & Log young-to-older stock ratio & 0.1131 \\
+Remotability conditioning & Paired coefficient difference & Log young-to-older stock ratio & 0.0510 \\
+Industry conditioning & Paired coefficient difference & Log young-to-older stock ratio & 0.0444 \\
+BA vs. non-BA & Paired coefficient difference & Log young-to-older stock ratio & 0.2014 \\
+Architecture comparisons & Paired coefficient differences & Log young-to-older stock ratio & 0.0609--0.1689 \\
+Adjacent exit & Q5--Q1 coefficient & Log conditional transition-rate ratio & 0.254 \\
+Adjacent occupation change & Q5--Q1 coefficient & Log conditional transition-rate ratio & 0.174 \\
+Adjacent entry destination & Q5--Q1 coefficient & Log conditional allocation ratio & 0.260 \\
+Twelve-month exit & Q5--Q1 coefficient & Log conditional transition-rate ratio & 0.336 \\
+Twelve-month occupation change & Q5--Q1 coefficient & Log conditional transition-rate ratio & 0.115 \\
+Twelve-month entry destination & Q5--Q1 coefficient & Log conditional allocation ratio & 0.282 \\
+\\bottomrule
+\\end{tabular}}
+\\tabnote{$\\mathrm{MDE}_{.80}=(1.96+0.84)\\widehat{SE}$ is the two-sided five-percent normal-theory minimum detectable quantity at 80-percent power. It summarizes resolution on the displayed scale; it
is not a wild-score critical value, equivalence margin, or acceptance threshold. Architecture entries report the observed range across seven beta-minus-alternative comparisons. Flow rows use
their own risk sets and should not be compared as if they estimated one common probability.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_pretrend_windows.tex b/paper/tables/r3_appendix_pretrend_windows.tex
new file mode 100644
index 0000000..be600e1
--- /dev/null
+++ b/paper/tables/r3_appendix_pretrend_windows.tex
@@ -0,0 +1,21 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Preperiod joint tests across fixed calendar windows}\\label{tab:app_pretrend_windows}
+\\resizebox{\\textwidth}{!}{%
+\\begin{tabular}{llrr}
+\\toprule
+Structure & Window & Equal to 2022Q4 & Equal within window \\
+\\midrule
+Pooled & 2017Q1--2022Q3 & 0.0171 & 0.0121 \\
+Pooled & 2017Q1--2019Q4 & 0.1365 & 0.0925 \\
+Pooled & 2021Q1--2022Q3 & 0.4285 & 0.3281 \\
+Pooled & Full, excluding 2020Q2--Q4 & 0.0772 & 0.0676 \\
+\\addlinespace
+Family by month & 2017Q1--2022Q3 & 0.0053 & 0.0641 \\
+Family by month & 2017Q1--2019Q4 & 0.0126 & 0.0744 \\
+Family by month & 2021Q1--2022Q3 & 0.0116 & 0.3264 \\
+Family by month & Full, excluding 2020Q2--Q4 & 0.0079 & 0.1048 \\
+\\bottomrule
+\\end{tabular}}
+\\tabnote{Entries are common-multiplier $p$-values. The third column tests every selected preperiod coefficient against the omitted 2022Q4 coefficient, which represents October--November 2022. The
fourth tests equality among coefficients inside the selected window and is invariant to the omitted reference. The two nulls therefore answer different questions. Windows were declared before
this result audit and are not selected for favorable $p$-values.}
+\\end{table}
diff --git a/paper/tables/r3_appendix_primitive_ds.tex b/paper/tables/r3_appendix_primitive_ds.tex
new file mode 100644
index 0000000..85c0dd3
--- /dev/null
+++ b/paper/tables/r3_appendix_primitive_ds.tex
@@ -0,0 +1,19 @@
+\\begin{table}[htbp]
+\\centering
+\\caption{Post-outcome exploratory direct and software-complement primitives}\\label{tab:appendix_primitive_ds}

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new file mode 100644
index 0000000..3b46fc1
--- /dev/null
+++ b/paper/tables/r3_appendix_webb_support.tex
@@ -0,0 +1,17 @@
\begin{table}[htbp]
\centering
\caption{Post-outcome exploratory Webb conditioning and support}\label{tab:appendix_webb_support}
\small
\setlength{\tabcolsep}{4pt}
\resizebox{\textwidth}{!}{%
\begin{tabular}{lrrrr}
\toprule
+Specification & Occupations & Beta coefficient & [95\% CI] & Same-support paired change [95\% CI] \\
\midrule
+Without Webb, fixed primary support & 468 &  $\beta = -0.134$  & [-0.219, -0.049] &  $\beta = -0.002$  & [-0.010, 0.007] \\
+With Webb, fixed primary support & 468 &  $\beta = -0.132$  & [-0.220, -0.044] & & Reference \\
+Without Webb, broader beta-valid support & 490 &  $\beta = -0.135$  & [-0.218, -0.051] & & Not paired \\
\bottomrule
\end{tabular}}
\end{table}
\tabnote{All entries are post-outcome exploratory. The first two rows hold occupations, beta cutoffs, and the outcome specification fixed; the displayed paired change is without-Webb minus with-Webb, uses common draws, and has  $\mathbf{MDE}_{\{80\}} = 0.013$ . Its interval containing zero is a nondetection, not equivalence. The final row relaxes Webb availability and therefore changes support and beta cutoffs; it has no paired interpretation. Webb conditioning is an occupational-software proxy, not realized software or AI adoption.}
\end{table}
diff --git a/paper/tables/r3_appendix_young_composition.tex b/paper/tables/r3_appendix_young_composition.tex
new file mode 100644
index 0000000..bebe15
--- /dev/null
+++ b/paper/tables/r3_appendix_young_composition.tex
@@ -0,0 +1,39 @@
\begin{table}[htbp]
\centering
\caption{Education, enrollment, and age composition of ages 22--25}\label{tab:app_young_composition}
\textit{Panel A: Annual enrollment and BA-or-higher shares}\par\smallskip
\resizebox{\textwidth}{!}{%
\begin{tabular}{lrrrrrrrr}
\toprule
+Year & Q1 & Q2 & Q3 & Q4 & Q5 & National population & National population (millions/month) \\
\midrule
+2017 & .135/.085 & .201/.239 & .191/.397 & .142/.473 & .171/.477 & .244/.275 & 17.376 \\
+2018 & .113/.089 & .207/.232 & .188/.439 & .149/.470 & .171/.484 & .239/.281 & 17.295 \\
+2019 & .121/.080 & .208/.238 & .168/.449 & .151/.489 & .162/.518 & .236/.293 & 16.865 \\
+2020 & .121/.098 & .190/.247 & .173/.463 & .125/.517 & .158/.562 & .232/.314 & 16.848 \\
+2021 & .118/.101 & .177/.245 & .167/.444 & .149/.474 & .164/.555 & .223/.301 & 16.541 \\
+2022 & .120/.088 & .186/.253 & .159/.459 & .130/.489 & .152/.538 & .225/.302 & 16.673 \\
+2023 & .118/.110 & .181/.245 & .154/.470 & .123/.480 & .152/.534 & .219/.306 & 17.659 \\
+2024 & .121/.109 & .195/.249 & .147/.469 & .129/.517 & .136/.559 & .221/.315 & 17.369 \\
+2025 & .121/.113 & .189/.269 & .157/.463 & .128/.493 & .151/.580 & .222/.316 & 17.638 \\
+2026 & .124/.096 & .186/.283 & .164/.470 & .127/.516 & .140/.575 & .222/.313 & 17.815 \\
\bottomrule
\end{tabular}}
+
\medskip
\textit{Panel B: Preperiod/postperiod age composition}\par\smallskip
\resizebox{\textwidth}{!}{%
\begin{tabular}{lrrrrrrrr}
\toprule
+Group & Mean age & YEAR--age proxy & Age 22 & Age 23 & Age 24 & Age 25 \\
\midrule
+Q1 & 23.482/23.486 & 1995.9/2000.8 & .257/.251 & .251/.250 & .245/.261 & .247/.238 \\
+Q2 & 23.481/23.469 & 1995.9/2000.9 & .256/.253 & .253/.257 & .244/.258 & .246/.232 \\
+Q3 & 23.642/23.638 & 1995.8/2000.7 & .198/.200 & .248/.241 & .267/.279 & .286/.280 \\
+Q4 & 23.690/23.681 & 1995.8/2000.6 & .190/.184 & .235/.245 & .270/.276 & .305/.295 \\
+Q5 & 23.657/23.674 & 1995.8/2000.6 & .194/.192 & .241/.236 & .280/.279 & .286/.293 \\
+National & 23.514/23.506 & 1995.9/2000.8 & .247/.246 & .248/.249 & .250/.258 & .255/.247 \\
\bottomrule
\end{tabular}}
\end{table}
\tabnote{Q1--Q5 condition on employment and the fixed 468-occupation support. "National population" uses every positive-WTFFINL CPS record ages 22--25 and assigns neither an occupation nor an exposure quintile to nonworkers. Entries in Panel A before the final column are enrollment share/BA-or-higher share; entries in Panel B are preperiod/postperiod. The preperiod is January 2017--November 2022 and the postperiod January 2023--July 2026; December 2022 is omitted, October 2025 is absent, and 2026 is partial. Annual national population is the person-weight sum divided by observed months. YEAR minus age is an approximate birth-year proxy, not an age-period-cohort coefficient. Complete annual age/cohort cells remain in the machine-readable composition files.}
\end{table}
diff --git a/paper/tables/r3_table1_data.tex b/paper/tables/r3_table1_data.tex
new file mode 100644
index 0000000..183bf15
--- /dev/null
+++ b/paper/tables/r3_table1_data.tex
@@ -0,0 +1,20 @@
\begin{table}[htbp]
\centering
\caption{Corrected CPS reconstruction and analysis support}\label{tab:data}
\begin{tabular}{lrr}
\toprule
+ & Count & Definition \\
\midrule
+Expected calendar months & 115 & Jan. 2017--July 2026 \\
+Observed source months & 114 & October 2025 absent \\
+Static analysis months & 113 & December 2022 excluded \\
+Eligible employed source records & 5,322,047 & ages 18--65, positive final weight \\
+Route-expanded rows & 6,188,956 & fractional descendants allowed \\
+Primary occupations & 468 & finite beta and Webb; positive pre-stock \\
+Occupation-month grid & 52,884 & 468 occupations  $\times$  113 months \\
+Positive-total fitted cells & 51,891 & one-sided zeros retained \\
+Both-age-zero cells & 993 & no likelihood contribution \\
\bottomrule
\end{tabular}
\end{table}
\tabnote{Source records are not independent observations for regression inference. Before 2020, one source record can generate several fractional crosswalk descendants. Counts, routing checks, named support, exclusions, and restricted-input hashes are generated by the corrected-calendar data audit.}
\end{table}
diff --git a/paper/tables/r3_table2_occupation.tex b/paper/tables/r3_table2_occupation.tex
new file mode 100644
index 0000000..8ceaf01
--- /dev/null
+++ b/paper/tables/r3_table2_occupation.tex
@@ -0,0 +1,13 @@
\begin{table}[htbp]
\centering
\caption{Pooled and broad-family exposure comparisons}\label{tab:occupation}
\begin{tabular}{lrrrr}
\toprule
+Specification & Coefficient & 95\% interval & Paired movement &  $\mathbf{MDE}_{\{80\}}$  \\
\midrule
+Pooled Q5--Q1 &  $\beta = -0.132$  &  $\beta = [-0.2206, -0.0437]$  & & 0.1266 \\
+SOC2  $\times$  young &  $\beta = -0.0217$  &  $\beta = [-0.1607, 0.1173]$  & & 0.1104 & 0.1998 \\
\bottomrule
\end{tabular}
\end{table}
\tabnote{The weighted-stock criterion is the grouped-binomial conditional likelihood for the two age-specific employment-stock counts. Coefficients parameterize log ratios of conditional means; no
```

observed log ratio is constructed, and one-sided zero-stock cells remain in the criterion. The second row changes the conditioning estimand. Intervals use 9,999 occupation wild-score draws. The paired movement is the second-row coefficient minus the first on common draws; its interval is $[0.0107, 0.2102]$ and its $\mathbf{MDE}_{\{80\}}$ is 0.1454. MDEs are normal-theory precision diagnostics, not rejection thresholds.)

```

\end{table}
diff --git a/paper/tables/r3_table2b_family_quintile_support.tex b/paper/tables/r3_table2b_family_quintile_support.tex
new file mode 100644
index 0000000..8c147c5
--- /dev/null
+++ b/paper/tables/r3_table2b_family_quintile_support.tex
@@ -0,0 +1,36 @@
\begin{table}[htbp]
\centering
\caption{Exposure-quintile support within broad occupation families}\label{tab:family_quintile_support}
\scriptsize
\setlength{\tabcolsep}{2.4pt}
\resizebox{\textwidth}{!}{%
\begin{tabular}{r l ccccc c}
\toprule
SOC2 & Occupation family & Q1 & Q2 & Q3 & Q4 & Q5 & Both tails & \\
\midrule
11 & Management & 0/0.00 & 0/0.00 & 15/4.85 & 12/8.45 & 2/0.25 & & \\
13 & Business and financial & 0/0.00 & 0/0.00 & 1/0.74 & 16/2.42 & 11/3.06 & & \\
15 & Computer and mathematical & 0/0.00 & 0/0.00 & 0/0.00 & 1/0.03 & 14/4.29 & & \\
17 & Architecture and engineering & 0/0.00 & 0/0.00 & 3/0.12 & 9/1.49 & 7/0.67 & & \\
19 & Life, physical, and social science & 0/0.00 & 0/0.00 & 12/0.86 & 6/0.34 & 3/0.05 & & \\
21 & Community and social service & 1/0.03 & 5/0.59 & 6/0.69 & 0/0.00 & 0/0.00 & & \\
23 & Legal & 0/0.00 & 0/0.00 & 1/0.91 & 2/0.35 & 1/0.08 & & \\
25 & Education and library & 0/0.00 & 3/3.52 & 2/0.85 & 0/0.00 & 2/0.16 & & \\
27 & Arts, media, and sports & 2/0.03 & 5/0.27 & 3/0.27 & 6/0.57 & 9/0.65 & Yes & \\
29 & Healthcare practitioners & 3/0.20 & 18/1.94 & 11/3.80 & 3/0.21 & 2/0.23 & Yes & \\
31 & Healthcare support & 4/1.58 & 6/1.45 & 3/0.61 & 0/0.00 & 1/0.03 & Yes & \\
33 & Protective service & 3/0.30 & 8/1.05 & 6/0.92 & 0/0.00 & 0/0.00 & & \\
35 & Food preparation and serving & 5/1.14 & 4/1.76 & 0/0.00 & 1/0.37 & 0/0.00 & & \\
37 & Building and grounds maintenance & 3/1.96 & 2/0.34 & 1/0.20 & 0/0.00 & 0/0.00 & & \\
39 & Personal care and service & 7/1.72 & 8/0.56 & 2/0.23 & 0/0.00 & 0/0.00 & & \\
41 & Sales and related & 1/0.03 & 2/1.53 & 3/2.75 & 4/3.48 & 7/2.28 & Yes & \\
43 & Office and administrative support & 0/0.00 & 5/0.80 & 3/0.04 & 7/2.60 & 32/7.77 & & \\
45 & Farming, fishing, and forestry & 3/0.09 & 2/0.05 & 0/0.00 & 0/0.00 & 0/0.00 & & \\
47 & Construction and extraction & 27/5.49 & 1/0.02 & 2/0.62 & 0/0.00 & 0/0.00 & & \\
49 & Installation, maintenance, and repair & 21/2.65 & 8/0.56 & 2/0.35 & 0/0.00 & 0/0.00 & & \\
51 & Production & 39/2.13 & 9/0.72 & 4/1.28 & 0/0.00 & 0/0.00 & & \\
53 & Transportation and material moving & 10/2.69 & 13/4.77 & 2/0.06 & 0/0.00 & 0/0.00 & & \\
\bottomrule
\end{tabular}}
\end{table}
\tabnote{Each Q cell is occupation count/national preperiod employment-stock share in percent. The producer forms each occupation's stock by summing monthly CPS-weighted young and older employment over the 71 observed months from January 2017 through November 2022. Because every occupation contributes the same months, these shares also equal shares based on average-month stock. They sum to 100 percent across the 110 cells; zero cells are shown rather than suppressed. The matrix uses the fixed 468-occupation support. Only four families contain both Q1 and Q5. The pooled regression connects other families through intermediate quintiles and common coefficients, but they do not supply direct within-family tail comparisons. Family totals, beta ranges, and information shares are in Online Appendix Table C.1.}
\end{table}
diff --git a/paper/tables/r3_table3_characteristics.tex b/paper/tables/r3_table3_characteristics.tex
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+++ b/paper/tables/r3_table3_characteristics.tex
@@ -0,0 +1,16 @@
\begin{table}[htbp]
\centering
\caption{Computer-use and broad-family conditioning on common support}\label{tab:characteristics}
\resizebox{\textwidth}{!}{%
\begin{tabular}{llrrrr}
\toprule
Family-month paths & Computer-use control & Occupations & Q5 by post [95\% CI] & Change from pooled baseline [95\% CI] & \\
\midrule
No & No & 455 &  $[-0.1807, -0.0108]$  & & \\
No & Yes & 455 &  $[-0.1966, -0.3161, -0.0771]$  &  $[-0.1009, -0.1772, -0.0245]$  & \\
Yes & No & 455 &  $[-0.0354, -0.0859, 0.1567]$  &  $[-0.1311, 0.0352, 0.2271]$  & \\
Yes & Yes & 455 &  $[-0.0467, -0.1821, 0.0888]$  &  $[-0.0642, 0.1624]$  & \\
\bottomrule
\end{tabular}}
\end{table}
\tabnote{All four post-outcome exploratory rows use the same 455 occupations, rebuilt exposure labels, outcome cells, calendar, and 9,999 common occupation multipliers. The computer-use rating is standardized with preperiod employment weights and interacted with young  $\times$  post. Family-month rows give every SOC2 family its own young-relative calendar-month path. The final column uses covariance-preserving paired draws. The four projections are not additive components, and no residual coefficient is a causal AI effect.}
\end{table}
diff --git a/paper/tables/r3_table4_inference.tex b/paper/tables/r3_table4_inference.tex
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\begin{table}[htbp]
\centering
\caption{Inference sensitivities for the core occupational comparison}\label{tab:inference}
\resizebox{\textwidth}{!}{%
\begin{tabular}{llrrr}
\toprule
Procedure and target & Pooled 95\% CI & Conditioned 95\% CI & Paired-movement 95\% CI & \\
\midrule
\multicolumn{5}{l}{\textit{Same pooled/SOC2-by-calendar-month targets}} \\
Occupation wild score, SOC2-by-month &  $[-0.221, -0.044]$  &  $[-0.161, 0.117]$  &  $[0.011, 0.210]$  & \\
Twenty-two-family Webb wild score &  $[-0.211, -0.053]$  &  $[-0.153, 0.110]$  &  $[0.006, 0.215]$  & \\
\addlinespace
\multicolumn{5}{l}{\textit{Sampling-oriented companion with regenerated preperiod labels}} \\
Household full refit, 399 draws &  $[-0.186, -0.076]$  &  $[-0.124, 0.066]$  &  $[0.035, 0.177]$  & \\
\bottomrule
\end{tabular}}
\end{table}
\tabnote{All rows report the reconstructed pooled, SOC2-by-calendar-month, and paired-movement coefficients. The 22-family row permits common shocks within broad occupational families. The household row uses 399 mean-one linked-household full refits and regenerates preperiod exposure labels; it is conditional on released CPS final weights and is not complete CPS design inference. The procedures target different perturbations and are not mechanically combined.}
\end{table}
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\begin{table}[htbp]
\centering
\caption{Corrected CPS flow margins}\label{tab:flows}
\resizebox{\textwidth}{!}{%
\begin{tabular}{llrrrr}
\toprule
Endpoint and margin & Raw link records at risk / entries & Coefficient & 95\% interval & Normal MDE{80} & \\
\midrule
Adjacent employment exit & 3,346,227 &  $0.132$  &  $[-0.042, 0.307]$  &  $0.254$  & \\
Adjacent reported-occupation change & 3,207,598 &  $0.003$  &  $[-0.118, 0.123]$  &  $0.174$  & \\
Adjacent entry-destination allocation & 96,981 &  $-0.070$  &  $[-0.250, 0.110]$  &  $0.260$  & \\
Twelve-month employment exit & 1,399,376 &  $0.123$  &  $[-0.111, 0.357]$  &  $0.336$  & \\
\end{tabular}}

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+Twelve-month reported-occupation endpoint change & 1,110,024 &  $-\$0.018$  &  $[-\$0.097, \$0.061]$  & 0.115 \\
+Twelve-month entry-destination allocation & 82,573 &  $-\$0.052$  &  $[-\$0.251, \$0.147]$  & 0.282 \\
+\bottomrule
+\end{tabular}}
+\tabnote{Counts are repeated endpoint-link records, not unique persons; an individual may contribute more than one eligible link. Coefficients are post-January-2023 changes in the ages-22--25 versus
ages-26--65, beta-Q5 versus beta-Q1 log rate ratio. Entry destination is a log allocation ratio conditional on an observed entry, not an employment-finding or employer-hiring rate. Primary
intervals use 9,999 occupation wild-score draws and official longitudinal weights.  $MDE_{\{80\}}=2.80\widehat{SE}$  uses the analytic occupation-cluster standard error and is a normal-theory
precision diagnostic, not the wild-score rejection threshold and not evidence that a smaller effect is absent.}
+\end{table}
diff --git a/paper/tables/r3_table6_lambda.tex b/paper/tables/r3_table6_lambda.tex
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+++ b/paper/tables/r3_table6_lambda.tex
@@ -0,0 +1,17 @@
+\begin{table}[\htbp]
+\centering
+\caption{Weighting direct and software-complement tasks}\label{tab:lambda}
+\resizebox{\textwidth}{!}{%
+\begin{tabular}{rrrr}
+\toprule
+ $\lambda$  in  $E_1+\lambda E_2$  & Q5--Q1 coefficient & Difference from  $\lambda=0.5$  [95% CI] & Stock changing group (\%) \\
+\midrule
+0.00 &  $-\$0.103$  &  $[-\$0.044, \$0.103]$  & 61.6 \\
+0.25 &  $-\$0.101$  &  $[-\$0.005, \$0.068]$  & 25.5 \\
+0.50 &  $-\$0.132$  & --- & 0.0 \\
+0.75 &  $-\$0.143$  &  $[-\$0.039, \$0.017]$  & 10.2 \\
+1.00 &  $-\$0.156$  &  $[-\$0.064, \$0.017]$  & 25.5 \\
+\bottomrule
+\end{tabular}}
+\end{table}
+\tabnote{All rows use 468 occupations, reconstructed preperiod weights, tie-preserving native groups, Webb software conditioning, and 9,999 common occupation draws. ‘‘Stock changing group’’ is
preperiod employment in occupations whose quintile differs from beta ( $\lambda=0.5$ ). The coefficient path is descriptive of the capability boundary; no paired beta comparison detects a
difference at five percent.}
+\end{table}

```